The 6th meeting of Board of Studies (BoS) was held on **8th October 2024 at 10.30 AM** in the Board Room with Dean in the Chair.

The following members were present for the BoS meeting

Board of Studies Members

Sl. No.	Name of the member with designation & official address	Members as per UGC norms
1.	**Prof.Dr. K. Vivek M.P.T. Dean I/C,** College of Physiotherapy- S.M.V.E.C., Madagadipet, Puducherry–605107. Email: deanphysio@smvec.ac.in Mobile: 9791718152	Chairman
2.	**Prof. Dr. K. Balu M.P.T.,** Professor, School of Physiotherapy, Aarupadai Veedu Medical College & Hospital, Puducherry – 607403. Email:balukhanphysio007@gmail.com Mobile:9486282929	External Member (University nominee)
3.	**Prof. Dr. A. Dinakaran M.P.T.,** Principal, Adhiparasakthi College of Physiotherapy Melmaruvathur– 603319. Email:apcopt1994@gmail.com Mobile:8248229285	External Member (Academic council nominee)
4.	**Prof. KSI. Dr. Murali Sankar M.P.T.,** Director, School of Physiotherapy, Aarupadai Veedu Medical College & Hospital, Puducherry – 607403. Email: Director.sptpdy@vmu.edu.in Mobile:9366666683	External Member (Academic council nominee)
5.	**Dr. S. Prabagaran B.P.T.,** Kiran Physio Clinic, Villupuram-605401. Email:selvaprabagaran@gmail.com Mobile:9840581118	Representative From Industry
6.	**Dr. K. Ezhilan M.P.T.,** Chief Physiotherapist, SMVMCH, Puducherry–605107. Email:ezhilan.mpt@gmail.com Mobile:9894857753	Special Member (Subject expert)

7.	**Dr. K. Manikandan M.S.(ortho.),** Associate Professor, Department of Orthopaedics, SMVMCH, Puducherry–605107. Email:manikandan7008@gamil.com	Special invitee (Subject expert)
8	**Dr. J. Sathyanarayanan M.D.,** Associate Professor, Department of Medicine, SMVMCH, Puducherry–605107. Email:drisathiyanarayannan@gmail.com Mobile:9944540888	Special invitee (Subject expert)
9.	**Dr. K. Karthikeyan, MD** Dean (Academic), SMVMCH. Email:deanaccd@smvmch.ac.in	Special invitee
10.	**Dr. M. Pragash,MBBS., D.Ortho., D.N.B.(Ortho),** Medical Superintendent SMVMCH. Email:ms@smvmch.ac.in	Special invitee
14.	**Dr. A. A. Arivalagar M.Tech. Ph.D.** Dean academics (other schools), Sri Manakula Vinayagar Engineering College, Puducherry-605107. Email:deanacademic1@smvec.ac.in	Special invitee
13.	**Dr. S. Anbumalar M.E. Ph.D.,** Dean Academics (Engineering), Sri Manakula Vinayagar Engineering College, Puducherry-605107. Email: deanacademic@smvec.ac.in	Special invitee
14.	**Dr. V. Velkumar M.P.T. (PhD).,** Assistant Professor, College of Physiotherapy- S.M.V.E.C., Madagadipet, Puducherry–605107.	Internal Members
15.	**Dr. V. Balakannan M.P.T.,** Assistant Professor, College of Physiotherapy- S.M.V.E.C., Madagadipet, Puducherry–605107.	Internal Members
16.	**Dr. R. Prabakaran M.P.T.,** Assistant Professor, College of Physiotherapy- S.M.V.E.C., Madagadipet, Puducherry–605107.	Internal Members
17.	**Dr. R. Murali M.P.T.,** Assistant Professor, College of Physiotherapy- S.M.V.E.C., Madagadipet, Puducherry–605107.	Internal Members

SMVEC

COLLEGE OF PHYSIOTHERAPY

(Approved by Government of Puducherry and Affiliated to Pondicherry University)

Madagadipet, Puducherry - 605 107

Venue: Board Room, SMVEC 08-10-2024

6th Board of Studies meeting Attendance Sheet

Board of Studies Members,

Sl. No.	Name of the member with designation & official address	Members as per UGC norms	Signature
1.	**Prof.Dr. K. Vivek M.P.T. Dean IC,** College of Physiotherapy- S.M.V.E.C., Madagadipet, Puducherry–605107. Email: deanphysio@smvec.ac.in Mobile: 9791718152	Chairman	
2.	**Prof. Dr. K. Balu M.P.T.,** Professor, School of Physiotherapy, Aarupadai Veedu Medical College & Hospital, Puducherry – 607403. Email:balukhanphysio007@gmail.com Mobile:9486282929	External Member (University nominee)	
3.	**Prof. Dr. A. Dinakaran M.P.T.,** Principal, Adhiparasakthi College of Physiotherapy Melmaruvathur–603319. Email:apcopt1994@gmail.com Mobile:8248229285	External Member (Academic council nominee)	9/10/24
4.	**Prof. KSI. Dr. Murali Sankar M.P.T.,** Director, School of Physiotherapy, Aarupadai Veedu Medical College &Hospital, Puducherry – 607403. Email: Director.sptpdy@vmu.edu.in Mobile:9366666683	External Member (Academic council nominee)	
5.	**Dr. S. Prabagaran B.P.T.,** Kiran Physio Clinic, Villupuram-605401. Email:selvaprabagaran@gmail.com Mobile:9840581118	Representative From industry	

6.	**Dr. K. Ezhilan M.P.T.,** Chief Physiotherapist, SMVMCH, Puducherry–605107. Email:ezhilan.mpt@gmail.com Mobile:9894857753	Special Member (Subject expert)	
7.	**Dr. K. Manikandan M.S.(ortho.),** Associate Professor, Department of Orthopaedics, SMVMCH, Puducherry–605107. Email:manikandan7008@gamil.com	Special invitee (Subject expert)	
8.	**Dr. J. Sathyanarayanan M.D.,** Associate Professor, Department of Medicine, SMVMCH, Puducherry–605107. Email:drisathiyanarayanan@gmail.com Mobile:9944540888	Special invitee (Subject expert)	
9.	**Dr. K. Karthikeyan, MD** Dean (Academic), SMVMCH. Email:deanaccd@smvmch.ac.in	Special invitee	
10.	**Dr. M. Pragash,MBBS., D.Ortho., D.N.B.(Ortho),** Medical Superintendent SMVMCH. Email:ms@smvmch.ac.in	Special invitee	
11.	**Dr. A. A. Arivalagar M.Tech. Ph.D.** Dean academics (other schools), Sri Manakula Vinayagar Engineering College, Puducherry-605107. Email:deanacademic1@smvec.ac.in	Special invitee	

Agenda of the Meeting	
BoS/2024/BPT/6.1	Confirmation of the minutes of the 5th BOS meeting held on 25th January 2024. ❖ **Syllabus and curriculum structure for semesters 7 and 8 were approved.** ❖ **Clinical internship and rotatory postings across different specialties were outlined.** ❖ **Introduction of external examiners and exam setting protocols was discussed.**
BoS/2024/BPT/6.2	Appraisal and Approval on the implementation of the 7th & 8th semester syllabus.
BoS/2024/BPT/6.3	Appraisal and Approval of the guidelines and duration for clinical postings in SMVMCH (Compulsory Rotatory Internship).
BoS/2024/BPT/6.4	Approval of external examiners and question paper setters for the academic year 2024-25.
BoS/2024/BPT/6.5	Guidelines for managing arrears in auxiliary courses.
BoS/2024/BPT/6.6	Any other point with permission of the chair.

Mr. Vivek. K Chair person, BoS opened the meeting by welcoming all the members and the meeting there after deliberated on Agenda items that had been approved by members of BoS.

6th Board of Studies Minutes of the Meeting

Sl.no.	Agenda	Discussion	Action taken
1	Welcome and Introductory session	Mr. K. Vivek, Chairperson, BoS opened the meeting by welcoming all the members and introduced the internal members to the external members there after deliberated on agenda items that had been approved by the Chairperson	
2	**BoS/2024/BPT/6.1** Confirmation of the minutes of the 5th BOS meeting held on 25th January 2024. ❖ **Syllabus and curriculum structure for semesters 7 and 8 were approved.** ❖ **Clinical internship and rotatory postings across different specialties were outlined.** ❖ **Approved, external & internal examiners list.**	To discuss and confirm the minutes of the 5th BoS meeting, which was held on 25-01-24. The members overviewed the previous discussions and approved the same.	Approved
3	**BoS/2024/BPT/6.2** Appraisal and Approval on the implementation of the 7th & 8th semester syllabus	The panel members discussed about the changes in 7th and 8th semester courses which has been approved for the implementation in R-2020 After deliberations, it is finally agreed to interchange sports physiotherapy with community physiotherapy.	Community physiotherapy from VII semester is interchanged with sports physiotherapy at VIII semester, Subject code remains the same. The details of 7th and 8th syllabus are available in the ***Annexure–A***
4	**BoS/2024/BPT/6.3** Appraisal and Approval of the guidelines and duration for clinical postings in SMVMCH (Compulsory Rotatory Internship).	The panel discussed and approved the guidelines proposed for the internship training and duration of the postings in the various department/specialties for 6 months, guidelines for the leave taken by the interns were framed (extension)	The duration of the posting in the various specialties are combined for clinical posting. The details of department/ specialties are available in the ***Annexure–B***

5	**BoS/2024/BPT/6.4** Approval of external examiners and question paper setters for the academic year 2024-25.	The members of BoS approved the panel of the examiner list summited by the chairman of the BoS.	Approved and the details are available in the ***Annexure–C***
6	**BoS/2024/BPT/6.5** Guidelines for managing arrears in auxiliary courses.	The BoS panel discussed and approved the examination guidelines for the auxiliary courses,	The candidate has to Re-register a course on the following semester. He/she needs to appear on the Internal evaluations exams - I & II, The marks scored in internal evaluative exams will be averaged, be submitted to the controller of examiner (COE) with due approval. The details are updated in ***Annexure -D***
7	**BoS/2024/BPT/6.6** Any other point with permission of the chair	End semester exam results of academic years 2024-2025 discussed, planning to introduce new revised syllabus as per New Education Policy 2020 for BPT program for batches from 2025 – 2026 onwards.	The panel members appreciated the End Semester Exam results. The BoS members gave their suggestions to improve the anatomy result in upcoming exams.

Annexure- A

Curriculum Structure for VII Semester

Core Courses					
Sl. No.	**Course Code**	**Course Name**	**Theory (in hours)**	**Practical (in hours)**	**Total (in hours)**
1.	U21BPTT738	Physiotherapy in Neurological conditions (Theory)	100	-	100
2.	U21BPTT739	Physiotherapy in obstetrics and Gynecological conditions (Theory)	65	15	80
3.	U21BPTT740	Sports Physiotherapy	45	15	60
4.	U21BPTT741	Clinical Cardiorespiratory conditions (Theory)	90	-	90
5.		Physiotherapy in neurological conditions (Practical)	-	90	90
Auxiliary Courses					
6.	U21BPTT742	Research methodology and Bio-statistics	45	-	45
7.	U21BPTT743	Veterinary Physiotherapy	15	-	15
8.	U21BPTT744	Case presentation	-	30	30
9.	U21BPTP745	Clinical training	-	90	90
Total hours:					600

Curriculum Structure for VIII Semester

Core Courses					
Sl. No.	**Course Code**	**Course Name**	**Theory (in hours)**	**Practical (in hours)**	**Total (in hours)**
1.	U21BPTT846	Physiotherapy in cardio–respiratory conditions (Theory)	90	-	90
2.	U21BPTT847	Community Physiotherapy (Theory)	60	-	60
3.	U21BPTT848	Rehabilitation medicine (Theory)	50	-	50
4.		Physiotherapy in cardio respiratory conditions (Practical)	-	90	90
5.		Community physiotherapy (Practical)	-	50	50
6.	U21BPTP849	Research project	-	70	70
Auxiliary Courses					
7.	U21BPTT850	Clinical reasoning and evidence based practice	30	-	30
8.	U21BPTP851	Case presentation		40	40
9.	U21BPTT852	Education technology	30	-	30
10.	U21BPTP853	Clinical training	-	90	90
Total hours:					600

COMMUNITY PHYSIOTHERAPY

COURSE CODE: U21BPTT847
TOTALHOURS: 110

COURSE DESCRIPTION:

Community Physiotherapy describes the roles & responsibilities of the Physiotherapist as an efficient member of the society. This component introduces the physiotherapist to a proactive preventive oriented philosophy for optimization &betterment of health.

OBJECTIVES:

1. Physiology of aging process and its influence on physical fitness.
2. National policies for the rehabilitation of disabled– role of PT.
3. The evaluation of disability and planning for prevention and rehabilitation.
4. Rehabilitation in urban and rural setup.
5. Be able to gain the ability to collaborate with other health professionals for effective service delivery & community satisfaction.

SYLLABUS

Sl. No.	**Topics**	**Didactic Hours**	**Practical Hours**	**Total Hours**
I	HEALTH PROMOTION: W.H.O. definition of health and disease. Health Delivery System –3 tier Physical Fitness: Definition and evaluation related to obesity. COMMUNITY: Definition of community, Multiplicity of community, the community-based approach, Community entry strategies, CBR and Community development, Community initiated versus community-oriented program.	8	10	18

Sl. No.	Topics	Didactic Hours	Practical Hours	Total Hours
II	COMMUNITY BASED REHABILITATION: Definition, historical review, concept of CBR, Need of CBR, objectives of CBR, scope of CBR, Members of CBR team, differences between institution based & community based rehabilitation. Agencies involved in rehabilitation of physical challenged, legislation for physically challenged. Role of physiotherapy in CBR.	8	10	18
III	DISABILITY: Definition of impairment, handicap and disability, difference between impairment, handicap and disability, causes for disability, types of disability, prevention of disability, disability in developed countries and in developing countries. Disability surveys: demography. Screening: early detection of disabilities and development disorders Prevention of disabilities - types and levels.	8	15	23

Sl. No.	Topics	Didactic Hours	Practicals Hours	Total Hours
	DISABILITY EVALUATION: Introduction, what, why and how to evaluate, quantitative versus qualitative data, uses of evaluation findings.			
IV	GERIATRICS: Senior citizens in India NGO, and health related legal rights and benefits for the elderly. Institutionalized &community dwelling elders. Theories of aging. Physiology of ageing: Musculoskeletal, neurological, cardiorespiratory, metabolic changes. Scheme of evaluation &role of PT in geriatrics.	9	15	24
V	INDUSTRIAL HEALTH Introduction to industrial health: definition, model of industrial therapy (Traditional medical &industrial model) Worker care spectrum: Ability management – job analysis - job description, job demand analysis, task analysis, ergonomics evaluation, injury prevention and employee fitness program.	12	15	27

Recommended textbooks

- T. Bhaskara Rao, “Textbook of community medicine & community rehabilitation”, paras’medical publisher,2013.
- DaleAvers, RitaA.wong “Guccione’s Geriatric physical therapy”, Elsevier publications, 4th edition,2020.
- GlendaL.Key“Industrialtherapy”,Mosby,1995.
- Pruthrish “Community based rehabilitation of person with disabilities”, Jaypeepublishers,2006.

Recommended Reference Books

1. K. Park “Textbook of preventive and social medicine”, Generic publisher, 22nd edition, 2013.
2. Helander, Einar “Training disabled people in the community: a manual on community based rehabilitation for developing countries”, world health organization, 1983.

Recommend web references

1. “Punarbhava”-National interactive web portal on disability.

www.physio-pedia.com

Annexure -B

The guidelines for the compulsory internship of 6months (180 days). It is stipulated evenly for all the speciality as per University Norms.

The candidate should undergo compulsory rotatory internship CRI training in the following departments/specialties of Physiotherapy for the duration prescribed against each

Sl. No.	Training Posting /Areas	Duration
1.	Physiotherapy & Rehabilitation	30 days
2.	Orthopaedics and Traumatology / Rheumatology	30 days
3.	Cardio-Pulmonary/ Critical care units	30 days
4.	General Medicine/ General & Plastic Surgery	30 days
5.	Obstetrics &Gynecology/ Paediatrics &Paediatrics surgery	15 days
6.	Neurology & Neurosurgery	15 days
7.	Community Physiotherapy	15 days
8.	Elective (any one of the above, or Sports Physiotherapy)	15 days

The guidelines as follows:

General Conduct and Professionalism:
- Adhere to punctuality, dress code, and patient confidentiality.
- Follow all hospital policies.

Attendance:
- Duration of CRI is 180 working days
- 100% attendance is mandatory, except for pre-approved leaves.
- Leaves should not exceed two days per month.

Academic Requirements:
- Maintain a detailed logbook and submit weekly/monthly log notes.
- Expect regular feedback.
- The students need to maintain the log book for entire CRI period, co-signature by clinicalin charge & CRI coordinator

Disciplinary Actions:
- Misconduct will result in disciplinary action, up to termination.

Leave Policy:
- One day of leave per month is allowed.
- Missed days must be compensated within the same posting.

Internship Start and Orientation:
- Internship starts 15 days after 8th-semester results.
- All interns must attend orientation before starting.
- Grading, attendance sheets, and log notes will be provided on orientation day.

Annexure -C

LIST OF EXAMINERS PROPOSED FOR END SEMESTER EXTERNAL EXAMINATIONS

Name of the Examiner	Institution Name	Designation	Years of Teaching Experience	Mobile No.	Mail Id
Dr. P. Ravichandran	Dept. of P.M.&R., Govt. Medical College & Hospital, Annamalai Nagar, Chidambaram.	Reader/Associate Professor	21 years	9894173032	ravichandranpandian@gmail.com
Dr. Arul Pragassame	Dept. of P.M.&R., Govt. Medical College & Hospital, Annamalai Nagar, Chidambaram.	Assistant Professor	15 years	8778163842	arulphysio77@gmail.com
Dr. V. Menaka	Dept. of P.M.&R., Govt. Medical College & Hospital, Annamalai Nagar, Chidambaram.	Lecturer	15 years	9942772207	menakampt@gmail.com
Dr. G. Velmurugan	Shanmuga College of Physiotherapy, Karaikal.	Professor cum Principal	13 years	8838185057	principalscpt@vmu.edu.in

LIST OF QUESTION PAPER SETTERS

Name of the Examiner	Institution Name	Designation	Years of Teaching Experience	Mobile No.	Mail id
Dr. Basavarajmotimath	KLE institute of Physiotherapy, Belagavi, Karnataka.	Professor	17 years	9886100461	bsmotimath@yahoo.co.in
Dr. S. Abarnadevi	ESIC Medical College, kk Nagar, Chennai.	Associate professor	8 years	9843678462	abarnadevis@gmail.com
Dr. A. Manikandan	Sri Venkateshwaraa Medical College, Puducherry.	Associate professor	11 years	7904532243	drmanikananadimoolam@gmail.com
Dr. S. Prathap	Saveetha College of Physiotherapy, Chennai, Tamil Nadu.	Professor	16Years	9840340195	Prathap.scpt@saveetha.com

Annexure -D

The guidelines to reappear for the arrears' exams in the Auxiliary courses,

1. The candidate must re-register for the course in the following semester. He/she is required to appear for two Internal Evaluative Examinations I & II.
2. The marks obtained in these exams will be averaged, and to pass, the studentmust secure 50% or above.
3. The final marks will be approved by the internal examiner and the Dean ofthe College of Physiotherapy.
4. This approval will proceed through the proper channels, first receivingconsent from the Dean of Academics, followed by the Director.
5. Once approved, the marks will be submitted to the Controller of Examinations (COE).

Table I: CURRICULUM STRUCTURE

I (a). FIRST YEAR (1150 hrs)

Semester – I

Sl. No.	Course Code	Course Name	Theory (in hours)	Practical (in hours)	Total (in hours)
Core Courses-Theory					
1.	U21BPTT101	Psychology (General and Health)	130	-	130
2.	U21BPTT102	Sociology	130	-	130
3.	U21BPTT103	Functional English	80	20	100
4.	U21BPTT104	Computer and its applications	40	60	100
Auxiliary Courses*					
5.	U21BPTT105	Physiotherapy Orientation	50	-	50
6.	U21BPTP106	Physical Education	-	40	40
				Total Hours	**550**
***Internals and Model Examination will be conducted at the Department and the Marks will be submitted to the Controller of Examination**					

Semester – II

Sl. No.	Course Code	Course Name	Theory (in hours)	Practical (in hours)	Total (in hours)
Core Courses-Theory					
1.	U21BPTT207	Anatomy (Systemic and Regional)	135	-	135
2.	U21BPTT208	Physiology	135	-	135
3.	U21BPTT209	Biomechanics and kinesiology I	90	30	120
Core Courses-Practical					
4.	U21BPTP210	Anatomy Practical	-	60	60
5.	U21BPTP211	Physiology Practical		60	60
Auxiliary Courses*					
6.	U21BPTT212	Nutrition	45	-	45
7.	U21BPTT213	Environmental Studies	45	-	45
				Total Hours	**600**
***Internals and Model Examination will be conducted at the Department and the Marks will be submitted to the Controller of Examination**					

I (b).SECOND YEAR (1200 hrs)

Semester – III

Sl. No.	Course Code	Course Name	Theory (in hours)	Practical (in hours)	Total (in hours)
Core Courses-Theory					
1.	U21BPTT314	Microbiology & Pathology	90	-	90
2.	U21BPTT315	Biochemistry & Pharmacology	90	-	90
3.	U21BPTT316	Biomechanics and Kinesiology II	90	-	90
4.	U21BPTT317	Exercise therapy - Basics & Soft tissue Manipulation	60	-	60
Core Courses-Practical					
5.	Biomechanics and Kinesiology Practical		-	60	60
6.	Exercise therapy - Basics & Soft tissue Manipulation Practical		-	120	120
Auxiliary Courses*					
7.	U21BPTT320	Therapeutic Yoga	15	30	45
8.	U21BPTP321	BLS and First aid	15	30	45
				Total Hours	**600**
***Internals and Model Examination will be conducted at the Department and the Marks will be submitted to the Controller of Examination**					

Semester – IV

Sl. No.	Course Code	Course Name	Theory (in hours)	Practical (in hours)	Total (in hours)
Core Courses-Theory					
1.	U21BPTT422	Exercise Therapy – Advanced	60	-	60
2.	U21BPTT423	Electrotherapy – I (LF & MF)	60	-	60
3.	U21BPTT424	Electrotherapy II (HF)	75	-	75
4.	U21BPTT425	Basic physics for Physiotherapy	45	-	45
Core Courses-Practical					
5.	Exercise Therapy – Advanced Practical		-	120	120
6.	Electrotherapy – I (LF & MF) Practical		-	90	90
7.	Electrotherapy II (HF) Practical		-	90	90
Auxiliary Courses*					
8.	U21BPTT429	Physiotherapy ethics	15	-	15
9.	U21BPTP430	Clinical Observation	-	45	45
				Total Hours	**600**
***Internals and Model Examination will be conducted at the Department and the Marks will be submitted to the Controller of Examination**					

I (c).THIRD YEAR (1155 hours)

Semester – V

Sl. No.	Course Code	Course Name	Theory (in hours)	Practical (in hours)	Total (in hours)
Core Courses-Theory					
1.	U21BPTT524	Clinical Orthopedic & Traumatology for Physiotherapy	120	-	120
2.	U21BPTT525	General Surgery, Plastic Surgery and OBG	90	-	90
3.	U21BPTT526	General Medicine, Paediatics and Psychiatry	120	-	120
	U21BPTT527	Community Medicine & Rehabilitation	90	-	90
Auxiliary Courses*					
5.	U21BPTT528	Diagnostic Imaging for Physiotherapy	30	15	45
6.	U21BPTP529	Clinical conditions	-	90	90
				Total Hours	**555**
***Internals and Model Examination will be conducted at the Department and the Marks will be submitted to the Controller of Examination**					

Semester – VI

Sl. No.	Course Code	Course Name	Theory (in hours)	Practical (in hours)	Total (in hours)
Core Courses-Theory					
1.	U21BPTT631	Physiotherapy in Orthopedic Conditions	90	-	90
2.	U21BPTT632	Physiotherapy in General Medicine and General Surgery	60	-	60
3	U21BPTT630	Clinical Neurology and Neurosurgery for Physiotherapy	120	-	120
Core Courses-Practical					
4.	Physiotherapy in Orthopedic Conditions Practical		-	90	90
5	Physiotherapy in General Medicine and General Surgery (Practical)		-	90	90
Auxiliary Courses*					
6.	U21BPTT635	Principles of Management	30	-	30
7.	U21BPTT636	Health Promotion and Fitness	15	15	30
8.	U21BPTP637	Clinical Training	-	90	90
				Total Hours	**600**
***Internals and Model Examination will be conducted at the Department and the Marks will be submitted to the Controller of Examination**					

I (d) FOURTH YEAR (1200 hrs)

Semester – VII

Sl. No.	Course Code	Course Name	Theory (in hours)	Practical (in hours)	Total (in hours)
Core Courses-Theory					
1.	U21BPTT738	Physiotherapy in Neurological conditions (Theory)	90	-	90
3.	U21BPTT739	Physiotherapy in Obstetrics and Gynecology	65	15	80
4.	U21BPTT740	Sports Physiotherapy	60	-	60
5.	U21BPTT741	Clinical Cardio-respiratory conditions	90	-	90
Core Courses-Practical					
6.	Physiotherapy in Neurological conditions (Practical)		-	90	90
Auxiliary Courses*					
7.	U21BPTT742	Research Methodology and Bio-statistics	45	-	45
8.	U21BPTT743	Veterinary Physiotherapy	15	-	15
9.	U21BPTP744	Case Presentation and Discussion	-	30	30
10.	U21BPTP745	Clinical Training	-	90	90
				Total Hours	**600**
***Internals and Model Examination will be conducted at the Department and the Marks will be submitted to the Controller of Examination**					

Semester – VIII

Sl. No.	Course Code	Course Name	Theory (in hours)	Practical (in hours)	Total (in hours)
Core Courses-Theory					
1.	U21BPTT846	Physiotherapy in Cardio-respiratory conditions (Theory)	90	-	90
2.	U21BPTT847	Community Physiotherapy (Theory)	45	15	60
3.	U21BPTT848	Rehabilitation Medicine	50	-	50
Core Courses-Practical					
4.	Physiotherapy in Cardio-respiratory conditions (Practical)		-	90	90
5.	Community Physiotherapy (Practical)		-	50	50
6.	U21BPTP849	Research Project	-	70	70
Auxiliary Courses*					
7.	U21BPTT850	Clinical reasoning and evidence-based practice	30	-	30
8.	U21BPTP851	Case/Journal Presentation	-	40	40
9.	U21BPTT852	Education Technology	30	-	30
10.	U21BPTP853	Clinical Training	-	90	90
				Total Hours	**600**
***Internals and Model Examination will be conducted at the Department and the Marks will be submitted to the Controller of Examination**					

TABLE - II

COMPULSORY ROTATORY INTERNSHIP

The candidates should undergo compulsory rotatory internship training in the following Departments/specialties* of Physiotherapy for the duration prescribed against each.

The modified Internship training areas as follows

Department/Speciality	Duration
Physiotherapy & Rehabilitation	30 days
Orthopaedics and Traumatology / Rheumatology	30 days
Cardio-Pulmonary/ Critical care units	30 days
General Medicine/ General & Plastic Surgery	30 days
Obstetrics &Gynecology/ Paediatrics &Paediatrics surgery	15 days
Neurology & Neurosurgery	15 days
Community Physiotherapy	15 days
Elective (any one, or Sports Physiotherapy)	15 days

*The above listed departments are not limited and it can be extended to any other advanced facilities available, which has to be decided by Dean, COPT.

Table- III

BACHELOR OF PHYSIOTHERAPY

SCHEME OF END SEMESTER EXAMINATION

(The End Semester Examination will be conducted for the duration of 3 hours)

Semester - I

Course Code	Course Name	Theory			Practical		Grand Total	
		Internal Marks	Passing Min.	Max. Marks	Passing Min.	Max. Marks	Passing Min.	Max. Marks
U21BPTT101	Psychology (General and Health)	25	38	75	-	-	50	100
U21BPTT102	Sociology	25	38	75	-	-	50	100
U21BPTT103	Functional English	25	38	75	-	-	50	100
U21BPTT104	Computer and its Applications	25	38	75	-	-	50	100
U21BPTT105	Physiotherapy Orientation *	25	38	75	-	-	50	100
U21BPTP106	Physical Education *	-	-	-	-	-	-	-
*Internals and Model Examination will be conducted at the Department and the Marks will be submitted to the Controller of Examination								

Semester – II

Course Code	Course Name	Theory			Practical		Grand Total	
		Internal Marks	Passing min.	Max. Marks	Passing min.	Max. Marks	Passing min.	Max. Marks
U21BPTT207	Anatomy (Systemic and Regional)	25	38	75	-	-	50	100
U21BPTT208	Physiology	25	38	75	-	-	50	100
U21BPTT209	Biomechanics and kinesiology I	25	38	75	-	-	50	100
U21BPTP210	Anatomy Practical	20	-	-	40	80	50	100
U21BPTP211	Physiology Practical	20	-	-	40	80	50	100
U21BPTT212	Nutrition*	25	38	75	-	-	50	100
U21BPTT213	Environmental Studies*	25	38	75	-	-	50	100
*Internals and Model Examination will be conducted at the Department and the Marks will be submitted to the Controller of Examination								

Semester – III

Course Code	Course Name	Theory			Practical		Grand Total	
		Internal Marks	Passing min.	Max. Marks	Passing min.	Max. marks	Passing min.	Max. Marks
U21BPTT314	Microbiology & Pathology	25	38	75	-	-	50	100
U21BPTT315	Biochemistry & Pharmacology	25	38	75	-	-	50	100
U21BPTT316	Biomechanics and Kinesiology II	25	38	75	-	-	50	100
U21BPTT317	Exercise therapy - Basics & Soft tissue Manipulation	25	38	75	-	-	50	100
	Biomechanics and Kinesiology Practical	20	-	-	40	80	50	100
	Exercise therapy - Basics & Soft tissue Manipulation Practical	20	-	-	40	80	50	100
U21BPTT320	Therapeutic Yoga*	25	38	75	-	-	50	100
U21BPTP321	BLS & First aid*	100	-	-	-	-	50	100

***Internals and Model Examination will be conducted at the Department and the Marks will be submitted to the Controller of Examination**

Semester – IV

Course Code	Course Name	Theory			Practical		Grand Total	
		Internal Marks	Passing Min.	Max. Marks	Passing Min	Max. Marks	Passing Min.	Max. Marks
U21BPTT422	Exercise Therapy – Advanced	25	38	75	-	-	50	100
U21BPTT423	Electrotherapy – I (LF & MF)	25	38	75	-	-	50	100
U21BPTT424	Electrotherapy II (HF)	25	38	75	-	-	50	100
U21BPTT425	Basic physics for Physiotherapy	25	38	75	-	-	50	100
	Exercise Therapy – Advanced Practical	20	-	-	40	80	50	100
	Electrotherapy – I (LF & MF) Practical	20	-	-	40	80	50	100
	Electrotherapy II (HF) Practical	20	-	-	40	80	50	100
U21BPTT429	Physiotherapy Ethics*	25	38	75	-	-	50	100
U21BPTP430	Clinical Training*	100	-	-	-	-	50	100
***Internals and Model Examination will be conducted at the Department and the Marks will be submitted to the Controller of Examination**								

Semester – V

Course Code	Course Name	Theory			Practical		Grand Total	
		Internal Marks	Passing Min.	Max. Marks	Passing Min.	Max. Marks	Passing Min.	Max. Marks
U21BPTT524	Clinical Orthopedic & Traumatology for Physiotherapy	25	38	75	-	-	50	100
U21BPTT525	General Surgery, Plastic Surgery and OBG	25	38	75	-	-	50	100
U21BPTT526	General Medicine, Paediatrics and Psychiatry	25	38	75	-	-	50	100
U21BPTT527	Community Medicine & Rehabilitation	25	38	75	-	-	50	100
U21BPTT528	Diagnostic Imaging for Physiotherapist*	25	38	75	-	-	50	100
U21BPTP529	Clinical conditions	100	-	-	-	-	50	100
***Internals and Model Examination will be conducted at the Department and the Marks will be submitted to the Controller of Examination**								

Semester – VI

Course Code	Course Name	Theory			Practical		Grand Total	
		Internal Marks	Passing Min.	Max. Marks	Passing Min.	Max. Marks	Passing Min.	Max. Marks
U21BPTT631	Physiotherapy in Orthopedic Conditions	25	38	75	-	-	50	100
U21BPTT632	Physiotherapy in General Medicine and General Surgery	25	38	75	-	-	50	100
U21BPTT630	Clinical Neurology and Neurosurgery for Physiotherapy	25	38	75	-	-	50	100
	Physiotherapy in Orthopedic Conditions Practical	20	-	-	40	80	50	100
	Physiotherapy in General Medicine and General Surgery Practical	20	-	-	40	80	50	100
U21BPTT635	Principles of Management*	25	38	75	-	-		
U21BPTT636	Health Promotion and Fitness*	25	38	75	-	-	50	100
U21BPTP637	Clinical Training	100	-	-	-	-	50	100
***Internals and Model Examination will be conducted at the Department and the Marks will be submitted to the Controller of Examination**								

Semester – VII

Course Code	Course Name	Theory			Practical		Grand Total	
		Internal Marks	Passing Min.	Max. Marks	Passing Min.	Max. Marks	Passing Min.	Max. Marks
U21BPTT738	Physiotherapy in Neurological Conditions(Theory)	25	38	75	-	-	50	100
U21BPTT739	Physiotherapy in Obstetrics and Gynecology conditions(Theory)	25	38	75	-	-	50	100
U21BPTT740	Sports Physiotherapy	25	38	75	-	-	50	100
U21BPTT741	Clinical Cardio-respiratory conditions	25	38	75	-	-	50	100
	Physiotherapy in Neurological Conditions (Practical)	20	-	-	40	80	50	100
U21BPTT742	Research Methodology and Bio-Statistics*	25	38	75	-	-	50	100
U21BPTT743	Veterinary Physiotherapy*	25	38	75	-	-	50	100
U21BPTP745	Clinical Training*	100	-	-	-	-	50	100

***Internals and Model Examination will be conducted at the Department and the Marks will be submitted to the Controller of Examination**

Semester – VIII

Course Code	Course Name	Theory			Practical		Grand Total	
		Internal Marks	Passing Min.	Max. Marks	Passing Min.	Max. Marks	Passing Min.	Max. Marks
U21BPTT846	Physiotherapy in Cardio-respiratory conditions	25	38	75	-	-	50	100
U21BPTT847	Community Physiotherapy	25	38	75	-	-	50	100
U21BPTT848	Rehabilitation Medicine	25	38	75	-	-	50	100
	Physiotherapy in Cardio-respiratory conditions Practical	20	-	-	40	80	50	100
U21BPTP849	Research Project	20	-	-	40	80	50	100
U21BPTT850	Clinical reasoning and evidence-based practice*	25	38	75	-	-	50	100
U21BPTT852	Education Technology*	25	38	75	-	-	50	100
U21BPTP853	Clinical Training*	100	-	-	-	-	50	100

***Internals and Model Examination will be conducted at the Department and the Marks will be submitted to the Controller of Examination**

I -Semester

Syllabus

PSYCHOLOGY (GENERAL AND HEALTH)

Course Code: U21BPTT101 **Instruction hours:** Theory – 130 hours

Course description: The course is designed to assist the students to acquire knowledge of fundamentals of psychology and develop an insight into behaviour of self and others. Further it is aimed at helping them to practice the principles of understanding the mental status and behaviour of patients in clinical settings.

General Psychology – 80 hours Health Psychology – 50 hours GENERAL

PSYCHOLOGY [80 hours]

Unit	Hrs (T)	Content	Teaching method
I	4	INTRODUCTION Definition Schools of Psychology Methods of Psychology Branches of Psychology	Lecture Discussion
II	6	HEREDITY AND ENVIRONMENT Twins Importance of heredity and environment Role in relation to physical characteristics Intelligence and personality Nature-nature controversy	Lecture Discussion
III	8	DEVELOPMENT AND GROWTH BEHAVIOUR Infancy, Childhood, Adolescence, Adulthood, Middle age, Old age.	Lecture Discussion
IV	6	INTELLIGENCE Definitions of Intelligence Quotient, Mental Age, List of various intelligence tests – WAIS, WISC, Bhatia's performance test, Raven's Progressive Matrices test.	Lecture Discussion
V	6	MOTIVATION Definitions of motive, drive, incentive and reinforcement, Basic information about primary needs: Hunger, Thirst, Sleep, Elimination activity, Air, Avoidance of pain, Attitude to sex. Psychological needs	Lecture Discussion
VI	6	EMOTIONS Definition, differentiate from feelings, physiological changes of emotion, role of RAS, hypothalamus, cerebral cortex, sympathetic nervous system, adrenal gland, heredity and emotion, Nature and control of anger, fear and anxiety.	Lecture Discussion

Unit	Hrs (T)	Content	Teaching method
VII	10	PERSONALITY Definition List of components: Physical characteristics, character, abilities, temperament interest, attitudes. Role of heredity, nervous system, physical characteristics, abilities, family and culture on personality development. Basic concepts of Freud: Unconscious, conscious, Id, Ego and Superego. List and the define 8 stages as proposed by Erickson Concepts of learning as proposed by Dollard and Miller; drive, cue, response and reinforcement. Personality assessment Projective tests	Lecture Discussion
VIII	6	LEARNING Definition Types of learning Effective ways to learn Role of language in learning	Lecture Discussion
IX	4	THINKING Definition & creativity Creativity: Steps, traits Delusions	Lecture Discussion
X	4	FRUSTRATION Definition, sources and solution. Conflicts	Lecture Discussion
XI	8	SENSATION, ATTENTION & PERCEPTION Senses: various senses and their functions Attention: Definition, factors determining attention Perception: Definition, principles. Illusion & hallucination: types	Lecture Discussion
XII	4	LEADERSHIP Qualities and types of leadership Attitude and its changes	Lecture Discussion
XIII	4	DEFENCE MACHANISMS Defence Mechanisms of the ego List of various defence mechanisms	Lecture Discussion
XIV	4	COMMUNITY PSYCHOLOGY Social psychology Community Psychology	Lecture Discussion

HEALTH PSYCHOLOGY [50 hours]

Unit	Hrs (T)	Content	Teaching method
I	4	PSYCHOLOGICAL REACTIONS OF A PATIENT Various Psychological reactions of a patient during admission in hospital and treatment.	Lecture Discussion
II	4	REACTIONS TO LOSS Reactions to loss, death and bereavement Stages of acceptance	Lecture Discussion
III	6	STRESS Physiological and psychological changes during stress Relations to health and sickness Relaxation methods	Lecture Discussion
IV	8	COMMUNICATIONS Types of communication Elements in communications, Barriers to good communications Developing effective communication, specific communication techniques	Lecture Discussion
V	6	COUNSELLING Definition and aims Guidance and counselling Principles in counseling Personality of counsellors	Lecture Discussion
VI	4	COMPLIANCE Nature of compliance Factors contributing to non-compliance Means to improve compliance	Lecture Discussion
VII	8	EMOTIONAL NEEDS Emotional needs and psychological factors in relation to unconscious patients, handicapped persons, bed-ridden patients, patients with chronical patients, cerebral palsy children, burns, leprosy, Parkinson’s disease, incontinence and mental illness.	Lecture Discussion
VIII	10	MISCELLANEOUS Geriatric psychology Paediatric psychology Behaviour modification in patients Personality styles of patients Substance abuse	Lecture Discussion

Recommended text books:

1. **Ramalingam& Bid (2009). Psychology for Physiotherapists. Jaypee Brothers, NewDelhi.**
2. **General Psychology by S.K. Mangal**
3. **Atkinson(1996). Dictionary of Psychology.**

SOCIOLOGY

Course Code: U21BPTT102 **Instruction hours:** Theory – 130 hours

Course description: The course is designed to introduce the basics of sociological concepts, principles and social process, social institutions in relation to individual, family and community in India and its relationship with health, illness and handicap.

Unit	Hrs (T)	Content	Teaching method
I	6	INTRODUCTION Definition Sociology – a science of society Application of sociology in physiotherapy	Lecture Discussion
II	14	SOCIOLOGY AND HEALTH Social factors affecting health status Social consciousness and meaning of illness Perception of illness Decision making in taking treatment Institutions of health and their role in the improvement of health of the people	Lecture Discussion
III	10	SOCIALISATION Meaning of socialisation Influence of social factors on personality Socialisation in hospitals Socialisation in rehabilitation of patients	Lecture Discussion
IV	10	SOCIAL GROUPS Concept of social group Influence of formal and informal groups on health on health and sickness Role of primary and secondary groups in the hospital and rehabilitation settings.	Lecture Discussion

V	16	FAMILY & COMMUNITY Influence of family on human personality Changes in the functions of a family Influence of the family on the individual's health, family and nutrition Effects of sickness on family, family and psychosomatic disease Concept of community Role of rural and urban communities in public health Role of community in determining beliefs, practices and home remedies in treatment	Lecture Discussion

Unit	Hrs (T)	Content	Teaching method
VI	10	CULTURE & CASTE SYSTEM Components of culture Impact of culture on human behaviour Cultural meaning & response of sickness, Choice of treatment Culture induced symptoms and disease Sub-culture of medical workers Caste system: Features of modern caste system & its trends	Lecture Discussion
VII	14	SOCIAL CHANGE Meaning of social change Factors of social change on human adaption, stress, deviance and health programmes Role of social planning in the improvement of health and rehabilitation.	Lecture Discussion
VIII	12	SOCIAL CONTROL Meaning of social control, Role of norms Folkways, customs, morals, religion, law and other means of social control in the regulation of human behaviour. Social deviance and disease.	Lecture discussion
IX	12	SOCIAL PROBLEMS OF THE DISABLED Consequences of the following social problems in relation to sickness and disability. Remedies to prevent the following problems: Population explosion, poverty and employment, beggary, juvenile delinquency, prostitution, alcoholism, problems of women in employment.	Lecture Discussion
X	16	SOCIAL SECURITY Social security and social legislation in relation to the disabled.	Lecture Discussion
XI	10	SOCIAL WORKER Role of a medical social worker	Lecture Discussion

Recommended Books:

1. **Bid D. (2006). Sociology for Physiotherapists. Jaypee Brothers, New Delhi.**
2. **Sachdeva and Vidyabushan: Introduction to the study of Sociology.**
3. **K. Parks Textbook of Preventive & Social Medicine.**
4. **Textbook of Preventive & Social Medicine – P.K. Mahajan & M.C.Gupta**

FUNCTIONAL ENGLISH

Course Code: U21BPTT103 **Instruction hours:** Theory – 80 hours Practical – 20 hours

Course description: The course is designed to enable to enhance ability to comprehend spoken and written English (and use English) required for effective communication in their professional work. Students will practice their skills in verbal and written English during clinical and classroom experiences.

Unit	Hrs (T+P)	Content	Teaching method
I	10+3	INTRODUCTION Study techniques Logical processes of analysis and synthesis Use of dictionary Effective diction	Lecture, Demonstration & Exercises to students
II	15+3	APPLIED GRAMMAR Review of grammar & correct usage Building vocabulary Structure of sentences & paragraphs Phonetics Public speaking	Lecture, Demonstration, Conversation& Public speaking
III	15+5	FORMS OF COMPOSITION Letter writing Note taking Précis writing Essay writing Anecdotal records Diary writing Reports Resume / Curriculum vitae etc.	Demonstration & Exercises to students
IV	15+3	COMMUNICATION Oral report Discussion Lecture / seminar Debate Summary Telephonic conversation	Demonstration & Exercises to students
V	10+3	READING COMPREHENSION Selected materials, articles, magazines, journals etc.	Demonstration & Exercises to students
VI	15+3	LISTENING COMPREHENSION Media, Audio, Video, Speeches etc.	Demonstration & Exercises to students

Recommended Books:

1. Communicative English for Engineers and Professionals. Author, Nitin Bhatnagar
2. **English for physiotherapy , Joanna Ciecierska**

COMPUTER AND ITS APPLICATIONS

Instruction hours: Theory – 40 hours
Course Code: U21BPTT104 Practical – 60 hours

Course description: This course is designed for students to develop basic knowledge offundamentals of computer and its application in Physiotherapy.

Unit	Hrs (T+P)	Content	Teaching method
I	4+0	INTRODUCTION TO COMPUTERS Concepts & features of computer Application areas of computers in health services Hardware and software	Lecture
II	8+15	HARDWARE Architecture of computers Types of storage devices Characteristics of disks, terminals, printers, network etc. Disk operating system: DOS, Windows Applications of networking concepts	Lecture Discussion Demonstration & Practicals
III	8+15	SOFTWARE Classification of software Application of software Operating system, computer system Computer virus: Precautions & dealing	Lecture Demonstration & Practicals
IV	10+20	PROGRAMMES MS – Word MS – Excel with pictorial presentations MS – Access MS – PowerPoint	Lecture Demonstration & Practicals
V	10+10	COMPUTER APPLICATIONS Multimedia: Types & uses Computer aided teaching & testing Use of internet: web pages & e-mail Principles in scientific research: Work processing, Health care systems, libraries, education, information system Application in Physiotherapy: E.M.G., Biofeedback, Exercise testing equipments, Spirometry, etc.	Visit, Demonstration & Practicals

Recommended Books:

1. **Basic Computer Knowledge Fundamental of Computer system- Reema Thareja**
2. **Basic Computer Knowledge - Maluth John Monyjok**
3. **A New Approach to Basic Computer Education - D. P. Nagpal**

PHYSIOTHERAPY ORIENTATION

Course Code: U21BPTT105 **Instruction hours:** Theory – 50 hours

Course description: The course is designed to help the students to develop an understanding of the philosophy, objectives and process of physiotherapy in various clinical settings. It is aimed at helping the students to acquire knowledge, understanding and skills in physiotherapy techniques in clinical settings.

Unit	**Hrs (T)**	**Content**	**Teaching method**
I	5	INTRODUCTION TO HEALTH Health and Health care delivery system	Lecture
II	10	INTRODUCTION TO HEALTH SCIENCE Overview of Health Science, Health Professions & their specialties	Lecture
III	10	PHYSIOTHERAPY PROFESSION History of Medical Therapeutics History of Physiotherapy Overview of impairment, disability, handicap Health – levels of prevention & Rehabilitation Physiotherapy in medical rehabilitation	Lecture, Demonstration
IV	10	PHYSIOTHERAPY IN MEETING HEALTH CARE NEEDS OF INDIA Needs versus demands Need for physiotherapy Scope of the profession Role of Physiotherapist in health care delivery system and prevention of disability	Lecture, Demonstration, Group discussions
V	15	PHYSIOTHERAPEUTIC METHODS Physical agents in therapy Exercise therapy Electrotherapy Specialties in physiotherapy Areas of physiotherapy services & training	Lecture Demonstration & Visit

PHYSICAL EDUCATION

Course Code: U21BPTT106 **Instruction hours:** – 40 hours Practical

Course description: The purpose of the course is to acquire knowledge and understand variouscomponents in physical fitness and training methods.

Unit	Hrs (P)	Content	Teaching method
I	5	INTRODUCTION Physical fitness	Lecture Discussion
II	15	TRAINING METHODS Definition Motor component Warming up Conditioning Cool down	Lecture Discussion & Demonstration.
III	20	PHYSICAL FITNESS & TRAINING Physical Fitness and Training of Motor components: Strength Speed Endurance Mobility Co-ordination	Lecture Discussion & Demonstration.

II-Semester

ANATOMY (SYSTEMIC AND REGIONAL)

Course code: U21BPTT207 **Instruction hours:** Theory –135hours
Practical –60 hours

Course description:

The course is designed to enable the students to acquire knowledge of normal structure of various human body systems particularly on musculoskeletal, nervous and cardio-pulmonary systems and understand their application in the practice of physiotherapy.

I - SYSTEMIC ANATOMY [70 HOURS]

Unit	Hrs (T+P)	Content	Teaching method
I	6+0	INTRODUCTION TO ANATOMICAL TERMS Definitions, subdivisions, systems of the body Cell: Structure, composition, function, cell division Tissues: Definition, types, characteristics, classification, location, functions. Genes and chromosomes.	Lecture Demonstration
II	8+0	CARDIO VASCULAR SYSTEM Structure of heart, blood vessels Blood and nervous supply of the heart Major blood vessels	Lecture Demonstration
III	2+0	LYMPHATIC SYSTEM Structure of Lymphatic organs and vessels Functional roles	Lecture Demonstration
IV	8+0	RESPIRATORY SYSTEM Structure of the organs of the respiratory system Muscles of respiration, tracheobronchial tree, bronchopulmonary segments	Lecture Demonstration
V	5+0	DIGESTIVE SYSTEM Structure of the alimentary tract and organs of digestive system Anatomy of the liver and pancreas	Lecture Demonstration
VI	3+0	GENITO-URINARY SYSTEM Structure of the organs of the genito-urinary system	Lecture Demonstration
VII	3+0	ENDOCRINE SYSTEM Structure of endocrine glands	Lecture Demonstration
VIII	20+0	NERVOUS SYSTEM Division of the nervous system and their organs Structure and functions of nerve cell Structure of brain, spinal cord and peripheral nerves (in detail) Structure & location of autonomic nervous system	Lecture Demonstration

Unit	Hrs (T+P)	Content	Teaching method
IX	5+0	OSTEOLOGY Definition and types of skeletal system Classification of bones Ossification: definition, types and process	Lecture Discussion, Demonstration
X	5+0	ARTHROLOGY Definition and classification of joint Functions of joints: mobility & stability	Lecture Discussion, Demonstration
XI	5+0	MYOLOGY Structure and types of muscles Skeletal muscles: classification, forms & groups Position, origin, insertion, nerve supply and action of skeletal muscles	Lecture Discussion, Demonstration

II - REGIONAL ANATOMY [Theory-65 Hours, Practical-60 Hours]

Unit	Hrs (T+P)	Content	Teaching method
XII	20+15	UPPER EXTREMITY Osteology, arthrology, myology of the following: Pectoral region Scapular region Axilla Shoulder girdle and arm Elbow and forearm Wrist and hand Nerves of upper limb Blood vessels of upper limb	Lecture Demonstration & Practicals
XIII	25+15	LOWER EXTREMITY Osteology, arthrology, myology of the following: Pelvic & gluteal region Hip & thigh region Knee & leg Ankle & foot Nerves & Blood vessels of lower limb	Lecture Demonstration & Practicals
XIV	10+15	TRUNK Osteology, Arthrology, myology and their relations of: Vertebral Column Thoracic cage Abdomen Pelvis	Lecture Demonstration & Practicals
XV	10+15	HEAD & NECK Musculoskeletal and neurovascular features of neck and cranium Cranial nerves	Lecture Demonstration & Practicals

PRACTICALS AND DEMONSTRATION:

1. **Upper extremity including surface Anatomy. Demonstration of the muscles ofthe upper extremity, movements in joints, identification of body prominences on inspection and by palpation, points of palpation of nerves and arteries. Identification of the bones of the upper extremity, side determination, parts, attachment of the muscles, nerves and vessels relation to bone.**

2. **Lower extremity including surface Anatomy. Demonstration of the muscles of the lower extremity, movements in joints, identification of body prominences oninspection and by palpation, points of palpation of nerves and arteries. Identification of the bones of the lower extremity, side determination, parts, attachment of the muscles and relation of nerves and vessels to bone.**

3. **Demonstration of the Head & Neck and Spinal cord & Brain including surface Anatomy.**

4. **Demonstration of the muscles of the back, pelvic girdle, pre and para vertebral muscles, movements in joints, identification of body prominences on inspection and by palpation.**

5. **Identification of the bones of the vertebral column (cervical, thoracic, lumbar, sacral and coccygeal) parts, attachment of the muscles and relation of nerves and vessels to bone.**

6. **Surface Markings of Various Organs and Bony Prominences**

7. **Radiographic Identification of Bone and Joints**

Recommended Text books:

1. **SNELL [Richard S], Clinical Anatomy for Medical students: Ed. 5. Little Brown andCompany Boston.**
2. **B.D Chaurasia's Human Anatomy – Regional and Applied; Volume I, Volume II andVolume III.**
3. **SINGH [Inderbir], Human Osteology. JP Brothers, New Delhi1990.**
4. **SINGH [Inderbir], Text book of Anatomy with colour atlas: Vol I, II, III.**
5. **SINGH [Inderbir], Essentials of Anatomy JP Brothers, New Delhi**
6. **Anatomy and Physiology - Dr. Minakshi Pathak**
7. **The carnial Nerves Anatomy, Pathology, Pathophysiology Diagnosis Treatment- M. Samii, P.J Jannetla**
8. **Anatomy by Vishram Singh**

Recommended Text books for Practical:

1. **ROMANES [G J], Cunningham manual of practical anatomy: Vol I, II, III**

Reference Books:

1. **PODAR - Handbook of Osteology: Ed. 11 Scientific book co.**

2. **Gray's Anatomy**
3. **McMinn – McMinn's color atlas of Human Anatomy.**

PHYSIOLOGY

Course Code: U21BPTT208

Instruction hours: Theory – 135hours
Practical –60 hours

Course description: The course is designed to assist the students to acquire knowledge of normal physiology of various human body systems and understand the alterations in physiology in diseases for physiotherapy practice.

Unit	Hrs (T+P)	Content	Teaching method
I	10+0	CELL PHYSIOLOGY Cell: Structure & functions of components Functions of membranes & glands	Lecture
II	10+6	CIRCULATORY SYSTEM Blood: Component and their functions, blood groups, coagulation, blood volume and its regulation Functions and regulations of the heart, cardiac cycle, cardiac output, E.C.G., heart sounds. Blood pressure: Maintenance and regulation. Effects of exercises on postural changes.	Lecture Discussion, Demonstration & Practicals
III	15+15	RESPIRATORY SYSTEM Functions of the respiratory organs Physiology of respiration Pulmonary ventilation, volume Mechanics of respiration Gaseous exchange in lungs Regulation of respiration Effects of exercises on respiration	Lecture Discussion, Demonstration & Practicals
IV	10+0	DIGESTIVE SYSTEM Functions of organs of digestive tract Movements of the alimentary tract Digestion in mouth, stomach, intestines Absorption of food Metabolism of carbohydrates, proteins and fat	Lecture Discussion, Demonstration
V	10+0	EXCRETORY SYSTEM Functions of organs of excretory tract Composition of urine Mechanism of urine formation & Micturition Functions of skin	Lecture Discussion, Demonstration
VI	10+0	ENDOCRINE SYSTEM Functions of the various endocrine glands Endocrine Hormones: Functions and their abnormalities.	Lecture Discussion, Demonstration

Unit	Hrs (T+P)	Content	Teaching method
VII	10+0	REPRODUCTIVE SYSTEM Functions of male reproductive systemFunctions of female reproductive system Outline of pregnancy, parturition, lactation Contraceptive measures Physiology of foetal growth	Lecture Discussion, Demonstration
VIII	15+10	NERVOUS SYSTEM Properties and functions of Neuron Mechanism of Stimulus and nerve impulse Functions of brain, spinal cord, cranial and spinal nerves. Synaptic transmission, reflexes, control of posturesand voluntary motor activity. Autonomic Nervous System	Lecture Discussion, Demonstration & Practicals
IX	5+10	SENSORY ORGANS Functions of the skin, eye, ear, nose and tongue	Lecture Discussion, Demonstration & Practicals
X	20+11	MUSCULAR SYSTEM Microscopic structure of muscle tissue, myoneuraljunction Physiology of Muscle contraction Exercisemetabolism Muscular activity based on metabolism and fatiguePhysiological changes on aging Exercise physiology	Lecture Discussion, Demonstration & Practicals
XI	20+8	APPLIED PHYSIOLOGY Heart and circulation: Normal ECG, blood pressure,cardiovascular compensation for postural and gravitational changes, determinants of cardiac performance. Neuromuscular system: Degeneration and re-generation of nerves, control of posture and voluntary movement, neuromuscular transmission,electrical phenomenon. Respiratory system: Normal breath sound, volume and compliance, effects of exercise on respiration, artificial respiration.	Lecture Discussion, Demonstration & Practicals

PRACTICALS AND DEMONSTRATION:

1. **Hematology:**
   a. **Study of Microscope and its uses**
   b. **Determination of RBC count**
   c. **Determination of WBC count**
   d. **Differential leukocyte count**
   e. **Estimation of hemoglobin**
   f. **Calculation of blood indices**

2. **Blood pressure– palpatory and auscultatory method: Variation of blood pressurein posture.**
   3. **Auscultation of Normal breath sound & heart sound**
   4. **Spirometry: Recording of Lung volumes & capacities.**
   5. **Breathe holding time**
   6. **Mercury column test (40 mm Hg test)**
   7. **Clinical Examination: Chest expansion, Pulse rate and Respiratory rate.**

3. **Central Nervous System:**
   1. **Testing of peripheral sensations and cranial nerves.**
   2. **Superficial and deep reflexes.**
   3. **Tests for Cerebral and Cerebella functions- Equillibrium and Nonequillibrium Tests**

4. **Graphs:**
   1. **Skeletal muscle-properties.**
   2. **Cardiac muscle-properties**

5. **Clinical examination-**
   Higher functions, memory, time, orientation, reflexes, motor & sensory system

Recommended text books:

1. **Anatomy and Physiology - Dr. Minakshi Pathak**
2. **Text book of medical physiology – Guyton Arthur**
3. **Essentials of medical physiology sembulingam**
4. **Concise medical physiology – Chaudhuri SujitK.**
5. **Human Physiology – ChatterjeeC.C.**
6. **Text book of practical Physiology –Ranade.**
7. **Text book of Physiology – A. K.Jain.**

Reference:

1. **Review of Medical Physiology – Ganong WilliamF.**
2. **Physiological basis of Medical practice – Best &Taylor**

BIOMECHANICS AND KINESIOLOGY I

Course Code: U21BPTT209 **Instruction hours: Theory** 90 hours

Practical 30 hours

Unit	Hrs T+P	Content	Teaching method
I	10+0	**Basic Concepts in Biomechanics: Kinematics and Kinetics** **a. Types of Motion** **b.** Location of Motion **c.** Direction of Motion **d.** Magnitude of Motion **e.** Definition of Forces f. Force ofGravity g. Reaction forces h. Equilibrium i. Objects in Motion j. force of friction k. Concurrent force systems l. Parallel force system m. Work n. Moment arm of force o. Force components p. Equilibrium of levers	Lecture Demonstration
II	10+6	**Joint structure and Function -** a. Joint design b. Materials used in human joints c. General properties of connective tissues d. Human joint design e. Joint function f. Joint motion g. General effects of disease, injury and immobilization.	Lecture Discussion, Demonstration & Practicals
III	10+6	**Muscle structure and function -** a. Mobility and stability functions of muscles b. Elements of muscle structure c. Muscle function d. Effects of immobilization, injury and aging	Lecture Discussion, Demonstration & Practicals

IV	8+3	**Biomechanics of the Thorax and Chest wall -** a. General structure and function b. Rib cage and the muscles associated with the rib cage c. Ventilatory motions: its coordination and integration d. Developmental aspects of structure and function e. Changes in normal structure and function in relation to pregnancy, scoliosis and COPD	Lecture Discussion, Demonstration & Practicals

	4+3	**The Temporomandibular Joint-** a. General features, structure, function and dysfunction	
V	16+4	**Biomechanics of Upper Limb** a. The shoulder complex: Structure and components of the shoulder complex and their integrated function	Lecture Discussion, Demonstration & Practicals
	16+4	b. The elbow complex: Structure and function of the elbow joint – humeroulnar and humeroradial articulations, superior and inferior radioulnar joints; mobility and stability of the elbow complex; the effects of immobilization and injury.	
	16+4	c. The wrist and hand complex: Structural components and functions of the wrist complex; structure of the hand complex; functional position of the wrist and hand.	

RECOMMENDED BOOKS

1 Joint structure and function – Cynthia Norkin

2 Textbook of Biomechanics by Subrata Pal

3 Kinesiology and Biomechanics by Bhartendu Kumar

NUTRITION

Course Code: U21BPTT212 **Instruction hours:** Theory – 45 hours

Course description: The course is designed to assist the students to acquire knowledge of nutrition for maintenance of optimum health and its application for different ages, activities in metabolic disorders.

Unit	Hrs (T)	Content	Teaching method
I	5	FOOD & NUTRITION Introduction Nutrition: Concepts & various aspects Role of nutrition in healthy body National nutritional policy Food: Role in nutritional & medicinal values Elements of nutrition: Macro & micro nutrients Calorie & Basal Metabolic Rate	Lecture Discussion
II	10	CARBOHYDRATES, PROTEINS, FATS Classification & caloric value Recommended daily allowance Dietary sources Functions Digestion, Absorption & Storage Malnutrition: Deficiencies & Over consumption	Lecture Discussion
III	5	WATER & ELECTROLYTES Water: Daily requirement, sources, regulation of water Metabolism Electrolytes: Types, sources, composition of body fluids	Lecture Discussion
IV	10	VITAMINS & MINERALS Classification Recommended daily allowance Dietary sources Functions Absorption and storage Deficiencies & Hypervitaminosis	Lecture Discussion
V	5	ENERGY Requirements of different categories of people Measurement of energy Body Mass Index and basic metabolism Basal Metabolic Rate – determination and factors affecting it	Lecture Discussion

Unit	Hrs (T)	Content	Teaching method
VI	10	BALANCED DIET Concept Recommended Daily Allowance Nutritive value of foods Planning balanced diets for different categories of people Budgeting of food	Lecture Discussion

Recommended text books:

1. Comprehensive Textbook of Nutrition for BSc Nurses- Rishi Avasthi

ENVIRONMENTAL STUDIES

Course Code: U21BPTT213 **Instruction hours:** Theory – 45 hours

Course description:

The course is designed as per the UGC regulation for all under graduate courses of branches of highereducation. The subject is designed to refresh the student regarding the multidisciplinary nature of the environment and conservation of the ecosystem.

Course Objectives: at the end of the course, the candidate should

1. **Know about the environment**
2. **Understand the surrounding**
3. **Know about the biotic interaction**

Unit	Hrs (T)	Content	Teaching method
I	3	THE MULTIDISCIPLINARY NATURE OF ENVIRONMENT STUDIES: • Definition, scope and importance • Need for public awareness.	Lecture Discussion
II	7	RENEWABLE AND NON-RENEWABLE RESOURCES • Forest resources: Use and over-exploitation, deforestation, case studies, timber extraction, mining, dams, and their effects on forests and tribal people. • Water Resources: Use and over-utilization of surface and ground water, floods drought, conflicts over water, dam benefits and problems. • Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies • Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer – pesticide problems, water logging, salinity, case studies.	Lecture Discussion

Unit	Hrs (T)	Content	Teaching method
		• Energy resources: Growing energy needs, renewable and non-renewable energy resources, use of alternate energy sources, case studies. • Land resources: Land as a resource, land degradation, man induces landslides, soil erosion and desertification Role of an individual in conservation of natural resources. Equitable use of resources for sustainable lifestyles.	Lecture Discussion
III	10	ECOSYSTEMS • Concept of an ecosystem • Structure and function of an Ecosystem • Producer, consumers, and decomposers • Energy flow in the ecosystem • Ecological succession Food chains, food webs and ecological pyramids Introduction, types, characteristic features, structure and function of the ecosystem Forest ecosystem, Grassland ecosystem, Desert ecosystem, Aquatic ecosystem (ponds, streams, lakes, estuaries)	Lecture Discussion
IV	10	BIODIVERSITY AND ITS CONSERVATION Introduction – Definition of genetics, species, and ecosystem diversity Biogeographical classification of India Value of Biodiversity: consumptive use, productive use, social, ethical aesthetic, and option views.	Lecture Discussion

Unit	Hrs (T)	Content	Teaching method
V	5	ENVIRONMENTAL POLLUTION: • Air pollution • Water Pollution • Soil Pollution • Marine Pollution • Noise Pollution • Thermal Pollution • Nuclear hazards Soil waste management: Causes, effects and control measures of urban and industrial wastes. Role of an individual in prevention of pollution. Pollution case studies. Disaster management: floods, earthquake, cyclone and landslides.	Lecture Discussion
VI	5	SOCIAL ISSUES AND THE ENVIRONMENT • From Unsustainable to Sustainable development Urban problems related to energy • Water conservation, rain water harvesting, watershed management • Resettlement and rehabilitation of people; its problems and concerns. Case Studies • Environmental ethics: Issues and possible solutions. • Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case Studies. • Wasteland reclamation. • Consumerism and waste products. • Environment Protection Act. • Air (Prevention and Control of Pollution) Act • Water (Prevention and control of Pollution) Act • Wildlife Protection Act • Forest Conservation Act • Issues involved in enforcement of environmental legislation. • Public awareness	Lecture Discussion

Unit	Hrs (T)	Content	Teaching method
VII	5	HUMAN POPULATION AND THE ENVIRONMENT • Population growth, variation among nations. • Population explosion – Family Welfare Programme. • Environment and human health. • Human Rights. • Value Education. • HIV/AIDS • Women and Child Welfare. • Role of Information Technology in Environment and human health • Case Studies.	Lecture Discussion

Recommended text books:

1. **Textbook of Environmental Studies for Undergraduate Courses-Erach Barucha**

Reference:

1. **Agarwal, K.C.2001 Environmental Biology, Nidhi Publications Ltd. Bikaner**
2. **Odum, EP.1971 Fundamentals of Ecology. W B Saunders Co**

III-Semester

SEMESTER III

MICROBIOLOGY

U21BPTT315 Instruction hours: Theory 37 hours

Practical 8 hours

Unit	Hrs (T+P)	Content	Teaching method	
I	8+0	**General Microbiology -** a. Definitions: infections, parasite, host, vector, fomite, contagious disease, infectiousdisease, epidemic, endemic, pandemic, Zoonosis, Epizootic, Attack rate. b. Normal flora of the human body. a. Routes of infection and spread; endogenous and exogenous infections; source atreservoir of infections. b. Bacterial cell. Morphology limited to recognizing bacteria in clinical samples Shape,motility and arrangement. Structures, which are virulence, associated. c. Physiology: Essentials of bacterial growth requirements. d. Sterilization, disinfection and universal precautions in relation to patient care and diseaseprevention. Definition of asepsis, sterilization, disinfection. e. Antimicrobials: Mode of action, interpretation of susceptibility tests, resistance spectrumof activity.	Lecture Discussion, Demonstration	
II	5+1	**Immunology -** a. Basic principles of immunity immunobiology: lymphoid organs and tissues. Antigen, Antibodies, antigen and antibody reactions with relevance to pathogenesis and serological diagnosis. b. Humoral immunity and its role in immunity Cell mediated immunity and its role in immunity. Imunology of hypersensitivity, measuring immune	Lecture Discussion, Demonstration & Practicals	

		functions.		
III.	9+3	**Bacteriology -** a. To be considered under the following headings b. Morphology, classification according to pathogenicity, mode of transmission, methods of prevention, collection and transport of samples for laboratory diagnosis, interpretation of laboratory reports. c. Staphylococci, Streptococci and Pneumococci. d. Mycobacteria: Tuberculosis, M.leprae, atypical mycobacteria, Enterobacteriaceae, e. Vibrois: V. cholerae and other medically important vibrios, Campylobacters and Helicobacters, Pseudomonas. f. Bacillus anthracis, Sporing and non-sporing anaerobes: Clostridia, Bacteroides and Fusobacteria.	Lecture Discussion, Demonstration & Practicals	
IV	4+0	**General Virology -** a. General properties: Basic structure and broad classification of viruses. Pathogenesis and pathology of viral infections. Immunity and prophylaxis of viral diseases. Principles of laboratory diagnosis of viral diseases. List of commonly used antiviral agents.	Lecture Discussion, Demonstration & Practicals	
V.	4+2	**Mycology -** a. General properties of fungi. Classification based on disease: superficial, subcutaneous, deep mycosel opportunistic infections including Mycotoxins, systemic mycoses. General principles of fungal diagnosis, Rapid diagnosis. Method of collection of samples. Antifungal agents.	Lecture Discussion, Demonstration & Practicals	

VI.	7+2	1. Clinical/Applied Microbiology - a. Streptococcal infections: Rheumatic fever and Rheumatic heart disease, Meningitis. b. Tuberculosis, c. Pyrexia of unknown origin, leprosy, d. Sexually transmitted diseases, Poliomyelitis, e. Hepatitis, f. Acute-respiratory infections, Central nervous System infections, Urinary tract infections, g. Pelvic inflammatory disease, Wound infection, Opportunistic infections, HIV infection, h. Malaria, Filariasis, Zoonotic diseases	Lecture Discussion, Demonstration & Practicals	

PRACTICAL

1. Demonstration of Microscopes and its uses
2. Principles, uses and demonstration of common sterilization equipment
3. Demonstration of common culture media
4. Demonstration of motility by hanging drops method
5. Demonstration of Gram Stain, ZN Stain
6. Demonstration of Serological test: ELISA
7. Demonstration of Fungus

PATHOLOGY

Course Code: U21BPTT316 **Instruction hours: Theory 40 hours**

Practical 5 hours

S. No	Hours T+P	Content	Teaching method
		GENERAL PATHOLOGY	
1	5+0	**INTRODUCTION** Introduction to Pathology, Reversible, irreversible injury (Necrosis) Pigments Cell injury and Adaptation , Inflammation [Acute &Chronic] Wound healing & Repair	Lecture Discussion
2	3+0	**IMMUNOLOGY** Hypersensitivity reactions Amyloidosis Autoimmune disease (SLE)	Lecture Discussion
2	3+0	**INFECTIONS** Mycobacterial disease – Tb,Leprosy,Syphilis Viral disease (Polio), Parasites (Amoebiasis, Malaria,Filaria) AIDS	Lecture Discussion
4	3+0	**HEMODYNAMICS** Edema and chronic venous congestion Infarction & Shock Thrombosis & Embolism	Lecture Discussion
5	1+0	**BLOOD BANKING** Blood Transfusion- Blood grouping, Typing, TTI, Transfusion Reaction, Blood components.	
6	3+0	**NEOPLASIA** – Definition, D/W benign & malignant, Metastasis Carcinogenesis, precancerous lesion	Lecture Discussion
7	1+0	**GENETICS** Down syndrome,Kleinfelter syndrome, Turner syndrome	Lecture Discussion
8	2+0	**NUTRITION** Nutritional deficiencies PEM, Vitamin A and D deficiency	Lecture Discussion
9	4+0	**BLEEDING AND PLATELET DISORDERS** Bleeding & Coagulation disorders Anaemia- Definition, Classification Hemolytic anaemias Leukemia (AML,CLL,CML), MM Lymphoma & Causes of splenomegaly	Lecture Discussion

		S SYSTEMIC PATHOLOGY	
10	2+0	**RESPIRATORY SYSTEM** Pneumonia, Tuberculosis Lung COPD, Lung cancer	Lecture Discussion
11	2+0	**CVS** Congenital heart disease, Rheumatic heart disease Atherosclerosis, IHD, Hypertensive heart disease	Lecture Discussion
12	3+0	**GIT** Gastritis, Peptic ulcer & Gastric Tumors Intestine- Polyps & Carcinoma Salivary gland & oesophageal lesion	Lecture Discussion
13	2+0	**HEPATOBILIARY** Jaundice & Hepatitis ALD, Cirrhosis, Liver Tumours	Lecture Discussion
14	1+0	**CNS**[central nervous system] Meningitis & CNS Tumour	Lecture Discussion
15	2+0	**ENDOCRINE** Exocrine & Endocrine lesions of pancreas Benign & Malignant lesions of Thyroid	Lecture Discussion
16	3+0	**BONE** Osteomyelitis, Bone Tumour Calcification, Intracellular accumulation Arthritis & Paget's disease	Lecture Discussion
		PRACTICALS	
17	0+5	Blood grouping & Typing Urine – Sugar & Ketone bodies Gross – Cholelithiasis, Cirrhosis,Hydatid cyst,Tb lung,Lipoma, Giant cell tumour M/E – Granuloma, Cavernous Hemangioma, Teratoma, Squamous cell carcinoma PS – (Iron deficiency anemia/Malaria/Leukemia)	Demonstration & Practicals

PRACTICALS:

Demonstration of Slides – The students may be demonstrated the common histopathological, hematological and cytological slides and specimens and charts and their interpretations

Bio chemistry

Course Code: U21BPTT317 Instruction hours: Theory 30 hours

Practical 15 hours

Unit	Hrs T	Content	Teaching method
I	2	**Cell:** Structure and function of Plasma membrane, Structure and functions of cell organelles, Membrane transport mechanisms	Lecture Discussion, Demonstration & Practicals
II	7	**Carbohydrates** Chemistry: Definition, Classification, functions of Carbohydrates, Composition and functions of Proteoglycans Metabolism: Glycolysis, TCA cycle (Krebs cycle), Gluconeogenesis, Glycogen metabolism, Glycogen storage disorders, Blood Glucose regulation, Diabetes Mellitus and its laboratory investigations.	Lecture Discussion, Demonstration & Practicals
III	6	**Lipid** Chemistry: Definition, Classification, biomedical importance of Lipids, Classification of Fatty acids, Essential fatty acid, Functions of Phospholipids, Structure and functions of Cholesterol, Names and functions of Bile salts, Types and functions Lipoproteins, Functions of prostaglandins	Lecture Discussion, Demonstration & Practicals
IV	4	**Proteins** Chemistry: Classification of amino acids, Essential amino acids, Functional classification of proteins, Plasma proteins and their function Metabolism: Urea cycle, Special products of Glycine, Phenylalanine, tyrosine, tryptophan, Arginine	Lecture Discussion, Demonstration & Practicals

V	2	**Muscle Biochemistry** Composition, Structure and disorders related to collagen, Elastin, Muscle contraction, Source of energy during fasting and fed state, Cori's cycle, Glucose-alanine cycle	Lecture Discussion, Demonstration & Practicals
VI	5	**Enzymes:** Definition, Classification, Co-enzymes, Factors affecting enzyme activity, Enzyme inhibition, Diagnostic and therapeutic uses of enzymes, Clinical enzymology	Lecture Discussion, Demonstration & Practicals
VII	10	**Vitamins and Minerals:** Sources, Daily intake (RDA), functions and deficiency manifestations of fat soluble vitamins. Sources, Daily intake (RDA), functions and deficiency manifestations of water soluble vitamins. Functions and deficiency manifestations of calcium, phosphorous, iron, zinc, copper, sodium, potassium, selenium, fluoride	Lecture Discussion, Demonstration & Practicals
VIII	2	**Electron Transport chain:** High energy phosphates, ATP-ADP cycle, Oxidative phosphorylation, Un Couplers	Lecture Discussion, Demonstration & Practicals
IX	3	**Nutrition:** Calorie and Calorific value, Respiratory quotient, Basal Metabolic rate, Specific dynamic action, Calculation of energy requirements, Dietary fiber, Glycemic index, Nitrogen balance, Limiting amino acid and mutual supplementation, Protein calorie malnutrition	Lecture Discussion, Demonstration & Practicals
X	3	**Acid base balance and imbalance:** Definition, normal range, regulation of pH, Acid base disorders	Lecture Discussion, Demonstration & Practicals
XI	1	**Free radicals and antioxidants** Definition & examples. Sources of free radical formation & Antioxidants Role of free radicals in the development of diseases	Lecture Discussion, Demonstration & Practicals

Recommended text books:

1. Concise textbook of Biochemistry for Paramedical Student-DM Vasudevan

PHARMACOLOGY

U21BPTT317 **Instruction hours: Theory** 45 hours

Unit	Hrs T	Content	Teaching method,
1	4	**General Pharmacology:** 1. Introduction (General Principles of pharmacology) 2. Pharmacokinetics 3. Pharmacodynamics 4. Adverse effects and factors modifying drug action	Lecture Discussion,
2	4	**Autonomic Nervous System:** 1. Introduction 2. Cholinergic drugs 3. Anticholinergics 4. Adrenergicdrugs 5. Antiadrenergics	Lecture Discussion,
3	5	**Cardiovascular System:** 1. Drugs for Hypertension 2. Ischemic heart disease 3. Drugs for the treatment of heart failure 4. Antiarrhythmic drugs	Lecture Discussion,
4	8	**Central Nervous System +Peripheral Nervous System:** 1. Sedatives 2. Mood disorders 3. Antipsychotics 4. Anti-parkinsonism and Anti-Alzheimer drugs 5. Skeletal muscle relaxants 6. Local anaesthetics 7. Alcohol and Antiepileptics 8. General anaesthetics	Lecture Discussion,
5	5	**Endocrine:** 1. Diabetes mellitus 2. Drugs for Thyroid disorders 3. Drugs for Ca^{2+}balance 4. Progesterone and oestrogen	Lecture Discussion,

		5. Androgens and anabolic steroids	
6	5	**Inflammation:** 1. NSAIDs 2. Glucocorticoids 3. Drugs for Rheumatoid arthritis 4. Drugs forOsteoarthritis and gout 5. Neuromuscular inflammatory disorders	Lecture Discussion,
7	2	**Respiratory System:** 1. Drugs for Bronchial asthma and COPD 2. Allergic rhinitis and antitussives	Lecture Discussion,
8	2	**Gastrointestinal Tract:** 1. Peptic Ulcer 2. Drugs for constipation and diarrhea	Lecture Discussion,
9	3	**Blood:** 1. Hypolipidemic drugs 2. Anticoagulants 3. Anaemia	Lecture Discussion,
10	1	**Geriatrics**	Lecture Discussion,
11	1	**Vitamins**	Lecture Discussion,
12	2	**Cancer Chemotherapy**	Lecture Discussion,
13	1	**Narrow spectrum antibiotics**	Lecture Discussion,
14	1	**Broad spectrum antibiotics**	Lecture Discussion,
15	1	**Topical antibiotics**	Lecture Discussion,

BIOMECHANICS AND KINESIOLOGY-II

Course Code: U21BPTT318 **Instruction hours:** Theory – 90 hours

Unit	Hrs T+P	Content	Teaching method
I	10	**Biomechanics of the vertebral column -** • General structure and function • Regional structure and function – Cervical region, thoracic region, lumbar region, sacral region • Muscles of the vertebral column • General effects of injury and aging	Lecture Discussion, Demonstration & Practicals
II	50	**Biomechanics of the peripheral joints -** a. The hip complex: structure and function of the hip joint; hip joint pathology- arthrosis, fracture, bony abnormalities of the femur: b. The knee complex: structure and function of the knee joint – tibiofemoral joint and patellofemoral joint; effects of injury and disease. c. The ankle and foot complex.: structure and function of the ankle joint, subtalar joint, talocalcaneonavicular joint, transverse tarsal joint, tarsometatarsal joints, metatarsophalangeal joints, interphalangeal joints, structure and function of the plantar arches, muscles of the ankle and foot, deviations from normal structure and function – Pes Planus and Pes Cavus.	Lecture Discussion, Demonstration & Practicals
III.	30	**Analysis of Posture and Gait –** Static and dynamic posture, postural control, kinetics and kinematics of posture, ideal posture analysis of posture, effects of posture on age, pregnancy, occupation and recreation; general features of gait, gait initiation, kinematics and kinetics of gait, energy requirements, kinematics and kinetics of the trunk and upper extremities in relation to gait, stair case climbing and running, effects of age, gender, assistive devices, disease, muscle weakness, paralysis, asymmetries of the lower extremities, injuries and malalignments in gait;Movement Analysis : ADL activities like sitting -- to standing, lifting, various grips , pinches.	Lecture Discussion, Demonstration & Practicals

EXERCISE THERAPY - BASICS & SOFT TISSUE MANIPULATION

Course Code: U21BPTT319 **Instruction hours:** Theory – 60 hours

Unit	Hrs	Content	Teaching method
I	3	**Introduction to Exercise Therapy** - The aims of Exercise Therapy, The techniques of Exercise Therapy, Approach to patient's problems, Assessment of patient's condition – Measurements of Vital parameters, Starting Positions – Fundamental positions & derived Positions, Planning of Treatment	Lecture Discussion, Demonstration & Practicals
II	6	**Methods of Testing** a. Functional tests b. Measurement of Joint range: ROM-Definition, Normal ROM for all peripheral joints & spine, Goniometer-parts, types, principles, uses, Limitations of goniometry, Techniques for measurement of ROM for all peripheral joints c. Tests for neuromuscular efficiency i. Electrical tests ii. Manual Muscle Testing: Introduction to MMT, Principles & Aims, Indications & Limitations, Techniques of MMT for group & individual: Techniques of MMT for upper limb / Techniques of MMT for lower limb / Techniques of MMT forspine. iii. Anthropometric Measurements: Muscle girth – biceps, triceps, forearm, quadriceps, calf iv. Static power Test v. Dynamic power Test vi. Endurance test d. Speed test Tests for Co-ordination	Lecture Discussion, Demonstration & Practicals

		e. Tests for sensation f. Pulmonary Function tests g. Measurement of Limb Length: true limb length, apparent limb length, segmental limblength h. Measurement of the angle of Pelvic Inclination.	
III.	5	**Relaxation** a. Definitions: Muscle Tone, Postural tone, Voluntary Movement, Degrees of relaxation, Pathological tension in muscle, Stress mechanics, types of stresses, Effects of stress onthe body mechanism, Indications of relaxation, Methods & techniques of relaxation-Principles & uses: General, Local, Jacobson's, Mitchel's, additional methods.	Lecture Discussion, Demonstration & Practicals
IV.	5	**Passive Movements** a. Causes of immobility, Classification of Passive movements, Specific definitions related to passive movements, Principles of giving passive movements, Indications, contraindications, effects of uses, Techniques of giving passive movements.	Lecture Discussion, Demonstration & Practicals
V.	5	**Active Movements** a. Definition of strength, power & work, endurance, muscle actions. b. Physiology of muscle performance: structure of skeletal muscle, chemical & mechanicalevents during contraction &relaxation, muscle fiber type, motor unit, force gradation. c. Causes of decreased muscle performance d. Physiologic adaptation to training: Strength & Power, Endurance. e. Types of active movements	Lecture Discussion, Demonstration & Practicals

VI.	5	**Free exercise:** Classification, principles, techniques, indications, contraindications, effects and uses	Lecture Discussion, Demonstration & Practicals
VII.	5	**Active Assisted Exercise:** principles, techniques, indications, contraindications, effects and uses Assisted-Resisted Exercise: principles, techniques, indications, contraindications, effects and uses Resisted Exercise: Definition, principles, indications, contraindications, precautions & techniques, effects and uses	Lecture Discussion, Demonstration & Practicals
VIII.	6	**Types of resisted exercises:** Manual and Mechanical resistance exercise, Isometric exercise, Dynamic exercise: Concentric and Eccentric, Dynamic exercise: Constant versus variable resistance, Isokinetic exercise, Open-Chain and Closed-Chain exercise.	Lecture Discussion, Demonstration & Practicals
IX.	20	a. History and Classification of Massage Technique b. Principles, Indications and Contraindications c. Technique of Massage Manipulations d. Physiological and Therapeutic Uses of Specific Manipulations	Lecture Discussion, Demonstration & Practicals

BIOMECHANICS AND KINESIOLOGY PRACTICAL

Course Code: U21BPTT320 **Instruction hours:** Practical – 60 hours

PRACTICAL- shall be conducted for various joint movements and analysis of the same. Demonstration may also be given as how to analyze posture and gait. The student shall be taught and demonstrated to analysis for activities of daily living – ADL – (like sitting to standing, throwing, lifting etc.) The student should be able to explain and demonstrate the movements occurring at the joints, the muscles involved, the movements or muscle action produced, and mention the axis and planes through which the movements occur. The demonstrations may be done on models or skeleton.

EXERCISE THERAPY - BASICS & SOFT TISSUE MANIPULATION PRACTICAL

Course Code: U21BPTT321 **Instruction hours:** Practical – 120 hours

1. Different test methods
2. Demonstrate relaxation techniques.
3. Demonstrate to apply the technique of passive movements
4. Demonstrate various techniques of Active movements
5. Demonstrate massage technique application according to body parts.

THERAPEUTIC YOGA

Course Code: U21BPTT322 **Instruction hours:** Theory – 15 hours

Practical 30 hours

Unit	Hrs	Content	Teaching method
I	5	**Foundations of Yoga** a. Introduction to Yoga and its philosophy b. Brief history, development of Yoga c. Philosophical foundations of Yoga d. Streams & types of Yoga	Lecture Discussion
II	5	**Yoga and Health** a. Concept of body in yoga – Panchakosha theory b. Concept of Health and Disease in yoga c. Stress management through yoga d. Disease prevention and promotion of positive health through yoga	Lecture Discussion
III.	5	**Physiological effects of Yoga practices** a. Physiological effects of Shat kriyas b. Physiological effects of Asanas c. Physiological effects of Pranayamas d. Physiological effects of Relaxation techniques and Meditation	Lecture Discussion

PRACTICAL - List of Practical / Demonstrations (30 hours)

IV.	3	**Sukshma Vyayama/Sithilikarna Vyayama and Surya Namaskar: (3 hours)** a. Loosening exercises of each part of the body particularly of the joints b. 12 step Surya namaskar with prayer and specific mantras	Demonstration & Practicals

V.	3	**Yogic kriyas [Observation/ demonstration only] (3 hours)** a. Neti (Jala Neti, Sutra Neti) b. Dhauti (Vamana Dhauti, Vastra Dhauti) c. Trataka d. Shankaprakshalana (Laghu & Deergha)	Demonstration & Practicals
VI.	18	**Yogasanas** a. **Standing postures (4 hours)** i. Tadasana (Upward stretch posture) ii. Ardha Chakrasana (Half wheel posture) iii. Ardha Katicakrasana (Half lumber wheel posture) iv. Utkatasana (Chair posture) v. Pada Hastasana (Hand to toes posture) vi. Trikonasana (Triangle posture) vii. Parshva Konasana (Side angle posture) viii. Garudasana (Eagle posture) ix. Vrikshasana (Tree posture) b. **Prone positions (4 hours)** i. Makarasana (Crocodile posture) ii. Bhujangasana (Cobra posture) iii. Salabhasana (Locust posture) iv. Dhanurasana (Bow posture) v. Naukasana (Boat posture) vi. Marjalasana (Cat posture) c. **Supine postures (4 hours)** i. Ardha halasana/ Uttana Padasana ii. Sarvangasana (All limb posture) iii. Pawana muktasana (Wind releasing posture) iv. Matsyasana (Fish posture) v. Halasana (Plough posture) vi. Chakrasana (Wheel posture) vii. Setu Bandhasana (Bridge posture) viii. Shavasana (Corpse posture) d. **Sitting postures (4 hours)** i. Parvatasana (Mountain posture) ii. Bhadrasana (Gracious posture) iii. Vajrasana (Adamantine posture) iv. Paschimottanasana (Back stretching posture) v. Janushirasana (Head to knee posture) vi. Simhasana (Lion posture) vii. Gomukhasana (Cow head posture) viii. Ushtrasana (Camel posture) ix. Ardha Matsyendrasana (Half matsyendra spine twist posture)	Demonstration & Practicals

		x. Vakrasana (Spinal twist posture) xi. Kurmasana (Turtle posture) xii. Shashankasana (Rabbit posture) xiii. Mandukasana (Frog Posture) e. **Meditative postures and Meditation techniques (2 hours)** i. Siddhasana (Accomplished pose) ii. Padmasana (Lotus posture) iii. Samasana iv. Swastikasana (Auspicious posture)	
VII.	4	**Pranayamas (4 hours)** a. The practice of correct breathing and Yogic deep breathing b. Kapalabhati c. Bhastrika d. Sitali e. Sitkari f. Sadanta g. Ujjayi h. Surya Bhedana i. Chandra Bhedana j. Anuloma-Viloma/Nadishodana k. Bhramari	Demonstration & Practicals
VIII.	2	**Relaxation Techniques (2 hours)** a. Shavasana b. Yoga Nidra	Demonstration & Practicals

FIRST AID AND BASIC LIFE SUPPORT

Course Code: U21BPTT323 **Instruction hours:** Theory – 15 hours

Practical 30 hours

Unit	Hrs T+P	Content	Teaching method
I	3+4	**Introduction:** Definition of first aid. Importance of first aid, Golden rules of first aid, Scope and concept of emergency	Lecture Discussion Demonstration & Practicals
II	3+6	**First aid emergencies:** 1. Wounds and Bleeding 2. Head injuries 3. Epilepsy 4. Burns & Scalds : Causes, Degrees of burns, First aid treatment, General treatment. 5. Poisoning: Classification (irritants, acid, alkali, narcotics), Signs and symptoms. First aid treatment, General treatment. 6. Trauma due to foreign body intrusion : Eye, ear, nose, throat, stomach and lungs. 7. Bites : First aid, signs, symptoms and treatment. Dog bite : rabies Snake bite : neurotoxin, bleeding diathesis	Lecture Discussion Demonstration & Practicals
III.	3+6	Skeletal injuries Definition: Types of fractures of various parts of the body. Causes, Signs and Symptoms. Rules of treatment, Transportation of patient with fracture and spinal cord injuries. First aid measures in dislocation of joints. Treatment of muscle injuries.	Lecture Discussion Demonstration & Practicals

		Respiratory emergencies : 1. Asphyxia : Etiology, Signs & Symptoms, rules of treatment. 2. Drowning : Definition and management. 3. Artificial respiration : Types and techniques.	
IV	3+4	Transportation of the injured Methods of transportation : Single helper, Hand seat, Stretcher Wheeled transport (ambulance). Precautions taken : Blanket lift, Air and Sea travel.	Lecture Discussion Demonstration & Practicals
V	3+10	**Basic Life Support** – AHA Guidelines	Lecture Discussion Demonstration & Practicals

Semester – IV

EXERCISE THERAPY – ADVANCED

Course Code: U21BPTT324 **Instruction hours:** Theory – 60 hours

Unit	Hrs	Content	Teaching method
I.	3	**Specific exercise regimens** a. Isotonic: de Lormes, Oxford, MacQueen, Circiut weight training b. Isometric: BRIME (Brief Resisted Isometric Exercise), Multiple Angle c. Isometrics Isokinetic regimens	Lecture Discussion Demonstration & Practicals
II.	5	**Proprioceptive Neuromuscular Facilitation** a. Definitions & goals b. Basic neurophysiologic principles of PNF: Muscular activity, Diagonals patterns ofmovement: upper limb, lower limb c. Procedure: components of PNF d. Techniques of facilitation e. Mobility: Contract relax, Hold relax, Rhythmic initiation f. Strengthening: Slow reversals, repeated contractions, timing for emphasis, rhythmicstabilization Stability: Alternating isometric, rhythmic stabilization g. Skill: timing for emphasis, resisted progression Endurance: slow reversals, agonistreversal	Lecture Discussion Demonstration & Practicals
III.	4	**Suspension Therapy** a. Definition, principles, equipments & accessories, Indications & contraindications, Benefits of suspension therapy b. Types of suspension therapy: axial, vertical, pendular Techniques of suspension therapyfor upper limb Techniques of suspension therapy for	Lecture Discussion Demonstration & Practicals

		lower limb	
IV.	3	**Functional Re-education** a. Lying to sitting: Activities on the Mat/Bed, Movement and stability at floor level; Sittingactivities and gait; Lower limb and Upper limb activities.	Lecture Discussion Demonstration & Practicals
V.	4	**Aerobic Exercise** a. Definition and key terms; Physiological response to aerobic exercise, Examination and evaluation of aerobic capacity – Exercise Testing, Determinants of an Exercise Program, The Exercise Program, Normal and abnormal response to acute aerobic exercise, Physiological changes that occur with training, Application of Principles of an Aerobic conditioning program for patients – types and phases of aerobic training.	Lecture Discussion Demonstration & Practicals
VI.	5	**Stretching** a. Definition of terms related to 7.stretching; Tissue response towards immobilization and elongation, Determinants of stretching exercise, Effects of stretching, Inhibition and relaxation procedures, Precautions and contraindications of stretching, Techniques of stretching.	Lecture Discussion Demonstration & Practicals
VII.	5	**Manual Therapy & Peripheral Joint Mobilization** a. Schools of Manual Therapy, Principles, Grades, Indications and Contraindications, Effects and Uses – Maitland, Kaltenborn, Mulligan b. Biomechanical basis for mobilization, Effects of joint mobilsation, Indications and contraindications, Grades of mobilization, Principles of mobilization, Techniques of mobilization for upper limb, lower limb, Precautions.	Lecture Discussion Demonstration & Practicals

VIII.	5	**Balance – Definition** a. Physiology of balance: contributions of sensory systems, processing sensory information, generating motor output b. Components of balance (sensory, musculoskeletal, biomechanical) c. Causes of impaired balance, Examination & evaluation of impaired balance, Activities for treating impaired balance: mode, posture, movement, Precautions & contraindications,Types Balance retraining.	Lecture Discussion Demonstration & Practicals
IX.	4	**Co-ordination Exercise** a. Anatomy & Physiology of cerebellum with its pathways Definitions: Co-ordination, Inco-ordination b. Causes for Inco-ordination, Test for co-ordination: equilibrium test, non-equilibrium testPrinciples of co-ordination exercise. c. Frenkel's Exercise: uses of Frenkel's exercise, technique of Frenkel's exercise,progression, home exercise.	Lecture Discussion Demonstration & Practicals
X.	5	**Posture** a. Definition, Active and Inactive Postures, Postural Mechanism, Patterns of Posture,Principles of re-education: corrective methods and techniques, Patient education.	Lecture Discussion Demonstration & Practicals
XI.	5	**Walking Aids** a. Types: Crutches, Canes, Frames; Principles and training with walking aids	Lecture Discussion Demonstration & Practicals
XII	7	**Basics in Manual Therapy & Applications with Clinical reasoning** a. Examination of joint integrity i. Contractile tissues ii. Non contractile tissues	Lecture Discussion Demonstration & Practicals

		b. Mobility - assessment of accessory movement & End feel c. Assessment of articular & extra-articular soft tissue status i. Myofascial assessment ii. Acute & Chronic muscle hold iii. Tightness iv. Pain-original & referred d. Basic principles, Indications & Contra-Indications of mobilization skills for joints & softtissues. i. Maitland ii. Mulligan iii. Mckenzie iv. Muscle Energy Technique v. Myofascial stretching vi. Cyriax vii. Neuro Dynamic Testing	
XIII.	3	**Hydrotherapy** a. Definitions, Goals and Indications, Precautions and Contraindications, Properties ofwater, Use of special equipment, techniques, Effects and uses, merits and demerits	Lecture Discussion Demonstration & Practicals
XIV.	2	**Individual and Group Exercises** a. Advantages and Disadvantages, Organization of Group exercises, Recreational Activitiesand Sports	Lecture Discussion Demonstration & Practicals

ELECTROTHERAPY I

Course Code: U21BPTT425 **Instruction hours:** Theory – 60 hours

Unit	Hrs	Content	Teaching method
I	3	**Basic types of current** a. Direct Current: types, physiological &therapeutic effects. b. Alternating Current	Lecture Discussion Demonstration & Practicals
II	3	**Types of Current used in Therapeutics** a. Modified D.C i. Faradic Current ii. Galvanic Current b. Modified A.C i. Sinusoidal Current ii. Diadynamic Current.	Lecture Discussion Demonstration & Practicals
III.	6	**Faradic Current:** Definition, Modifications, Techniques of Application of Individual, Muscle and Group Muscle stimulation, Physiological & Therapeutic effects of Faradic Current, Precautions, Indications & Contra-Indications, Dangers.	Lecture Discussion Demonstration & Practicals
IV.	6	**Galvanic Current:** Definition, Modifications, Physiological & Therapeutic effects of Galvanic Current, Indications & Contra-Indications, Dangers, Effect of interrupted galvanic current on normally innervated and denervated muscles and partially denervated muscles.	Lecture Discussion Demonstration & Practicals
V.	2	1. Sinusoidal Current & Diadynamic Current in Brief. 2. HVPGS – Parameters & its uses	Lecture Discussion Demonstration & Practicals
VI.	5	1. Ionization / Iontophoresis: Techniques of Application of Iontophoresis, Indications, Selectionof Current, Commonly used Ions (Drugs) for pain, hyperhydrosis, would healing. 2. Cathodal / Anodal galvanism. 3. Micro Current & Macro Current	Lecture Discussion Demonstration & Practicals
VII.	5	**Types of Electrical Stimulators** a. NMES- Construction component. b. Neuro muscular diagnostic stimulator-construction component.	Lecture Discussion Demonstration & Practicals

		c. Components and working Principles d. Principles of Application: Electrode tissue interface, Tissue Impedance, Types of Electrode, Size& Placement of Electrode – Waterbath, Unipolar, Bi-polar, Electrode coupling, Current flow intissues, Lowering of Skin Resistance.	
VIII.	3	**Nerve Muscle Physiology:** Action Potential, Resting membrane potential, Propagation of Action Potential, Motor unit, synapse, Accommodation, Stimulation of Healthy Muscle, Stimulation of Denervated Muscle, and Stimulation for Tissue Repair.	Lecture Discussion Demonstration & Practicals
IX.	5	**TENS:** Define TENS, Types of TENS, Conventional TENS, Acupuncture TENS, Burst TENS,Brief & Intense TENS, Modulated TENS. Types of Electrodes & Placement of Electrodes, Dosage parameters, Physiological & Therapeutic effects, Indications & Contraindications.	Lecture Discussion Demonstration & Practicals
X	2	**Pain:**Define Pain, Theories of Pain (Outline only), Pain Gate Control theory in detail.	Lecture Discussion Demonstration & Practicals
XI.	10	**Electro-diagnosis** 1. FG Test 2. SD Curve: Methods of Plotting SD Curve, Apparatus selection, Characters of Normally innervated Muscle, Characters of Partially Denervated Muscle, Characters of Completely denervated Muscle, Chronaxie & Rheobase. 3. Nerve conduction velocity studies 4. EMG: Construction of EMG equipment. 5. Bio-feedback.	Lecture Discussion Demonstration & Practicals
XII.	10	**Medium Frequency** 1. Interferential Therapy: Define IFT, Principle of Production of IFT, Static Interference System,Dynamic Interference system, Dosage Parameters for IFT, Electrode placement in IFT, Physiological & Therapeutic effects, Indications & Contraindications. 2. Russian Current 3. Rebox type Current	Lecture Discussion Demonstration & Practicals

ELECTROTHERAPY II

Course Code: U21BPTT426 **Instruction hours:** Theory – 75 hours

Unit	Hrs	Content	Teaching method
I	2	Electro Magnetic Spectrum.	Lecture Discussion Demonstration & Practicals
II	10	**SWD:** a) Define short wave, Frequency & Wavelength of SWD, Principle of Production of SWD,Circuit diagram & Production of SWD, Methods of Heat Production by SWD treatment, Typesof SWD Electrode, Placement & Spacing of Electrodes, Tuning, Testing of SWD Apparatus, Physiological & Therapeutic effects, Indications & Contraindications, Dangers, Dosage parameters. b) Pulsed Electro Magnetic Energy: Principles, Production & Parameters of PEME, Uses of PEME.	Lecture Discussion Demonstration & Practicals
III.	5	**Micro Wave Diathermy:** Define Microwave, Wave length & Frequency, Production of MW, Applicators, Dosage Parameters, Physiological & Therapeutic effects, Indications & Contraindications, Dangers of MWD.	Lecture Discussion Demonstration & Practicals
IV.	10	**Ultrasound Therapy** Define Ultrasound, Frequency, Piezo Electric effects: Direct, Reverse, Production of US, Treatment Dosage parameters: Continuous& Pulsed mode, Intensity, US Fields: Near field, Far field, Half value distance, Attenuation, Coupling Media, Thermal effects, Non-thermal effects, Principles & Application of US: Direct contact, Water bag, Water bath, Solid sterile gel pack method for wound. Uses of US, Indications & Contraindications, Dangers of Ultrasound. Phonophoresis: Define Phonophoresis, Methods of application, commonly used drugs, Uses. Dosages of US.	Lecture Discussion Demonstration & Practicals
V.	5	**IRR:** Define IRR, wavelength & parameters, Types of IR generators, Production of IR, Physiological &	Lecture Discussion Demonstration & Practicals

		Therapeutic effects, Duration & frequency of treatment, Indication & Contraindication.	
VI.	10	**UVR**Define UVR, Types of UVR, UVR generators: High pressure mercury vapour lamp, Water cooled mercury vapour lamp, Kromayer lamp, Fluorescent tube, Theraktin tunnel, PUVA apparatus. Physiological & Therapeutic effects. Sensitizers & Filters. Test dosage calculation. Calculation of E1, E2, E3, E4 doses. Indications, contraindications. Dangers. Dosages for different therapeutic effects, Distance in UVR lamp	Lecture Discussion Demonstration & Practicals
VII.	10	**LASER:** Define LASER. Types of LASER. Principles of Production. Production of LASER by various methods. Methods of application of LASER. Dosage of LASER. Physiological & Therapeutic effects of LASER. Safety precautions of LASER. Classifications of LASER. Energy density & power density	Lecture Discussion Demonstration & Practicals
VIII.	18	**Superficial heating Modalities** 1. Wax Therapy: Principle of Wax Therapy application – latent Heat, Composition of Wax Bath Therapy unit, Methods of application of Wax, Physiological & Therapeutic effects, Indications & Contraindication, Dangers. 2. Contrast Bath: Methods of application, Therapeutic uses, Indications & Contraindications. 3. Moist Heat Therapy: Hydro collator packs – in brief, Methods of applications, Therapeutic uses, Indications & Contraindications. 4. Cyclotherm: Principles of production, Therapeutic uses, Indications & Contraindications. 5. Fluidotherapy: Construction, Method of application, Therapeutic uses, Indications & Contraindications. 6. Whirl Pool Bath: Construction, Method of Application, Therapeutic Uses, Indications & Contraindications. 7. Magnetic Stimulation, Principles, Therapeutic uses, Indications & contraindication	Lecture Discussion Demonstration & Practicals
IX.	5	**Cryotherapy:** Define- Cryotherapy, Principle- Latent heat of fusion,Physiological & Therapeutics effects, Techniques of Applications, Indications & Contraindications, Dangers, Methods of application with dosages.	Lecture Discussion Demonstration & Practicals

BASIC PHYSICS FOR PHYSIOTHERAPY

Course Code: U21BPTT427 **Instruction hours:** Theory – 45 hours

Unit	Hrs	Content	Teaching method
		Physical principles a. Structure and properties of matter - solids, liquids and gases, adhesion, surface tension,viscosity, density and elasticity. b. Structure of atom, molecules, elements and compound c. Electricity: Definition and types. Therapeutic uses. Basic physics of construction. Working d. Importance of currents in treatment. e. Static Electricity: Production of electric charge. Characteristic of a charged body. f. Characteristics of lines of forces. Potential energy and factors on which it depends.Potential difference and EMF. g. Current Electricity: Units of Electricity: farad, Volt, Ampere, Coulomb, Watt h. Condensers: Definition, principle, Types-construction and working, capacity & uses. i. Magnetism: Definition. Properties of magnets. Electromagnetic induction. Transmissionby contact. Magnetic field and magnetic forces. Magnetic effects of an electric field. j. Conductors, Insulators, Potential difference, Resistance and intensity k. Ohm's law and its application to DC and AC currents. Fuse: construction, working andapplication. l. Transmission of electrical energy through solids, liquids, gases and vacuum. m. Rectifying Devices-Thermionic valves, Semiconductors, Transistors, Amplifiers,transducer and Oscillator circuits. n. Display devices and indicators-analogue and digital.	Lecture Discussion Demonstration & Practicals

		o. Transformer: Definition, Types, Principle, Construction, Eddy current, working uses p. Chokes: Principle, Construction and working, Uses	
II.	5	**Effects of Current Electricity** a. Chemical effects-Ions and electrolytes, Ionisation, Production of an EMF by chemicalactions. b. Ionization: Principles, effects of various technique of medical ionization. c. Electromagnetic Induction. d. Electromagnetic spectrum.	Lecture Discussion Demonstration & Practicals
III.	10	**Electrical Supply** a. Brief outline of main supply of electric current b. Dangers-short circuit, electric shocks: Micro/ Macro shocks c. Precaution-safety devices, earthing, fuses etc. d. First aid and initial management of electric shock e. Burns: electrical & chemical burns, prevention and management	Lecture Discussion Demonstration & Practicals
IV.	10	**Various agents** a. Thermal agents: Physical Principles of cold, Superficial and deep heat. b. Ultrasound: Physical Principles of Sound c. Electro- magnetic Radiation: Physical Principles and their Relevance to PhysiotherapyPractice d. Electric Currents: Physical Principles and their Relevance to Physiotherapy Practice.	Lecture Discussion Demonstration & Practicals

EXERCISE THERAPY ADVANCED PRACTICAL

Course Code: U21BPTT428 **Instruction hours:** Practical – 120 hours

The students of exercise therapy are to be trained in Practical Laboratory work for all the topics discussed in theory. The student must be able to evaluate and apply judiciously the different methods of exercise therapy techniques on the patients. They must be able to

1. Demonstrate the technique of measuring using goniometry
2. Demonstrate muscle strength using the principles and technique of MMT
3. Demonstrate the techniques for muscle strengthening based on MMT grading
4. Demonstrate the PNF techniques
5. Demonstrate exercises for training co-ordination – Frenkel's exercise
6. Demonstrate the techniques of massage manipulations
7. Demonstrate techniques for functional re-education
8. Assess and train for using walking aids
9. Demonstrate mobilization of individual joint regions
10. Demonstrate to use the technique of suspension therapy for mobilizing and strengthening jointsand muscles
11. Demonstrate the techniques for muscle stretching
12. Assess and evaluate posture and gait
13. Demonstrate techniques of strengthening muscles using resisted exercises
14. Demonstrate techniques for measuring limb length and body circumference.

ELECTROTHERAPY I PRACTICAL

Course Code: U21BPTT429 **Instruction hours:** Practical – 90 hours

The student of Electrotherapy must be able to demonstrate the use of electrotherapy modalities applying the principles of electrotherapy with proper techniques, choice of dosage parameters and safety precautions.

1. Demonstrate the technique for patient evaluation – receiving the patient and positioning thepatient for treatment using electrotherapy.
2. Collection of materials required for treatment using electrotherapy modalities and testing of theapparatus.
3. Demonstrate placement of electrodes for various electrotherapy modalities
4. Electrical stimulation for the muscles supplied by the peripheral nerves
5. Faradism under Pressure for UL and LL
6. Plotting of SD curve with chronaxie and rheobase
7. Demonstrate FG test

ELECTROTHERAPY II PRACTICAL

Course Code: U21BPTT430 **Instruction hours:** Theory – 90 hours

The student of Electrotherapy must be able to demonstrate the use of electrotherapy modalities applying the principles of electrotherapy with proper techniques, choice of dosage parameters and safety precautions.

Demonstrate the technique for patient evaluation – receiving the patient and positioning the patient for treatment using electrotherapy.

1. Collection of materials required for treatment using electrotherapy modalities and testing of theapparatus.
2. Application of Ultrasound for different regions-various methods of application
3. Demonstrate treatment techniques using SWD, IRR and Microwave diathermy
4. Demonstrate the technique of UVR exposure for various conditions – calculation of test dose
5. Demonstrate treatment method using IFT for various regions
6. Calculation of dosage and technique of application of LASER
7. Technique of treatment and application of Hydrocollator packs, cryotherapy, contrast bath, waxtherapy
8. Demonstrate the treatment method using whirl pool bath
9. Winding up procedure after any electrotherapy treatment method.

Equipment care -

1. Checking of equipments
2. Arrangement of exercise therapy and electro therapy equipment.
3. Calibration of equipment
4. Purchase, billing, document of equipment.
5. Safety handling of equipments.
6. Research lab equipment maintenance.
7. Stock register, movement register maintenance

PHYSIOTHERAPY ETHICS

Course Code: U21BPTT431 **Instruction hours:** Theory – 15 hours

Legal and ethical considerations are firmly believed to be an integral part of medical practice in planning patient care. Advances in medical sciences, growing sophistication of the modern society's legal framework, increasing awareness of human rights and changing moral principles of the community at large, now result in frequent occurrences of healthcare professionals being caught in dilemmas over aspects arising from daily practice.

Medical/ Physiotherapy ethics has developed into a well based discipline which acts as a "bridge" between theoretical bioethics and the bedside. The goal is "to improve the quality of patient care by identifying, analyzing, and attempting to resolve the ethical problems that arise in practice". Doctors are bound by, not just moral obligations, but also by laws and official regulations that form the legal framework to regulate medical practice. Hence, it is now a universal consensus that legal and ethical considerations are inherent and inseparable parts of good medical practice across the whole spectrum. Few of the important and relevant topics that need to focus on are as follows:

Unit	Hrs	Content	Teaching method
I	1	Medical ethics versus medical law - Definition - Goal - Scope	Lecture Discussion
II	2	1. Introduction to Code of conduct 2. Basic principles of medical ethics – Confidentiality 3. Malpractice and negligence - Rational and irrational drug therapy 4. Autonomy and informed consent - Right of patients	
III.	2	1. Care of the terminally ill- Euthanasia 2. Organ transplantation 3. Medical diagnosis versus physiotherapy diagnosis.	
IV.	4	**Medico legal aspects of medical records** – Medico legal case and type- Records and documentrelated to MLC - ownership of medical records - Confidentiality Privilege communication - Release of medical information - Unauthorized disclosure - retention of medical records - othervarious aspects.	
V.	2	1. Professional Indemnity insurance policy 2. Development of standardized protocol to avoid near miss or sentinel events	

		3. Obtaining an informed consent. 4. Biomedical ethical principles	
VI.	4	1. Code of ethics for physiotherapists 2. Ethics documents for physiotherapists 3. Laws affecting physiotherapy practice	

V-Semester

CLINICAL ORTHOPAEDICS & TRAUMATOLOGY FOR PHYSIOTHERAPY

COURSE CODE: U21BPTT524 **TOTAL HOURS:** 120hrs

COURSE

This course intends to familiar with principles of orthopaedics along with familiarization with terminology and abbreviation for efficient and effective chart reviewing and documentation. The purpose of this course is to make physiotherapy students aware of various orthopaedics condition so these can be physically managed effectively both pre and postoperatively.

COURSE OBJECTIVE:

At the end of the course, the candidate will

a) Be able to discuss the aetiology, pathophysiology, clinical manifestation and conservative/ surgical management of various orthopaedics conditions.
b) Gain the skill of clinical examination and interpretation of the pre-operative cases and post-operative cases.
c) Be able to read and interpret salient features of the X-ray and correlate the radiologicalfindings with the clinical findings.
d) Be able to interpret pathological studies pertaining to orthopaedic condition.

Unit	Topics	Didactic Hours	Clinical hours	Total hours
I	1. Introduction to orthopaedics: i. Orthopaedic terminologies ii. Clinical examination iii. Common investigative procedures iv. General principles of management	7		7
II	1. Healing of fractures 2. Fractures & dislocations of upper limb: i. Causes, clinical features, mechanism of injury, complication, conservative & surgical management for following injuries. a) Fracture of clavicle & scapula. b) Fracture of greater tuberosity and neck of humerus. c) Fracture of shaft of humerus. d) Supracondylar fracture of humerus. e) Fracture of capitulum, radial head, olecranon, coronoid & epicondyles. f) Fracture both bone of forearm. g) Colle's fracture h) Smith fracture	15	4	19

	i) Scaphoid fracture j) Fracture of metacrapals k) bennet's fracture l) fracture of the phalanges m) Dislocation of shoulder & elbow 3. Fracture & dislocation of lower limb: i. Causes, clinical features, mechanism of injury, complication, conservative & surgical management for following injuries. a) Fracture neck, shaft and condyles of femur b) Fracture Patella c) Fracture of tibial condyles & bothbone leg fracture. d) Fracture calcaneum. e) Dislocation of hip & knee.	15	5	20
III	1. Deformities & Anomalies: i. Deformities of the spine. ii. Deformation of the lower limb 2. Degeneration of inflammatory condition: i. Osteo-orthosis/ arthritis ii. Spondylosis iii. Spondylolisthesis iv. Rheumatoid arthritis v. Ankylosing spondylitis. 3. Infective conditions i. Septic arthritis ii. Osteomyelitis iii. Tuberculosis spine 4. Arthoplasty: i. THR ii. TKR 5. Arthroscopy	6 6 9	4 4 4	33
IV	1. Soft tissue lesions: i. Sprains & strains ii. Capsulitis iii. Bursitis iv. Tenosynovitis v. Tendonitis	15	2	17

V	1. Peripheral nerve lesions: i. Radial, median & ulnar nerve lesions ii. Sciatic nerve lesions iii. Lateral popliteal nerve lesions. 2. Hand injuries: Tendon & joint injury 3. Amputation: Definition,levels of amputation, indications & complications	18	6	24

	4. Spine: Cervical Compressive myelopathy with quadriparesis, post traumatic paraplegia & pott's paraplegia.			

Recommended Text Books:

1. Mayil Vahanan Natarajan, "Textbook of Orthopaedics and Traumatology", 8th edition, Wolters Kluwer Pvt. Ltd., 2018.
2. Maheswari & Mhaskar, "Essential Orthopaedics", 7th edition, Jaypee publishers, 2022.
3. Adam's "outline of fractures", 12th edition, Elsevier, 2020
4. Apley's "System of Orthopaedics and fractures", 9th edition, CRC, 2017

General Surgery, Plastic Surgery and O.B.&G.

Course code: U21BPTT525 **Total: 90** Hours

COURSE DESCRIPTION:

This course intends to familiarize students with principles of General surgery including various specialties like cardiovascular,thoracic, neurology and plastic surgery as well as to provide introduction towomen's health which includes problems related to pregnancy and otherdisorders specific to women. It familiarizes the students with terminology andabbreviations for efficient and effective chart reviewing and documentation. Itexplores various conditions needing attention, focusing on epidemiology,pathology as well as primary and secondary clinical characteristics and theirsurgical and medical management. The purpose of this course is to makephysiotherapy students aware of various surgical conditions general surgery and specialty surgeries so these can be physically managed

OBJECTIVES:

effectively both pre aswell as postoperatively.

At the end of the course, the candidate will be able to:

a) Describe the effects of surgical trauma & anaesthesia in general.
b) Describe pre-operative evaluation, surgical indications in various surgical approaches, management and post operative care in above mentioned areas with possible complications.
c) Read & interpret findings of the relevant investigations.
d) Normal and abnormal physiological events, complications and management of pregnancy (Pregnancy, Labour, Puerperium).
e) Normal and abnormal physiological events, complications and management of uro- genital dysfunction (Antenatal, Postnatal and during menopause).

Units	Topics	Didactic Hours	Clinical Hours	Total Hours
	GENERAL SURGERY	**30**	**10**	**40**
I	1.Infection and inflammation: Acute & chronic signs, symptoms, complications &management. 2.Wounds and ulcers : Classification, healing & management,	3 2		
II	1. General : i. Anaesthesia types, Effect, indications , contraindications and postoperative complications ii. Haemorrhage and Shock, classification, iii. Water & Electrolyte imbalance	12	10	

	iv. Mastectomy and oncosurgery– approach, complications & management v. Amputation – types, sites, complications & management. vi. Burns – causes, complications, classification & management vii. Varicose veins and PVD 2. Neurosurgery: i. Head Injury – management ii. Intra cranial & Spinal tumors iii. Intracranial Aneurysm &AV malformation iv. Post operative Neurosurgical care 3. Abdominal surgeries i. Surgical anatomy of anterior abdominal wall. ii. Surgical approaches. iii. Common abdominal surgeries like cholecystectomy, gastrectomy, hernias, appendectomy, nephrectomy 4. E.N.T. Surgery i. Tracheostomy – indications, surgical approach & management ii. Surgical procedures in VIIth cranial nerve palsy Vertigo	4 5 3 1		
III	**CARDIO –THORACIC SURGERY&PLASTIC SURGERY**	**15**	**10**	**25**
	Cardio – thoracic surgery : 1. Introduction, Cardiorespiratory resuscitation, cardiopulmonary bypass, Special investigation procedures in cardiac surgery, Basic techniquesin cardiac surgery approach, incisions, types of operation, complications of cardiac surgery, Lines, drains and tubes. 2. Brief description of indications, surgery,Complications for following surgeries; i. Surgeries of the lung ii. Surgeries of pleura and pericardium	10	5	

	iii. Surgery for coronary artery disease iv. Valvular surgeries v. Surgery for Congenital Heart Disease vi. Peripheral arterial disorder, Burger's disease, Raeynaud's disease and Aneurysm Gangrene, Amputation, DVT Plastic surgery : Skin grafts & flaps – Types, indicationswith special emphasis to burns, wounds i. Ulcers, complications and postoperativecare ii. Tendon transfers, with special emphasis tohand, foot & facial paralysis, & repair of Flexor & Extensor Tendon Injuries iii. Keloid & Hypertrophied scar management iv. Reconstructive surgery of peripheral nerves Micro vascular surgery- reimplantation & revascularization			
IV	**OB&G**	**10**	**5**	**15**
	1) physiology of pregnancy: i. Development of the foetus, Normal/ Abnormal / multiple gestations, ii. Common Complications during pregnancy: a. Anaemia, b. P I H c. Eclampsia d. Diabetes, e. Hepatitis, TORCH infection or HIV 2) physiology of labour: i. Normal – Events of Ist, IInd & IIIrd Stages of labour ii. Complications during labour& management iii. Caesarean section- elective/ emergency & post operative care			

V	1) Post natal period: i. Puerperium & Lactation ii. Complications of repeated child bearing with small gaps iii. Methods of contraception 2) Uro-genital dysfunction: Uterine prolapse – Classification & Management (Conservative /Surgical) Cystocoele, Rectocoele, Enterocoele, Urethrocoele			

RECOMMENDED TEXT BOOKS

1. Bailey & Love's "Short Practice of Surgery", 28th edition, CRC Press, 2022.
2. S. Das "A Concise Textbook of Surgery", 6th edition, Dr. S. Das, 2010.

GENERAL MEDICINE, PAEDIATRICS AND PSYCHIATRY

Subject code: U21BPTT526 **Theory hours:** 120 hours

COURSE DESCRIPTION:

This subject follows the basic science subjects to provide the knowledge about relevant aspects of General Medicine, Pediatric and Psychiatry. The student will have a general understanding of the diseases the therapist would encounter in their practice. The objective of this course is that discussion the student will be able to list the etiology, pathology, clinical features and treatment methods for various medical conditions.

COURSE OBJECTIVES:

At the end of the course, the candidate will:

1. Be able to describe Etiology, Pathophysiology, Signs & Symptoms & Management of the Endocrinal, Metabolic, Geriatric & Paediatric & Psychiatric, Nutrition Deficiency conditions.
2. Be able to describe Etiology, Pathophysiology, Signs & Symptoms, Clinical Evaluation & Management of the various Rheumatologic Cardiovascular & Respiratory Conditions.
3. Acquire skill of history taking and clinical examination of Musculoskeletal, Respiratory, Cardio- vascular & Neurological System as a part of clinical teaching.
4. Be able to interpret auscultation findings with special emphasis to pulmonary system.
5. Study Chest X-ray, Blood gas analysis, P.F.T. findings & Haematological studies, for Cardiovascular, Respiratory, Neurological & Rheumatological conditions.
6. Be able to describe the principles of Management at the Intensive Care Unit.
7. Be able to acquire the skills of Basic Life Support. 8. Acquire knowledge of various drugs used for each medical condition to understand its effects and its use during therapy.

Unit	Topics	Didactic Hours	Clinical Hours	Total Hours
I	**GENERAL MEDICINE** **INFECTIONS:** Mode of spread, Effects of Infection in the body – and preventive measures of the following diseases: Bacteria - Tetanus Viral - Herpes simplex, Herpes Zoster, Varicella, Measles, Hepatitis B, AIDS. Protozoal – Filaria.	10	8	18

	POISONING: Clinical features – general management – common agents in poisoning – pharmaceutical agents – drugs of misuse – chemical pesticides – Envenomation. **FOOD AND NUTRITION:** Assessment – Nutritional and Energy requirements. Deficiency diseases – vitamins (A, B, C, D& K) and Minerals (iron, calcium phosphorus, iodine) clinical features and treatment. **METABOLIC & ENDOCRINE DISEASES:** Definition, aetiology, types, complications & management of: Diabetes mellitus Diseases of thyroid & parathyroid, adrenal and pituitary glands, Obesity. **HEMATOLOGY**: Clinical aspect of Anemia: iron deficiency, B12 & Folic acid deficiencies, Types of bleeding diathesis,. Clinical features and management of Hemophilia.			
II	**GIT:** • **Clinical manifestations of gastrointestinal disease –** Etiology, clinical features, diagnosis, complications and treatment of the following conditions: Reflux Oesophagitis, Achlasia Cardia, Carcinoma of Oesophagus, GI bleeding, Peptic Ulcer disease, Carcinoma of Stomach, Pancreatitis, Malabsorption Syndrome, Ulcerative Colitis, Peritonitis, Infections of Alimentary Tract . • **Clinical manifestations of liver diseases –** Aetiology, clinical features, diagnosis, complications and treatment of the following conditions : Viral Hepatitis, Wilson's Disease, Alpha1-antitrypsin deficiency, Tumors of the Liver, Gall stones, Cholycystitis. **DERMATOLOGY:** Examination and clinical manifestations of skin diseases; Causes, clinical features and management of the following skin conditions: Leprosy, Psoriasis, Pigmentary Anomalies, Vasomotor disorders, Dermatitis, Coccal and Fungal Parasitic and Viral infections.	31	10	41

	RESPIRATORY DISEASE: Definition, Etiology, Clinical features, signs and symptoms, complications, management and treatment of following lung disease: • Chronic bronchitis, Emphysema • Asthma • Bronchiectasis, cystic fibrosis • upper respiratory tract infections • Pneumonia, tuberculosis, lung abscess, fungal diseases, interstitial lung disease • Disease of pleura (Pleural effusion, Pneumothorax, Hydropneumothorax, Emphysema) • Disease of diaphragm and chest wall deformities • Respiratory failure – definition, types ,causes ,clinical features ,diagnosis and management. **NERVOUS SYSTEM:** • CVA – thrombosis, embolism, hemorrhage • Extrapyramidal lesion – Parkinsonism, athetosis, chorea, dystonia. • Disorders of cerebellar infarction • Disorders of muscle – myopathy • Disease of Anterior horn cells: SMA,MND, Syringomyelia ,poliomyelitis . • Multiple sclerosis • Infections of nervous system – encephalitis, neurosyphilis, meningitis. • Tetanus, transverse myelitis, tabes dorsalis, TB spine • Epilepsy **CARDIO-VASCULAR SYSTEM :** Definition, aetiology, clinical features & management of: cardiac failure, rheumatic fever, infective endocarditis, ischaemic heart disease, hypertension, pulmonary embolism & pulmonary infarction and deep vein thrombosis. • Congenital heart disease – ASD, VSD, Fallot's tetralogy, PDA.			

III	**RENAL CONDITION:** Acute & Chronic Renal Failure Urinary tract infections. **RHEUMATOLOGICAL CONDITION:** Introduction to autoimmune disease Systemic lupus erythematosis, Polymyositis, Dermatomyositis, Polyarthritis nodasa, Scleroderma, Rheumatoid arthritis, Osteoarthritis. **ENT:** Otitis media, otosclerosis, functional achonia and deafness, facial palsy – classification, medical and surgical management. **OPTHALMOLOGY:** Ophthalmologic surgical conditions, refractions, conjunctivitis, glaucoma, corneal ulcer, iritis, cataract, detachment of retina, defects of extra -ocular muscles surgical management. **GERIATRIC:** Physiology of aging /degenerative changes – musculoskeletal /neuromotor /cardio-respiratory /metabolic, endocrine, cognitive, immune systems, osteoporosis, alzhemimers, dementia List disease commonly encountered in the elderly population and causing disability	16		16
UNIT	**<u>PAEDIATRICS</u>**			
IV	**GROWTH AND DEVELOPMENT** [from child birth to 12 years • Normal uterine development of foetus . • Growth and development of child – motor, mental, language and social. **HIGH RISK PREGNANCY:** Perinatal, Postnatal problems and management (Birth injuries) Neck, shoulder dystocia, Brachial plexus injury,			

	Fractures, Birth Asphyxia, Inherited disease, Maternal infection (viral; bacterial) Gestational diabetes, Pregnancy induced hypertension, Epilepsy, sepsis. **DEVELOPMENTAL DELAY**: Etiology, pathophysiology, classification, clinical signs & symptoms. **CHILD AND NUTRITION :** • Significance of breast feeding. • Problems and management of LBW infants, • Nutritional requirements, malnutrition syndrome, vitamin deficiency disorders. **COMMON INFECTIOUS DISEASE IN CHILDREN:** Tetanus, diphtheria, mycobacterial measles, chicken pox, gastroenteritis, HIV. **ORTHOPEDIC AND NEUROLOGICAL DISORDERS:** • Cerebral palsy (Define and briefly outline etiology of prenatal, perinatal and postnatal causes: briefly mention pathogenesis, types of cerebral palsy (classification) findings on examination; general examination of C.N.S. Musculoskeletal and respiratory system and management). • Meningitis, encephalitis • Hydrocephalus • Ataxia • Arnold-chiari malformation, dandy walker syndrome • Basilar impression & cerebral malformations • Floppy infant • GBS • Poliomyelitis • Epilepsy • Neural tube defects • Neuropathy, infantile hemilegia. • Still’s disease • Limping child • DDH, perthes disease, CTEV, torticollis.	30	10	40

	RESPIRATORY DISEASES : Bronchiectasis, Bronchopneumonia, Bronchial asthma, Lung Abscess Tuberculosis, SIDS, cystic fibrosis, Acute pediatric respiratory distress syndrome. **CARDIOVASCULAR DISODERS :** Rheumatic heart disease ,subacute bacterial endocarditis, congenital heart defects -VSD ,ASD, PDA ,TOF ,Transposition of Great Vessels RHD. **GENETIC DISORDERS :** • Types of genetic disorder • Downs syndrome • Muscular dystrophy -Types Mode of inheritance Clinical manifestation Progression and prognosis of disease, Evaluation (Genetic screening test), Treatment. • Common Genetic disorders. **HAEMATOLOGICAL DISORDER :** Iron deficiency anemia, Aplastic anemia, sickle cell disease, hemophilia, thalassemia, leukemia. **MISCELLANEOUS :** • Feeding and communication difficulty, • Sensory disorders – problems due to loss of vision and hearing • Learning and behavioural problems - Attention Deficit Hyperactivity Disorder & Autism, Mental retardation. **CLINICAL ASSESSMENTS:** • NICU-Overview. • Neonatal assessment including neonatal reflex. • Pre-term baby assessment. • Low birth weight baby assessment. • Examination of the nervous system • Examination of respiratory system • Examination of cardiovascular system • Community programmes: International (WHO),			

	national and local for prevention measures for disease. Outline the immunization schedule for children			
UNIT	**<u>PSYCHIATRY</u>**			
V	• Psychiatric History, classification and mental status examination. • Organic mental disorders (delirium, dementia, organic amnestic syndrome and other organic mental disorders). • Mood disorders (manic, depressive episodes, bipolar mood disorders). • Neurotic stress related and somatoform disorders (Anxiety disorder, phobic anxiety disorders, obsessive compulsive disorders, adjustment disorders, dissociative disorders, somatoform disorders post-traumatic stress Disorder.	5		5

RECOMMENDED REFERENCE BOOK:

1. Davidson's "Principles and practice of Medicine", 23rd edition, Elsevier, 2018.
2. O.P. Ghai, "Essential Paediatrics", 9th edition, CBS Publishers, 2019.
3. Graham Davey, Nick Lake "Clinical Psychology", 2nd edition, Taylor & Francis group, 2015.
4. M.S. Bhakia "Short textbooks of Psychiatry", 7th edition, CBS Publishers, 2019.

COMMUNITY MEDICINE AND REHABILITATION

SUBJECT CODE: U21BPTT527 **TOTAL HOURS:** 90Hours

COURSE DESCRIPTION:

This course is designed to assist the students to acquire knowledge of the disease and help the students to understand the limitation imposed by the diseases.

COURSE OBJECTIVES:

At the end of the course the candidate shall be able to understand the contents given in the syllabus.

Units	Topics	Didactic Hours	Clinical Hours	Total Hours
I	1. INTRODUCTION i. Natural history of diseases ii. Influence of social, economic, cultural aspect of health and diseases iii. Measures of prevention for disease with disability a. Methods of intervention for disease with disability. 2. HEALTH CARE DELIVERY SYSTEM AND PUBLIC HEALTH ADMINISTRATIVE SYSTEM i. National level ii. State level 3. NATIONAL HEALTH PROGRAMME i. Role of social, economic, cultural factors in theimplementation of National programmes ii. Primary health care a. Objectives and implementation.	15		15

II	**1.** OCCUPATIONAL HEALTH i. Definition ii. Scope iii. Occupational diseases iv. Prevention of occupational diseases and hazards v. Role of E.S.I. a. Employee state b. insurance scheme and its benefit. 2.SOCIAL SECURITY MEASURES	15		15

	i. Protection of occupational hazards, accidents and diseases ii. Workmen compensation act iii. Environmental safety.			
III	1. FAMILY WELFARE PROGRAMME i. Objectives of family welfare programme ii. Family planning methods iii. General idea of Advantages and disadvantages iv. Concepts of planned pregnancy a. Population dynamics. 2. COMMUNICABLE DISEASES (With reference to reservoir, mode of transmission, route ofentry and levels of prevention) i. Poliomyelitis ii. Meningitis iii. Encephalitis iv. Tuberculosis v. Filariasis vi. Leprosy vii. Tetanus viii. Measles ix. Malaria Universal immunization programme-ARI, Diarrhea &polio control programme. 3. EPIDEMOLOGY OF i. Rheumatic heart disease ii. Chronic degenerative disease iii. Cerebro vascular accident iv. Blindness v. Accident vi. Cancer	25		25
IV	1. MENTAL HEALTH i. Community aspect of mental health. a. Role of physiotherapist in mental health problem incerebral palsy, mental retardation. 2. HEALTH EDUCATION i. Philosophy ii. Main principles and objectives iii. Methods and tools of communication iv. Health education versus health legislation v. Education versus propaganda vi. Role of community leader in health education vii. Role of health professionals in health education	20		20

	Element of planning a health education programme with special emphasis on community participation 3. COMMUNITY BASED REHABLITATION VERSUS INSTITUTIONAL BASED REHABILITATION			
V	1. REVIEW OF i. Beliefs, values, norms, habits, taboos a. Their importance in learning and change process 2. REVIEW OF i. Concept of perception ii. Attitudes iii. Socialization process iv. Learning and theories of learning v. Social change and change process a. Motivation needs 3. ITAL HEALTH STATISTICS i. Basic concept ii. Mortality and morbidity rate iii. Period, age, causes of specific death rate a. Role of this rate as b. indicator of health and c. disease.	15		15

RECOMMENDED TEXT BOOK:

1. K. Park “Textbook of Preventive and Social Medicine”, 22nd edition, Generic Publisher, 2013.

DIAGNOSTIC IMAGING FOR PHYSIOTHERAPY

COURSE CODE: U21BPTT528 **TOTAL HOURS:** 45hrs

COURSE DESCRIPTION:

This course covers the study of common diagnostic and therapeutic imaging tests. At the end of the course students will be aware of the indications and implications of commonly used diagnostic imaging tests as they pertain to patient's management. The course will cover that how X-Ray, CT, MRI, Ultrasound and Other Medical Images are created and how they help the health professionals to save lives.

COURSE OBJECTIVE:

The objectives is to train Physiotherapy students to become a skilled and competent to interpret various diagnostic / interventional imaging studies both conventional and advanced imaging

Units	Topics	Didactic Hours	Clinical Hours	Total Hours

I	**1. IMAGE INTERPRETATION** i. Introduction to Radiology ii. Evolution of X Rays iii. Role of Medical Imaging iv. Radiography(x-rays) v. Fluoroscopy vi. Computed Tomography (CT) vii. Magnetic Resonance Imaging (MRI) viii. Ultrasound	2	1	3
	1. RADIOGRAPHY i. Equipment components ii. Procedures for Radiography iii. Benefits versus Risks and Costs iv. Indications and Contraindications.	3	2	5
	2. MAMMOGRAPHY i. Equipment components ii. Procedures for Mammography iii. Benefits versus Risks and Costs iv. Indications and contraindications.	3	2	5

II	**3. FLUOROSCOPY** i. What is Fluoroscopy? ii. Equipment used for fluoroscopy iii. Indications and Contra indications iv. The Findings in Fluoroscopy v. Benefits versus Risks and Costs.	**3**	**2**	**5**
	4. COMPUTED TOMOGRAPHY (CT) i. What is Computed Tomography? ii. Equipment used for Computed Tomography iii. Indications and Contra indications iv. The findings in Computed Tomography v. Benefits versus Risks and Costs.	5	2	7
III	**MAGNETIC RESONANCE IMAGING (MRI)** i. What is MRI? ii. Equipment used forMRI iii. Indications and Contra indications iv. The Findings in MRI v. Benefits versus Risks and Costs vi. Functional MRI.	5	2	7
IV	**ULTRASOUND** i. What is Ultrasound? ii. Equipment used for ultrasound iii. Indications and Contra indications iv. The findings in ultrasound v. Benefits versus Risks and Costs. vi. Echocardiography? Indications and contraindication	5	2	7
V	**NUCLEAR MEDICINE** i. What is Nuclear medicine? ii. Equipment used for Nuclear medicine iii. Indications and Contra indications **iv.** Benefits versus Risks and Costs	3	3	6

RECOMMENDED REFERENCE BOOKS

1. Brant and Helms, Fundamentals of diagnostic radiology, 5th edition, Wolters Kluwer, 2019.
2. Chapman & Nakielny's "Aids to Radiological Differential Diagnosis", 7th edition, 2019.

CLINICAL TRAINING

Course code: U21BPTP529 **Total:** 90hrs

Course Description

The student will learn the Subjective data taking procedure.

Course Objective

At the end of the course the students will able to

1. Understand the history taking procedure
2. Various procedures in patient's assessments
3. Learn the various treatment procedures

Sl. No	TOPIC	HOURS
1.	Demographic data Collection Socioeconomic	
2.	history collection	
3.	Social behavior and its influence on health	
4.	Schedule the patients treatment according to their condition	

SCHEME OF EXAMINATION

IA	FINAL EXAM	TOTAL
50	END SEMESTER EXAMINATION 50	100

Sl. No	Components	Marks
1	2 case presentation	25 ×2 = 50

VI Semester

PHYSIOTHERAPY IN ORTHOPEDIC CONDITIONS

Course Code: U21BPTT631 **Total:** 90 Hours

Course Description:

The subject serves to integrate the knowledge gained by the students in orthopedics with skills to apply these in clinical situations of dysfunction and musculoskeletal pathology.

Physiotherapy in Orthopedic condition focuses on maximizing functional independence and well-being. The course uses a patient-centered model of care with multi-system assessment, evidence-based interventions and a significant patient education component to promote a healthy, active lifestyle and community-based living.

Course Objectives:

The objective of the course is that after the specified hours of lectures and demonstrations the student will be able to identify disabilities due to musculoskeletal dysfunction, plan and set treatment goals and apply the skills gained in exercise therapy and electrotherapy in these clinical situations to restore musculoskeletal function.

SYLLABUS

Unit	Topic	Didactic Hours	Clinical Hours	Total Hours
I	**1. Fractures & Dislocations** i. Fractures and dislocation of the Spine & Extremities – classification, complications, PT management of following: a) Upper limb fractures & dislocations. b) Lower limb fractures and dislocations including pelvis. c) Spinal fractures ii. PT management in complications - early & late - shock, compartment syndrome, VIC, fat embolism, delayed and mal-union, RSD, myositis ossificans, AVN, pressure sores etc. iii. Dislocation of Shoulder, Elbow, Hip & Knee	15	15	30

II	**1. Soft Tissue Injuries** (i) Synovitis (ii) Capsulitis (iii) Tendonitis (iv) Rupture of tendons (v) Cartilage & meniscal injuries (vi) Fasciitis (vii) Burns **2. Deformities** (i) Congenital: CTEV, CDH, Torticollis, pes planus, pes cavus and other common deformities. (ii) Acquired: scoliosis, kyphosis, coxa vara, genu varum, valgum and recurvatum.	15	15	30
III	**1. Infectious diseases of the bone & joints** (i) Osteomyelitis – acute and chronic (ii) Septic arthritis and Pyogenic arthritis (iii) TB spine and major joints - knee and hip **2. Degenerative and Inflammatory Conditions** (i) Osteoarthritis - emphasis on knee, hip and hand (ii) Rheumatoid Arthritis, Ankylosing spondylitis (iii) Gout, Perthes disease	15	15	30
IV	**1. Peripheral Nerve Injury** Physiotherapy management for peripheral nerve injuries. **2. Amputation** Definition, levels, indications, types, PT management for various levels of amputation.	10	10	20
	1. Regional Conditions **(i) Spine Complex** (a) Cervical & Lumbar spondylosis (b) Intervertebral disc prolapses (c) Spinal canal stenosis (d) Spondylolysis, Spondylolisthesis (e) Coccydynia (f) Sacro-iliac joint dysfunction (g) Sacralisation, Lumbarisation			

V	**(ii) Shoulder Complex** Shoulder instabilities, Thoracic outlet syndrome (TOS), Reflex sympathetic dystrophy (RSD), and Impingement syndrome - conservative and post- operative PT management. Total shoulder replacement and Post operative PT management. Acromio clavicular joint injuries - rehabilitation. Rotator cuff tears-conservative and surgical repair. Subacromial decompression - Post operative PT Management, Subacromial bursitis. **(iii) Elbow Complex** Tennis and Golfer's elbow. Excision of radial head- Post operative PT management. Total elbow arthroplasty- Post operative PT management. **(iv) Wrist and Hand Complex** (a) Wrist sprains (b) Dequervain's Tenosynovitis, Trigger and Mallet finger (c) Repair of ruptured flexor and extensor tendons (d) Carpal tunnel syndrome. **(v) Hip Complex** Joint surgeries - hemi and total hip replacement - Post operative PT management Management. **(vi) Knee Complex** (a) ACL, PCL and MCL reconstruction surgeries - Post operative rehabilitation (b) Meniscectomy and meniscal repair - Post operative management. (c) Pre patellar. (d) Anterior Knee pain: PFPS, Plica syndrome, patellar dysfunction and Hoffa's syndrome etc. - conservative management (e) TKR- rehabilitation protocol (f) Patellar tendon ruptures and Patellectomy- rehabilitation **(vii) Ankle and foot Complex** a. Ligamentous tears- Post operative management b. TA rupture c. Plantar fasciitis, Metatarsalgia, hammer toe	35	35	70

Recommended Text Books:

1. Jayant Joshi, Prakash Kotwal, “Essentials of Orthopaedic & Applied Physiotherapy”, 3rd edition, Elsevier Publications, 2016.
2. David J. Magee, Robert C. Manske, “Orthopaedic Physical Assessment”, 7th edition, Elsevier Publications, 2021.
3. Carrie M. Hall, Lori Thein Brody “Therapeutic Exercise: Moving towards Function”, 4th edition, Wolters Kluwer, 2017.

PHYSIOTHERAPY IN GENERAL MEDICINE AND GENERAL SURGERY

Course Code: U21BPTT632 **Total: 60+**90 Hours

Course Description:

Physiotherapy in General Medicine & General surgery emphasizes the management of techniques based on the best available evidence. Physiotherapy strategies for assessment and treatment address structural & functional impairments and activity limitation of individuals. The therapeutic approach is patient and family focused.

Course Objective:

At the end of the course the candidate will be able to:

1. Describe physiotherapy management of the various metabolic & nutrition deficiency condition.
2. Describe physiotherapy management of the various cardiovascular and respiratory condition.
3. Describe the principles of management at the MICU & SICU.

Units	Topics	Didactic Hours	Clinical Hours	Total Hours
	GENERAL SURGERY			

I	1. Physiotherapy in mother and child care – ante and post- natal management, early intervention and stimulation therapy in child care (movement therapy) 2. Geriatrics – handling of old patients and their problems. Complication common to all operations 3. Abdominal incisions. 4. Physiotherapy in pre and post-operative stages. Operations on upper G.I.T.- oesophagus, stomach, duodenum 5. Operations on large and small intestine – Appendisectomy, cholecystectomy, partial colectomy, ileostomy, hernia and herniotomy, hernioraphy, hernioplasty. 6. Physiotherapy in dentistry	12	20	
II	Burns and its treatment – physiotherapy burns, skin grafts, and reconstructive surgeries.	10	15	

III	1. Management of wound ulcers- Care of ulcers and wounds - Care of surgical scars-U.V.R. and other electro therapeutics for healing of wounds, prevention of hyper- granulated scars keloids, 2. Electrotherapeutics measures for relief of pain during mobilization of scars tissues.	12	15	
IV	1. Physiotherapy in dermatology - Documentation of assessment, treatment and follow up skin conditions. U.V.R therapy in various skin conditions; Vitiligo; Hair loss; Pigmentation; Infected wounds ulcers. Faradic foot bath for Hyperhydrosis. 2. Massage maneuvers for cosmetic purpose of skin; use of specific oil as medium; Care of anesthetic hand and foot; Evaluation, planning and management of leprosy- prescription, fitting and training with prosthetic and orthotic devices.	12	25	
V	1. Physiotherapy intervention in the management of Medical, surgical and radiation oncology cases 2. ENT – sinusitis, non-suppurative and chronic suppurative otitis media, osteosclerosis, labrynthitis, mastoidectomy, chronic rhinitis, laryngectomy, pharyngeo – laryngectomy, facial palsy.	14	15	

RECOMMENDED TEXT BOOKS:

1. O' Sullivan "Physical Rehabilitation", 7th edition, F.A. Davis, 2019.
2. William E. Prentice, "Therapeutic Modalities", 6th edition, MC Graw hill, 2021.
3. Joan E. Cash, "Textbook of Physiotherapists", 4th edition, Mosby, 1987.

CLINICAL NEUROLOGY AND NEUROSURGERY FOR PHYSIOTHERAPY

Course cord: U21BPTT630 **Total:** 120 hours

COURSE DESCRIPTION:

This subject follows the fundamental science subjects to provide the knowledge about relevant aspects of neurology & neurosurgery. The student will have a basic knowledge of the diseases the therapist would encounter in their practice. The objective of this course is that after 90 Hours of lectures and discussion the student will be able to list the etiology, pathology, clinical features and ways for treating various neurological conditions.

COURSE OBJECTIVES:

At the end of the course, the candidate will:

1. Be able to describe Aetiology, Pathophysiology, signs & Symptoms & Management of the various Neurological Conditions (or) disease.
2. Acquire skill of history taking and clinical examination of Neurological patients as a part of clinical teaching.
3. Be able to describe neuromuscular, musculoskeletal, cardio-vascular & immunological conditions, nutritional deficiencies, infectious diseases, & neuro transmitted conditions.
4. Acquire skill of clinical examination of a neonate / child with respect to the neurological condition

Unit	Topic	Didactic Hours	Clinical Hours	Total Hours

I	**1. GENERAL NEUROLOGICAL ASSESSMENT** i. Nervous system- history and examination ii. Conscious level assessment iii. Higher nutral function iv. Cranial nerve examination v. Motor, sensory, cerebellar & gait examination vi. Examination of the unconscious patients vii. The neurological observation charts **2. NEUROLOGICAL INVESTIGATIONS** i. investigation of the central and peripheral nervous systems ii. skull x-ray iii. computerized tomography scanning {CT} iv. magnetic resonance imaging {MRI}			

	v. ultrasound vi. angiography vii. radionuclide imaging viii. electroencephalography {EEG} ix. intracranial pressure monitoring x. evoked potentials xi. lumbar puncture xii. cerebrospinal fluid {CSF} xiii. electromyography/ nerve conduction studies xiv. neuro - otological test **3. CEREBRO –VASCULAR ACCIDENTS:** Define stroke, TIA, RIA, stroke in evolution, multi infarct dementia and Lacunar infarct. Classification of stroke – Ischemic, hemorrhagic, venous infarcts. Risk factors, cause of ischemic stroke, causes of hemorrhagic stroke. Classification of hemorrhagic stroke, classification of stroke based on symptoms, stroke syndrome, investigations, differential diagnosis, medical and surgical management. **4. HEAD INJURY:** Etiology, classification, clinical signs & symptoms, investigations, differential diagnosis, medical management, surgical management and complications. **5. BRAIN TUMORS AND SPINAL TUMORS:** Classification, clinical features, investigations, medical and surgical management. **6. INFECTIONS OF BRAIN AND SPINALCORD:** Etiology, pathophysiology, classification, clinical signs & symptoms, investigations, differential diagnosis, medical management, surgical management and complications of following disorders: i. Meningitis, Encephalitis ii. Neurosyphilis, HIV infection iii. Herpes, Leprosy, Tetanus iv. Poliomyelitis and Post-polio syndrome	30		30

	7. COMPLICATIONS OF SYSTEMIC INFECTIONS ON NERVOUS SYSTEM Septic encephalopathy, AIDS, Rheumatic fever, Brucellosis, tetanus, and pertussis. **8. MOVEMENT DISORDERS:** Definition, etiology, risk factors, pathophysiology, classification, clinical signs & symptoms, investigations, differential diagnosis, medical management, surgical management and complications of following disorders: i. Parkinson's disease ii. Dystonia, Chorea iii. Ballism's, Athetosis iv. Tics, Myoclonus v. Wilson's disease **9. CEREBELLAR AND COORDINATION DISORDERS:** Etiology, pathophysilogy, classification, clinical signs & symptoms, investigations, differential diagnosis, management. i. Congenital ataxia ii. Friedreich's ataxia iii. Ataxia talengiectasia iv. Metabolic ataxia v. Hereditary cerebellar ataxia vi. Tabes dorsalis vii. Syphilis			
II	**1. POLYNEUROPATHY:** Classification of Polyneuropathies, Hereditary motor sensory neuropathy, hereditary sensory and Autonomic neuropathies, Amyloid neuropathy, acute idiopathic Polyneuropathies. Guillain-Barre syndrome – Causes, clinical features, management of GBS, Chronic Idiopathic Polyneuropathies, diagnosis of polyneuropathy, nerve biopsy. **2. DISORDERS & DISEASES OF MUSCLE:** Classification, etiology, signs & symptoms, investigations, imaging methods, Muscle biopsy, genetic			

	counselling &management of muscle diseases emphasis i. Muscular dystrophy ii. Myotonic dystrophy iii. Myopathy iv. Non-dystrophic myotonia **3. MOTOR NEURON DISEASES:** Etiology, pathophysiology, classification, clinical signs & symptoms, investigations, differential diagnosis, medical management, and complications of following disorders: i. Amyotrophic lateral sclerosis ii. Spinal muscular atrophy iii. Hereditary bulbar palsy iv. Neuromyotonia v. Post-irradiation lumbosacral polyradiculopathy **4. MULTIPLE SCLEROSIS:** Etiology, pathophysiology, classification, clinical signs & symptoms, investigations, differential diagnosis, medical management, and complications **5. DISORDERS OF MYONEURAL JUNCTION:** Etiology, classification, signs & symptoms, investigations, management of following Disorders: i. Myasthenia gravis ii. Eaton-Lambert syndrome iii. Botulism **6. SPINAL CORD DISORDERS:** Anatomy and Functions of tracts, definition of spinal cord disorders, etiology, risk factors, pathophysiology, classification, clinical signs & symptoms, investigations, differential diagnosis, medical management, surgical management and complications of following disorders: i. Spinal cord injury ii. Inter vertebral disc prolapse iii. Spinal epidural abscess iv. Transverse myelitis v. Viral myelitis, syringomyelia vi. Spina bifida vii. Sub-acute combined degeneration of the cord viii. Hereditary spastic paraplegia ix. Radiation myelopathy	35		35

	x. Progressive encephalomyelitis xi. Conus medullaris syndrome xii. Bladder & bowel dysfunction xiii. Sarciodosis.			
III	**1. DISORDERS OF LOWER CRANIAL NERVES & SPECIAL SENSES** Etiology, clinical features, investigations, and management of following disorders i. lesions in trigeminal nerve ii. trigeminal neuralgia iii. trigeminal sensory neuropathy iv. lesions in facial nerve v. facial palsy vi. bell's palsy vii. hemi facial spasm viii. Glossopharyngial neuralgia ix. lesions of Vagus nerve x. lesions of spinal accessory nerve xi. lesions of hypoglossal nerve xii. Dysphagia – swallowing mechanisms, causes of dysphagia, symptoms, examination, and management of dysphagia. **2. HIGHER CORTICAL, NEURO PSYCHOLOGICAL AND NEUROBEHAVIOURAL DISORDERS** i. Epilepsy, physiology, classification, clinical features, investigations, medical& surgical management of following disorders – non-epileptic attacks of childhood, Epilepsy in childhood, seizures and epilepsy syndromes in adult. ii. Classification & clinical features of dementia, Alzheimer's disease. iii. Causes & investigations of coma, criteria for diagnosis of brain death. **3. PSYCHIATRY IN NEUROLOGY:** i. Psychiatric History, classification and mental status examination ii. Organic mental disorders (delirium, dementia, organic amnestic syndrome and other organic mental disorders) iii. Mood disorders (manic, depressive episodes, bipolar mood disorders) iv. Schizophrenia, delusional disorders and	35		35

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	schizoaffective disorders v. Child psychiatry (mental retardation, developmental disorders, attention deficit, hyperkinetic disorder, enuresis, conduct disorders) vi. Disorders of adult personality and behavioor (specific personality disorders, habit and impulse disorders, gender identity disorders) vii. Stress, psychosomatic disorders, suicide, **4. PERIPHERAL NEUROPATHY:** Clinical diagnosis of focal neuropathy, neurotmesis, Axonotmesis, Neuropraxia. Etiology, risk factors, classification, neurological signs & symptoms, investigations, management, of following disorders – RSD, Nerve tumors, Brachial plexus palsy, Thoracic outlet syndrome, Lumbosacral plexus lesions, Phrenic & Intercostal nerve lesions, Median nerve palsy, Ulnar nerve palsy, Radial nerve palsy, Musculocutaneous nerve palsy, Anterior & Posterior interosseous nerve palsy, Axillary nerve palsy, Long thoracic nerve palsy, Suprascapular nerve palsy, Sciatic nerve palsy, Tibial nerve palsy, Common peroneal nerve palsy, Femoral nerve palsy, Obturator nerve palsy, Pudental nerve palsy.			
IV	**1. TOXIC, METABOLIC AND ENVIRONMENTAL DISORDERS:** Etiology, risk factors, classification, neurological signs & symptoms, investigations, management, of following disorders – Encephalopathy, Alcohol toxicity, Recreational drug abuse, Toxic gases & Asphyxia, Therapeutic & diagnostic agent toxicity, Metal toxicity, Pesticide poisoning, Environmental & physical insults, Plant & Fungal poisoning, Animal poisons, & Complications of organ transplantation. **2. INTRODUCTION, INDICATIONS AND COMPLICATIONS OF FOLLOWING NEURO SURGERIES:** Craniotomy, Cranioplasty, Stereotactic surgery, Deep brain stimulation, Burr-hole, Shunting, Laminectomy, Hemilaminectomy, Rhizotomy, Microvascular decompression surgery, Endarterectomy, Embolization, Pituitary surgery, Ablative surgery -	15		15

	Thalamotomy and Pallidotomy, coiling of aneurysm, Clipping of aneurysm, and Neural implantation. **3. DEAFNESS, VERTIGO, AND IMBALANCE:** Physiology of hearing, disorders of hearing, examination & investigations of hearing, tests of vestibular function, vertigo, peripheral vestibular disorders, central vestibular vertigo. **4. NEURO-OPHTHALMOLOGY:** Assessment of visual function – acuity, field, colour vision, Pupillary reflex, accommodation reflex, abnormalities of optic disc, disorders of optic nerve, tract, radiation, occipital pole, disorders of higher visual processing, disorders of pupil, disorders of eye movements, central disorders of eye movement			
V	**1. PEDIATRICS IN NEUROLOGY:** i. Normal development & growth ii. Breast feeding and immunization iii. Perinatal, Postnatal problems and management (Birth injuries) Neck, shoulder dystocia, Brachial plexus injury, Fractures iv. Congenital abnormalities and its management Problems and management of LBW infants **2. DEVELOPMENTAL DELAY:** Etiology, pathophysiology, classification, clinical signs & symptoms, investigations, differential diagnosis, medical management, surgical management and complications **3. RESPIRATORY CONDITIONS OF CHILDHOOD:** Pneumonia – Bacterial & Tubercular, Empyema, Asthma **4. ORTHOPEDIC AND NEUROLOGICAL DISORDERS IN CHILDHOOD, CLINICAL FEATURES AND MANAGEMENT:** **i.** Cerebral palsy ii. Meningitis, Encephalitis	5		5

	iii. Hydrocephalus iv. Ataxia v. Arnold-Chiari malformation, Dandy walker syndrome vi. Basilar impression & Cerebral malformations vii. Down's syndrome viii. Floppy infant ix. GBS x. Poliomyelitis xi. Epilepsy xii. Neural tube defects xiii. Muscular dystrophies xiv. Neuropathy **5. NUTRITIONAL DISORDERS OF CHILDHOOD** Rickets and scurvy, PEM (Kwashiorkor and Marasmus) INFECTIONS – Congenital & Neonatal, Mental retardation Coma in Paediatrics and Acute rheumatic fever				

Reference Books:

1. Davidson's "Principles and practice of Medicine", 23rd edition, Elsevier, 2018.
2. Sandhya A.Kamath, "API Textbook of Medicine", 12th edition, Jaypee Publishers, 2022.
3. P.J. Mehta's, "Practical Medicine", 21st edition, National Publications, 2018.
4. O.P. Ghai, "Essential Paediatrics", 9th edition, CBS Publishers, 2019.

PRINCIPLES OF MANAGEMENT

Course Code: U21BPTT635 **Total:** 30 Hours

Course Description:

This course is designed to students to acquire understanding the principles & methods of management in physiotherapy services & educational Programmes.

Course Objective:

At the end of the course the student will

1. Learn the management basics in fields of clinical practice, teaching, research and Physiotherapy practice in the communicating.
2. Acquire communications skills in relation with patients, seniors and other professionals.

Unit	Topics	Didactic Hours	Practical Hours	Total Hours
I	**Introduction to Management** Definition, Science or Art, Manager Vs. Entrepreneur, types of managers, managerial roles and skills, Evolution of management, Principles, Functions of management, Current trends and issues in management.	6	-	6
II	**Management Process** Planning, types of planning, motivation theories, leadership styles, Formal and informal organization, Line and staff authority, communication, process of communication, barrier in communication, effective communication, communication and IT.	6	-	6
III	**Management of Physiotherapy Services** (i) Planning: Patient care units, emergency management (ii) Human resources management: Recruiting, selecting, deploying, retaining, promoting. Patient classification system, Staff development & welfare (iii) Budgeting: Proposal, staff, equipment & supplies requirements	6	-	6

	(iv) Material management : Procurement, inventory control, Auditing & maintenance. (v) Directing & Leading: Delegation, participatory management, staff development, discipline maintenance (vi) Controlling / Evaluation: Physiotherapy rounds & visits, Quality assurance model, document of records & reports, performance appraisal			
IV	**Organizational Behaviour & Human Relations** (i) Human Resource Management, HR Planning, Recruitment, selection, Training and Development & Group dynamics. (ii) Public relations: Professional, Clinical & Social, Trade Unions, Collective bargaining.	6	-	6
V	**Management of Educational Institutions** (i) Physiotherapy institution (ii) College & Hostel: structure, committees & Management of equipments, Clinical & transport facilities, institutional records & reports	6	-	6

Recommended Text Books:

1. C.B. Gupta, "Business Management", Sultan Chand Sons, 9th Edition, 2018.
2. L. M. Prasad, "Principles and Practice of Management", Sultan Chand & Sons, 10th Edition 2019.
3. Koontz O'Donnell, "Essentials of Management", Tata McGraw Hill, 9th Edition, 2012.
4. P C Tripathi, “Principles of Management”, McGraw Hill, 6th Edition, 2017.
5. Catherine G. Page, “Management in physical Therapy Practices”, F. A. Davis, 2nd Edition, 2015

HEALTH PROMOTION AND FITNESS

SUBJECT CODE: U21BPTT636 **TOTAL HOURS: 30**

COURSE DESCRIPTION:

This course includes discussion on the theories of health and wellness, including motivational theory, locus of control, public health initiative, and psycho-Social, spiritual and cultural consideration. Health risks, screening, and assessment considering epidemiological principles are emphasized. Risk reduction strategies for primary and secondary prevention, including programs for special populations are covered.

COURSE OBJECTIVES:

a. Students will able to know the principles and components of primary health care and the national health policies to achieve the goal of 'Health for All'.

b. Learning Physiological changes occurring in various conditions in Women's Health relevant to Physiotherapy, and plan physiotherapy management for fitness

c. Advice with clinical reasoning at urban, rural and community level for achieve physical fitness.

d. Exploring physical fitness with students using this lesson plan can create personal fitness plan.

e. Students willable to explain the benefits of physical activity and nutrition for health, wellness, quantity & quality of life.

Units	Topics	Didactic Hours	Clinical Hours	Total Hours
I	**1. PREVENTION PRACTICE:** A holistic perspective for physiotherapy i. Define Health ii. Predictions of Health Care iii. Comparing Holistic Medicine and Conventional Medicine iv. Distinguishing Three Types of Prevention Practice. **2. HEALTHY PEOPLE:** i. Definition of healthy people ii. Health education Resources iii. Physiotherapist role for a healthy community.	4	4	8
II	**1. KEY CONCEPTS OF FITNESS :** i. Defining & Measuring Fitness ii. Assessment of Stress with a Survey iii. Visualizing Fitness iv. Screening for Mental and Physical Fitness v. Body Mass Index calculations. vi. Exercise testing. vii. Agility test. **2. FITNESS TRAINING :** i. Physical Activities Readiness ii. Physical Activities Pyramid iii. Exercise Programs iv. Evidence-Based Practice.	4	4	8
III	**1. CHILD AND WOMENS HEALTH:** i. Health, fitness, and wellness issues during childhood and adolescence, Health, fitness, and wellness during adulthood. ii. Women's health issues: focus on pregnancy	3	3	6

IV	**PREVENTION:** i. Resources to optimize health and wellness, Health protection, Prevention practice for older adults. ii. Prevention practice for musculoskeletal conditions, cardiopulmonary conditions	2	2	4
V	**PREVENTION:** i. Prevention practice for neuromuscular conditions, integumentary disorders. ii. Prevention practice for individuals with developmental disabilities, Marketing health and wellness	2	2	4

RECOMMENDED TEXT BOOK:

1. Roberta L. Duyff, "Academy of nutrition and dietetics complete food and nutrition guide", 5th edition, Harvest Publisher, 2017.
2. Nicholas Bjorn "Fitness Nutrition", 5th edition, Lulu Publisher, 2019.
3. Randall R. Cottrell, "Principles of Health Education and Promotion", Jones & Bartlett Publishers, 8th Edition, 2021
4. James F. Mckenzie, Brad L. Neiger, "Health Promotion Programs", Jones & Bartlett Publishers, 8th Edition, 2022

CLINICAL TRAINING

Course code: U21BPTP637 **Total:** 90 hours

Course Description

The Student will learn the subjective and objective data taking procedure.

Course Objective

End of the course the students will able to

1. Understand the history taking procedure
2. Various procedures in patient's assessments
3. Learn the various treatment procedures

Sl. No	TOPIC	HOURS
1.	Demographic data Collection	
2.	Socioeconomic history collection	
3.	Social behavior and its influence on health	
4.	Schedule the patients treatment according to their condition	

SCHEME OF EXAMINATION

IA	FINAL EXAM	TOTAL
50	50	100

END SEMESTER EXAMINATION

Sl. No	Components	Marks
1	2 case presentation	25 ×2 = 50

Note: Each candidate has to document minimum 10 cases in the log book.

VII-Semester

PHYSIOTHERAPY IN NEUROLOGICAL CONDITIONS

COURSE CODE: U21BPTT738 TOTAL HOURS: 180 hours

COURSE DESCRIPTION:

The subject serves to integrate the knowledge gained by the students in neurology with skills to apply these in clinical situations of dysfunction and nervous system pathology.

Physiotherapy in neurological condition focuses on maximizing functional independence and well- being. The course uses a patient-centered model of care with multi-system assessment, evidence based interventions and a significant patient education component to promote a healthy, active lifestyle andcommunity-based living.

COURSE OBJECTIVES:

The objective of the course is that after the specified hours of lectures and demonstrations the studentwill be able to identify disabilities due to neurological dysfunction, plan and set treatment goals and applythe skills gained in exercise therapy and electrotherapy in these clinical situations to restore neurological functions.

Unit	Topic	Didactic Hours	Clinical Hours	Total Hours
I	NEURO ANATOMY AND NEURO PHYSIOLOGY 1. Cerebral hemispheres 2. Cerebellum 3. Spinal cord 4. Peripheral nerves 5. Pyramidal system 6. Extra pyramidal system 7. Cranial nerves	10	10	20
II	DEFINITION, CLINICAL FEATURES AND PT MANAGEMENT FOR THE FOLLOWING CONDITIONS. CONGENINTAL AND CHILDNOOD DISORDERS 1. Hydrocephalus 2. Cerebral palsy 3. Spina bifida 4. Craniovertebral junction anomalies 5. Arnold chiari malformation 6. Dandy walker syndrome.	10	10	20

III	1. PHYSIOTHERAPY MANAGEMENT FOR CEREBROVASCULAR ACCIDENTS i. General classification, thrombotic, embolic, haemorrhagic, inflammatory strokes. ii. Gross localization and sequelae. iii. Detailed rehabilitative programme 2. PHYSIOTHERAPY MANAGEMENT FOR TRAUMA i. Head injury ii. Spinal cord injury	23	23	46
Unit	**Topic**	**Didactic Hours**	**Clinical Hours**	**Total Hours**

Unit	Topic	Didactic Hours	Clinical Hours	Total Hours
IV	1. PHYSIOTHERAPY MANAGEMENT FOR DISEASE OF THE SPINAL CORD i. Syringomyelia ii. Spinal arachnoiditis iii. T.B. Spine iv. Tumors in spinal cord 2. PHYSIOTHERAPY MANAGEMENT FOR DEMYELINATING DISEASES (CENTRAL AND PERIPHERAL) i. Guillain - barre syndrome ii. Acute disseminated encephalomyelitis. iii. Transverse myelitis iv. Multiple sclerosis 3. PHYSIOTHERAPY MANAGEMENT FOR DISEASES OF THE MUSCLE i. Myopathies ii. Muscular dystrophy iii. Spinal muscular atrophy 4. PHYSIOTHERAPY MANAGEMENT FOR DEGENERATIVE DISORDERS i. Parkinson' disease ii. Dementia iii. Alzheimer’s disease	25	25	50
V	1. PHYSIOTHERAPY MANAGEMENT FOR PERIPHERAL NERVE DISORDERS i. Peripheral nerve injuries : localization and management ii. Entrapment neuropathies iii. Peripheral neuropathies	10	10	20
Unit	**Topic**	**Didactic Hours**	**Clinical Hours**	**Total Hours**

	2. MISCELLANEOUS i. Epilepsy: Definition, classification and PT management. ii. Myasthenia gravis: Definition, course PT and management. iii. Intracranial tumors: Broad classification, signs, symptoms and PT management. iv. PT management for motor neuron disease. v. Deep vein thrombosis (DVT): Definition, symptoms & PT management.	12	12	24

RECOMMENDED TEXT BOOKS

1) Partricia A, Cash's "Text Book of Neurology for Physiotherapist," Jaypee Brothers Publishers, 4th edition, 1993.
2) Darcy Umphred, "Neurological Rehabilitation," Elsevier Publishers, 6th edition, 2012.
3) Susan B O'Sullivan, "Physical Rehabilitation," F A Davis Company Publishers, 6th edition, 2013.
4) Patricia M.Davies, "Right in the Middle," Springer -Verlag Publishers, 2nd edition, 2006.
5) Berta Bobath, "Adult Hemiplegia Evaluation and Treatment," Elsevier Publishers, 2nd edition, 2009.
6) Adams and Victor's, "Principles of Neurology," M C Graw Hill Publishers, 11thedition,2019.

RECOMMENDED REFERENCE TEXT BOOKS

1) Ida Bromley, Tetraplegia and Paraplegia, "A guide for physiotherapists," Churchill Livingstone Publishers, 7th edition, 2006.
2) Suzann K Campell, "Paediatric Neurologic Physical Therapy," Elsevier Publishers, 2nd edition, 1984.
3) Martin Kessler, "Neurologic Interventions for Physical Therapy," Elsevier Publishers, 4th edition, 2020.
4) Susan Edwards, "Neurological Physiotherapy," Elsevier Publishers, 2nd edition, 2001.
5) Patricia M. Davies, "Steps to Follow," Springer -Verlag Publishers, 1st edition, 2000.
6) Karen Whalley Hammell, "Spinal Cord Injury Rehabilitation," Springer-Verlag Publishers, 1st edition, 2014.

RECOMMENDED WEB REFERENCE

1) www.Physiospot.com
2) www.connectneurophysiotherapy.com
3) www.physio.co.uk/treatments.com

4) www.neuroconnecttrehab.co.uk/com
5) www.physiotherapy-treatment.com
6) www.mobilephysiotherapyclinic.in/stroke-physiotherapy.com
7) www.cambridge.org/core/books/stroke-prevention-and-treatment.com

PHYSIOTHERAPY IN OBSTETRICS AND GYNAECOLOGY

COURSE CODE: U21BPTT739 TOTAL HOURS: 80

COURSE DESCRIPTION:

At the end of the course of obstetrics and gynecology, the students will be able to provide proper care in managing women's health including pregnancy, labour and puerperium and thereby to ensure maternal, neonatal health and well-being.

COURSE OBJECTIVE:

At the end of the course, the candidate,

a) To obtain knowledge, expertise and skills towards the physiotherapy interventions in Women's.
b) To understand the role of physiotherapy for various women's health disorders.
c) To gain knowledge about the various physiotherapy interventions at child bearing age.
d) To understand the effective strategies for clinical management of various conditions in older women's. Educate patients regarding the prevention and post-operative care after gynecological surgeries.

SYLLABUS

Unit	**Topics**	**Didadic Hours**	**Clinical/ practical Hours**	**Total Hours**

I	PHYSIOTHERAPY ASSESSMENT IN OBG i. Pre and post-operative assessment for gynecological surgery. ii. Assessment of pelvic pain, bowel dysfunction assessment, menstrual disorders assessment. iii. Pelvic floor assessment: measurement of pelvic floor muscle function and strength. iv. Assessment of urinary dysfunction, Assessment of urinary incontinence. v. Antenatal physiotherapy assessment. vi. Pre natal and post natal physiotherapy assessment.	10	2	12

Unit	Topics	Didadic Hours	Clinical/ practical Hours	Total Hours
II	PHYSIOTHERAPY MANAGEMENT IN ADOLESCENCE i. Physiotherapy management for amenorrhea and dysmenorrhea ii. Physiotherapy management in urinary incontinence.	15	3	18
III	PHYSIOTHERAPY IN ANTENATAL PERIOD i. Importance of prenatal exercise, ii. Antenatal classes and benefits of exercise in pregnancy. iii. Indications, contraindications and precautions. iv. Various exercises during pregnancy flexibility, strengthening and conditioning exercises. v. Relaxation technique in prenatal education. Physiologic basis for relaxation training, Fitness in childbearing year, Psycho analgesic methods of pain control,	20	5	25

	Physiotherapy management of musculoskeletal dysfunction during pregnancy. vi. Physiotherapy management for high risk pregnancy.			
Unit	**Topics**	**Didadic Hours**	**Clinical/ practical Hours**	**Total Hours**
	vii. Physiotherapy management in different stages of pregnancy. viii. Physiotherapy intervention for gestational diabetes and hypertension. ix. Effects of aerobic exercise during pregnancy.			
IV	PHYSIOTHERAPY MANAGEMENT IN POST NATAL PERIOD i. Obstetric anal sphincter injury-anal sphincter exercises, pelvic floor muscle exercises. ii. Postpartum physical/mental condition postnatal care, Educational principles for individual learning of exercises, ergonomics. iii. Postnatal home exercise, exercise classes in the community, Kegels exercises functional exercises, therapeutic modalities for post-operative pain management. iv. Immediate post natal complications and its management, late post natal complications and its management	10	5	15
	PHYSIOTHERAPY INTERVENTION AFTER GYNECOLOGICAL SURGERIES.			

V	i. Hysterectomy ii. After delivery with episiotomy iii. After birth with symphysiolysis,caesarean section iv. After surgical intervention in ectopic pregnancy v. oncological conditions vi. Stress incontinence	10		10

Recommended text books:

1. Jill mantle, Jeanette haslam, Margaret polden “physiotherapy in obstetrics and gynecology” 2nd edition, 2004.
2. Carolyn Kisner, “Therapeutic Exercise Foundations and Technique”, jaypee 7th edition, 2018.
3. Rebecca G. Stephenson, Linda J. O’ Connor, “Obstetric and Gynecologic Care in Physical Therapy”, slack books, 2nd edition, 2016.
4. O'Sullivan, Susan, “Physical rehabilitation: Assessment and treatment” A. Davis 7th edition 2020.

Recommended Reference books:

1. Bo, K., Bergman’s, L.C.M., and Van Kampen, Evidence-based physical therapy for the pelvic floor: Bridging science and clinical practice, Churchill Livingstone 1st edition, 2015.

2. Hacker Moore Gambone, Essential of obstetrics and gynecology, Elsevier, 6th edition 2016.

3. Sapsford R, Markwell, Bullock–Saxton J “Women’s Health: A Textbook for Physiotherapists” WB Saunders Company Ltd 1st edition.

Recommended web references:

1. Labour physiopedia.
2. www.sciencedirect.com physiotherapy in obstetrics and gynecology.

SPORTS PHYSIOTHERAPY

COURSECODE: U21BPTT740 TOTALHOURS:60

COURSE DESCRIPTION:

The subject serves as specialized branch of physiotherapy that focuses on the whole well-being of a person who engages in sports. The course Demonstrates advanced competencies in the promotion of safe physical activity participation, provision of advice, and adaptation of rehabilitation and training interventions. It also extents the purposes of preventing injury, restoring optimal function, and contributing to the enhancement of sports performance, while ensuring a high standard of professional and ethical practice

COURSE OBJECTIVES:

The objective of the course is to enable the student understand about principles of sports physiotherapy, causes for injuries clinical reasoning and therapeutic skills to assess and diagnose sports-related injuries. Furthermore, they are skilled in designing, implementing, evaluating and modifying evidence-based interventions that allow for a safe return to the athlete's optimal level of performance in their specific sport or physical activity.

Unit	**Content**	**Didactic Hours**	**Practical Hours**	**Total Hours**

I	EXERCISE PHYSIOLOGY 1. Training the aerobic and anaerobic energy system 2. Physiological responses, changes & adaptations to various exercises - aerobic exercises & anaerobic exercises in pulmonary, cardiovascular, neuromuscular system, hormones 3. Detraining effects of cardiovascular, musculoskeletal and nervous system 4. Sports specific training and cross training.	12		12
II	SPORTS INJURIES 1. Pre–participation examination Causes &mechanism of sports injuries Prevention of sports injuries to various structures. 2. Common acute, chronic and overuse injuries in various sports at injuries to peripheral nerves, muscles, ligament, tendon, bone, synovial joint structures (with physiological response to injury)	11		11
III	SPORTING EMERGENCIES & FIRST AID 1. Cardio pulmonary resuscitation, shock management, internal and external bleeding, splinting, stretcher use–handling and transfer Management of cardiac arrest, acute asthma, epilepsy, drowning, burn 2. Medical management of mass participation. Heat stroke and heat illness.	12		12
IV	ASSESSMENT IN SPORTS PERFORMANCE 1. Various methods to estimate body composition: water displacement method, under water weighing method, skin fold method, surface anthropometry, bioelectrical impedance analysis, ultrasound assessment of fat, arm x-ray assessment of fat, CT assessment of fat 2. Evaluation of physical fitness: assessment of strength, power, endurance (muscular &cardiac), VO2max, flexibility, 3. Reaction time and pulmonary function.	15		15
V	PHYSIOTHERAPY TREATMENTTECHNIQUES Physiotherapy treatment by using electrotherapy modalities, sports massage, therapeutic exercises and taping techniques for sports injuries	10		10

RECOMMENDED TEXTBOOKS

Brukner and Khan,"*Clinical Sports Medicine*",McGrawHillPublishers, 4th Edition, 2017

2. ... osem cDonald,"*PocketbookofTapingTechniques*",ElsevierHealthSciences,1st Edition 2009

3. Devinder K Kansal, "*Textbook of Applied Measurement, Evaluation &Sports Selection*", Sport &SpiritualSciencePublisher, 2nd Edition, 2008.
4. Mc Ardle, Katch and Katch, "*Essentials of Exercise Physiology*", LippincottWilliams & WilkinsPublisher,3rd Edition,2006.

RECOMMEDED REFERENCE BOOKS

1. BernhardtDonna,"*SportPhysicalTherapy*",ChurchillLivingstonePublisher,1st Edition, 1986.
2. Bird, Black and Newton , "*Sports Injuries: Causes, Diagnosis, Treatment and Prevention*", Cengage Learning; 2nd Edition,1997

3. Brownstein and Bronner, "*Functional Movement in Orthopaedic and Sports Physical Therapy:Evaluation,TreatmentandOutcomes*",ChurchillLivingstonePublisher,1st Edition 1997

4. William Prentice, "*Rehabilitation Techniques in Sports Medicine*", SLACKIncorporated Publisher,7thEdition,2020.

RECOMMEDEDWEBREFERENCE

1. www.SportsPhysiotherapyForAll.org/publications/

CLINICAL CARDIO RESPIRATORY CONDITIONS

COURSE CODE: U21BPTT741 TOTAL HOURS: 90

COURSE DESCRIPTION:

Following the basic science and clinical science courses, this course introduces the student to the cardio – respiratory conditions which commonly cause disability.

COURSE OBJECTIVE:

The objective of this course is after 90 hours of lectures & demonstrations, in addition to clinics, the student will be able to demonstrate an understanding of cardio – respiratory condition causing liability and their management.

SYLLABUS

UNIT	TOPICS	DIDADIC/ THEORY HOURS	CLINICAL/ PRACTICAL HOURS	TOTAL HOURS
	1.Review Of Anatomy & Physiology : i. Anatomy of the lungs, bronchi and Broncho pulmonary segments. ii. the relationship of the bony thorax and lungs iii. abdominal contents and lungs iv. movements of the thorax respirations			

I	v. muscles of respirations vi. Anatomy of the heart and its blood supply and electrical activity of the myocardium and normal ECG. vii. Physiological control of respiration (respiratory center & receptors) viii. Mechanism for maintenance of blood pressure. ix. Cough reflex. x. The mechanical factor involved in breathing. xi. factors affecting diffusion of O^2 and CO^2 in the lungs.	15	-	15

II	1.Cardio Vascular System i. Myocardial Infarction ii. Heart Failure iii. Cardiac Arrest {Definition, Causes Pathogenesis, Signs & symptoms ,Complication ,Management} 2.Rheumatic Fever {Definition, Causes Pathogenesis, Signs & symptoms ,Complication ,Management.} 3.Congenital & Acquired Heart Diseases i. A S D ii. V S D iii. P D A iv. Tetralogy of Fallot's v. Transposition of great vessels vi. AV Malformation vii. Mitral stenosis viii. Mitral regurgitation ix. Aortic stenosis x. Aortic regurgitation. {Definition, Causes Pathogenesis, Signs & symptoms ,Complication ,Management} 4. Ischemic Heart Diseases {Definition, Causes Pathogenesis, Signs & symptoms, Complication, Management.}	20	-	20

III	1. Pulmonary Embolism i. Etiology ii. Clinical features iii. Management 2. Introduction To Cardiac Surgery i. Types of incision ii. Pre- and post-operative assessment iii. Complications iv. Management	10	-	10
IV IV	1. Cardiac Surgery: i. List the cardiac conditions requiring closed heart surgery and briefly describe the following: open heart surgery {OHS} AND closed heart surgery {CHS} a. thoracotomy – median sternotomy b. Heart lung machine c. Angioplasty d. Coronary artery bypass graft e. Percutaneous transluminal coronary angioplasty f. Valve replacement g. valvotomy h. Acquired heart diseases i. Mitral stenosis j. Aortic stenosis k. congenital heart diseases l. Patent ductus arteriosus m. coarctation of aorta	15	-	15
V	1. The clinical features and management of the following: i. Fracture ribs ii. Flail chest iii. Stove in chest iv. Pneumothorax v. Haemothorax vi. Haemopneumo thorax vii. Lung contusion & laceration viii. Injury to heart, great vessels & bronchus ix. Carcinoma of lung. 2. Post-operative management for the following: i. Segmentectomy ii. Lobectomy iii. Bilobectomy			

	iv. Pneumonectomy v. pleuropneumonectomy vi. tracheostomy vii. pleurodesis viii. thoracoplasty ix. lung transplant i. Post extubation care.	30	-	30

Recommended text books:

1. Davidson's, "Principles and Practice of Medicine",Elsevier health sciences, 24th edition, 2022.
2. Harrison's, "Internal Medicine", McGraw Hill / Medical, 21st edition. 2022.
3. Bailey and Love's, "Short Practice of Surgery" CBS publishers and distributors Pvt.Ltd, 28th edition, 2023.
4. Ellen Hilligoss, "Essentials of cardiopulmonary physical therapy" Elsevier health sciences, 5th edition. 2022
5. Gowalla's, "medicine for students", Jaypee brothers' medical publishers, 25th edition, 2017.
6. Downie, patricia.A, "Cash text book of cardio respiratory condition for physiotherapy", Elsevier health sciences, 2005

Recommended references books:

1. Mariya jiandani, "Physiotherapy in cardiopulmonary condition", Jaypee brothers' medical publishers, 2022
2. Eleanor main, Linda Denehy, "Cardiorespiratory physiotherapy", Elsevier health sciences, 5th edition, 2023.
3. Eleanor main, Linda Denehy, "Cardiorespiratory physiotherapy adults and pediatrics", Elsevier health sciences, 5th edition, 2022
4. Fish man's "pulmonary diseases and disorder", 2- volume set., McGraw Hill / Medical., 6th edition, 2022

Recommended web-References:

1. Wiley online library
2. www. Escardio.org/education/textbooks/general-cardiology

RESEARCH METHODOLOGY& BIO-STATISTICS

COURSE CODE: U21BPTT742 TOTAL HOURS: 45

COURSE DESCRIPTION:

The course is designed to enable the students to develop an understanding of basic concepts of research, research process and statistics in professional practice.

OBJECTIVES:

At the end of the course, the candidate should Gain knowledge of the basic concepts of biostatistics & its need for professional practice, research design & methodology of an experiment or survey, sampling & interpretation of date.

SYLLABUS

UNIT	CONTENT	DIDADIC/ THEORY HOURS	CLINICAL/ PRACTICAL HOURS	TOTAL HOURS
I	1. Introduction to Research Methodology i. Meaning of research ii. Objectives of research iii. Types of research & research approaches iv. Criteria for good research v. Problems encountered by researchers in India. vi. Ethics in research 2. Sampling Design i. Criteria for selecting sampling procedure ii. Steps in sampling design iii. Characteristics of good sample design iv. Different types of sample design	10		10
II	1. Methods of data collection i. Collection of primary data ii. Collection of data through questionnaires & schedules	10		10

	iii. Difference between questionnaires & schedules. 2. Testing of hypothesis i. Define hypothesis ii. Basic concepts concerning testing of hypothesis iii. Procedure of hypothesis testing iv. Measuring the power of hypothesis test, v. Tests of hypothesis			
III	1. Introduction to Biostatistics i. Definition ii. Importance of biostatistics iii. Branches of biostatistics 2. Tabulation of Data i. Basic principles of graphical representation ii. Types of diagrams – histograms, frequency polygons, smooth frequency polygon, cumulative frequency curve iii. Normal probability curve.	8		8
IV	1. Measure of Central Tendency i. Definition and calculation of mean, median, mode. ii. Comparison of mean, median and mode 2. Probability And Standard Distributions i. Meaning of probability of standard distribution ii. The binominal distribution iii. The normal distribution iv. Divergence from normality – skewness, kurtosis.	10		10
V	1. Sampling Techniques i. Need for sampling – Criteria for good samples ii. Procedures of sampling and sampling designs errors iii. Sampling variation and tests of significance. 2. Statistical Significance i. Parametric tests: - t test, ii. Non parametric tests: - chi square test, Mann-whitney U test, Z test, Wilcoxons matched pair test	7		7

	iii. Correlations			

Recommended text books:

1. B. K. Mahajan, “Methods in Biostatistics”. Jaypee brothers, 6th edition.
2. C R Kothari “Research methodology methods and techniques”, new age international publishers, 4th edition, 2019.
3. R. pannerselvam, “Research methodology”, 2nd edition, PHI learning, 2014.

Recommended references books:

1. K. Visweshwar Rao, “Biostatistics A manual of Statistics Methods”, Jaypee Brothers Medical Publishers, 2nd edition 2009.
2. Hicks, “Research methodology and biostatistics for physical therapists”, Elsevier Health Sciences, 5th edition, 2009
3. J. Richard, P. S. S. Sundar Rao, “An Introduction to Biostatistics and research methods”, PHI Learning, 2006

Recommended web-References:

1. Www. Researchgate.Net
2. Wiley Online Library

VETERINARY PHYSIOTHERAPY

Course Code: U21BPTT743 TOTAL HOURS: 15 HOURS

COURSE DESCRIPTION:

The course is designed to enable the students to have knowledge of the role of physiotherapy in treating the various disorders in animals and to develop the professional work.

COURSE OBJECTIVE:

At the end of the course, student will understand

1. Various dysfunctions in pet animals.
2. Gain knowledge in order to apply physiotherapy to pet animals.
3. Various animal welfare measures.

SYLLABUS

Unit	Topic	Didactic Hours	Clinical Hours	Total Hours
I	INTRODUCTION TO VETERINARY PHYSIOTHERAPY History of veterinary Physiotherapy & rehabilitation Practice issues in veterinary physiotherapy concepts of veterinary physiotherapy & rehabilitation.	3	-	3
II	BASICS OF VETERINARY MEDICINE Basic anatomy & physiology in view to pet animals.	3	-	3
III	ASSESSMENT 1. Osteoarthritis in canines 2. Physical examination 3. Muscle strength & functions 4. Gait analysis	3	-	3

Unit	Topic	Didactic Hours	Clinical Hours	Total Hours
IV	PHYSIOTHERAPY FOR 1. Orthopaedic conditions 2. Neurologic conditions 3. Geriatric & arthritic conditions 4. Critically injured animals 5. Development of rehabilitation facility for small animals	3	-	3
V	ANIMAL WELFARE MEASURES 1. Blue cross society 2. Medico-legal aspects	3	-	3

Recommended text books:

1. M. M Ansari, “Principles of Veterinary physiotherapy,” Write and Print Publications, 1st edition, 2016.
2. Gemma del Pueyo Motesinous, “Veterinary Physiotherapy and Rehabilitation”, Servet Publications, 14^{th} edition, 2012.

Recommended reference books:

1. Catherine Mc Gowan, “Animal Physiotherapy”, Wiley Blackwel Publications, 2^{nd} edition, 2016.

CLINICAL TRAINING

Course code: U21BPTP745 **Total:** 90 hours

Course Description

The Student will learn the subjective and objective data taking procedure.

Course Objective

End of the course the students will able to

1. Understand the history taking procedure
2. Various procedures in patient’s assessments
3. Learn the various treatment procedures

Sl. No	TOPIC	HOURS
1.	Demographic data Collection	
2.	Socioeconomic history collection	
3.	Social behavior and its influence on health	
4.	Schedule the patients treatment according to their condition	

SCHEME OF EXAMINATION

IA	FINAL EXAM	TOTAL
50	50	100

END SEMESTER EXAMINATION

Sl. No	Components	Marks
1	2 case presentation	25 ×2 = 50

Note: Each candidate has to document minimum 10 cases in the log book.

VIII -Semester

PHYSIOTHERAPY IN CARDIO RESPIRATORY CONDITIONS

COURSE CODE: U21BPTT846 TOTAL HOURS**:** 180

COURSE DESCRIPTION:

The course is designed to enable the students to integrate the knowledge in clinical cardio-respiratory conditions with the skills gained in exercise therapy, electrotherapy and therapeutic massage and thus enabling to apply in clinical situations of dysfunction due to cardio-respiratory pathology.

COURSE OBJECTIVE**:**

The objective of this course is that after 90 hours of lectures, demonstrations, practical and clinics the student will be able to identify cardio-respiratory dysfunction, set treatment goals and apply their skills in exercise therapy, electrotherapy and massage in clinical situation to restore cardio-respiratory function

UNIT	**TOPICS**	**DIDADIC/ THEORY HOURS**	**CLINICAL/ PRACTICAL HOURS**	**TOTAL HOURS**

I	1. Review Of Basic Applied Anatomy & Physiology i. Pulmonary Anatomy & Physiology ii. Cardiac anatomy & Physiology iii. Cardiac and Respiratory Pharmacology iv. Biomechanics of Thorax (Revision) 2. Investigation And Exercise Testing i. Investigation & Clinical Implication - X-ray, PFT, ABG, ECG, ABI, claudication time, pulses, auscultation, postural hypotension ii. Stress testing i. 6 Minute Walk test & Harward Step test Skill & Interpretation ii. Shuttle Walk Test & Modified Bruce Protocol (should be interpretation only) 3. Exercise Physiology i. Nutrition (Bioenergetics) ii. Total energy expenditure (MET) sources iii. Acute and chronic adaptation to exercise iv. Complication of bed rest/ Immobilization & prevention v. Aerobic & Anaerobic Training, vi. f. Principles of Exercise Prescription	20	20	40

II	1. Physiotherapy Skills i. Bronchial Hygiene Therapy- Postural Drainage, Forced Expiratory Technique,ACBT,Autogenic Drainage ii. Adjunct Therapy – Flutter & PEP Therapy iii. Therapeutic positioning to improve ventilation & perfusion matching. iv. Therapeutic positioning to alleviate dyspnea v. Breathing exercises vi. Nebulization vii. Humidification viii. Lung Expansion Therapy ix. Neurophysiological facilitation of respiration x. Electrotherapeutic modalities for pain, swelling, &wound healing. xi. Strengthening program for respiratory muscle xii. Applied Yoga in Cardio-respiratory conditions	20	20	40
III	1. PHYSIOTHERAPY MANAGEMENT in: i. Medical & Surgical Cardiovascular Diseases a. Hypertension b. I.H.D., Myocardial Infarction c. Valvular Heart Disease d. Other Arterial disorders ii. Obstructive & Restrictive Respiratory disorders a. Bronchitis b. Emphysema c. Bronchial Asthma d. Cystic Fibrosis e. Occupational lung diseases f. Interstitial Lung Diseases iii. General Respiratory Infection a. Tuberculosis b. Pneumonia c. Lung Abscess d. Bronchiectasis	20	20	40

	e. Pneumothorax f. Hydropneumothorax g. Atelectasis h. Pleuritis i. Pleural Effusion j. Empyema & other Pleuralk k. Disorders			
IV	1. Cardiac Rehabilitation (A.H.A./A.C.S.M. guidelines) i. Definition, ii. Indications, Contraindications iii. Phases (I,II,III,& IV) iv. Outcome Measures 2. Pulmonary Rehabilitation (A.A.C.V.P.R. /A.T.S. guidelines) i. Definition, ii. Indications iii. Contraindications iv. Components of management v. Outcome measures 3. I.C.U. Evaluation & Management i. Basic evaluation ii. Principles of ICU Monitoring iii. Mechanical Ventilator modes iv. Suctioning v. Humidification vi. Therapeutic intervention in a. Tetanus b. Head Injury c. Pulmonary Oedema d. neuromuscular disease e. Smoke Inhalation f. Poisoning g. Aspiration near Drowning h. A.R.D.S. i. Shock j. Spinal Cord Injury & Other Acute respiratory Disorders	15	15	30

V	1. Introduction To Functional Scales i. Generic and disease specific ii. b. Patient"s perception of his disability and functioning and correlating the same with therapist evaluation	15	15	30

Recommended text books:

1. Downie, Patricia.A, "Cash"s textbook for physiotherapists in chest, heart & vasculardiseases", Mosby, 4th edition, 1993
2. Downie, Patricia.A, "Cash"s text book in general medicine & surgical conditions forphysiotherapists", Mosby; 2nd edition, 1990
3. Pryor and prasad, "Physiotherapy in respiratory and cardiac problem" Elsevier Health –,4th edition, 2013.
4. Colin Mackenzie, "Chest physiotherapy in intensive care", Williams & Wilkins 1989
5. Pierce, "Management of mechanical ventilation", W B Saunders Co Ltd; 2nd edition, 2006
6. Subin Solomen & Pravin Aaron, "Techniques in cardiopulmonary physiotherapy", PEEPEEpublishers.

Recommended references books:

2. Irwin Scott, "Cardiopulmonary physical therapy", CBS Publishers, 4th edition, 2004.
3. GB Madhuri, "Textbook of physiotherapy for cardio-respiratory cardiac surgery andthoracic surgery condition", Jaypee bothers, 1st edition, 2008.
4. GB Madhuri, "Textbook of physiotherapy for clinical cardiothoracic condition", Jaypeebothers, 2nd edition, 2022.
5. Hilligoss and sodosky, Saunders, "Essential of cardio pulmonary physical therapy",Elsevier Health, 4th edition, 2016.
6. McArdle, Lippincott, "Exercise physiology, energy, nutrition and human performance",Williams & Wilkins, 8th edition, 2014.
7. Skinner, Lippincott, "Exercise testing and prescription", Williams and Wilkins,3rd edition, 2005.

Recommended web-References:

1. wiley online library
2. www. escardio.org/education/textbooks/general-cardiology

COMMUNITY PHYSIOTHERAPY

COURSE CODE: U21BPTT847
TOTALHOURS: 110

COURSE DESCRIPTION:

Community Physiotherapy describes the roles & responsibilities of the Physiotherapist as an efficient member of the society. This component introduces the physiotherapist to a proactive preventive oriented philosophy for optimization &betterment of health.

OBJECTIVES:

6. Physiology of aging process and its influence on physical fitness.
7. National policies for the rehabilitation of disabled– role of PT.
8. The evaluation of disability and planning for prevention and rehabilitation.
9. Rehabilitation in urban and rural setup.
10. Be able to gain the ability to collaborate with other health professionals for effective service delivery & community satisfaction.

SYLLABUS

Sl. No.	Topics	Didactic Hours	Practical Hours	Total Hours
I	HEALTH PROMOTION: W.H.O. definition of health and disease. Health Delivery System –3 tier Physical Fitness: Definition and evaluation related to obesity. COMMUNITY: Definition of community, Multiplicity of community, the community-based approach, Community entry strategies, CBR and Community development, Community initiated versus community-oriented program.	8	10	18

Sl. No.	Topics	Didactic Hours	Practical Hours	Total Hours
II	COMMUNITY BASED REHABILITATION: Definition, historical review, concept of CBR, Need of CBR, objectives of CBR, scope of CBR, Members of CBR team, differences between institution based & community based rehabilitation. Agencies involved in rehabilitation of physical challenged, legislation for physically challenged. Role of physiotherapy in CBR.	8	10	18
III	DISABILITY: Definition of impairment, handicap and disability, difference between impairment, handicap and disability, causes for disability, types of disability, prevention of disability, disability in developed countries and in developing countries. Disability surveys: demography. Screening: early detection of disabilities and development disorders Prevention of disabilities - types and levels.	8	15	23

Sl. No.	Topics	Didactic Hours	Practicals Hours	Total Hours
	DISABILITY EVALUATION: Introduction, what, why and how to evaluate, quantitative versus qualitative data, uses of evaluation findings.			
IV	GERIATRICS: Senior citizens in India NGO, and health related legal rights and benefits for the elderly. Institutionalized &community dwelling elders. Theories of aging. Physiology of ageing: Musculoskeletal, neurological, cardiorespiratory, metabolic changes. Scheme of evaluation &role of PT in geriatrics.	9	15	24
V	INDUSTRIAL HEALTH Introduction to industrial health: definition, model of industrial therapy (Traditional medical &industrial model) Worker care spectrum: Ability management – job analysis - job description, job demand analysis, task analysis, ergonomics evaluation, injury prevention and employee fitness program.	12	15	27

Recommended textbooks

- T. Bhaskara Rao, "Textbook of community medicine & community rehabilitation", paras'medical publisher,2013.
- DaleAvers, RitaA.wong "Guccione's Geriatric physical therapy", Elsevier publications, 4th edition,2020.
- GlendaL.Key"Industrialtherapy",Mosby,1995.
- Pruthrish "Community based rehabilitation of person with disabilities", Jaypeepublishers,2006.

Recommended Reference Books

3. K. Park "Textbook of preventive and social medicine", Generic publisher, 22nd edition, 2013.
4. Helander, Einar "Training disabled people in the community: a manual on community based rehabilitation for developing countries", world health organization, 1983.

Recommend web references

2. "Punarbhava"-National interactive web portal on disability.

www.physio-pedia.com

REHABILITATION MEDICINE

COURSE CODE: U21BPTT848 TOTAL HOURS: 50

COURSE DESCRIPTION:

This course is designed to enable the students to understand the principles of rehabilitation and the role of physiotherapy in the rehabilitation team.

COURSE OBJECTIVES:

The objectives of this course are that after 50 hours of lectures and demonstrations, in addition to clinic, the student will be able to demonstrate an understanding of:

a. The concept of team approach in rehabilitation will be discussed and implemented, through practical demonstration with contributions from all members of the team.
b. Observation and identification of diagnostic features in physical conditions will be practiced through clinical demonstration.
c. Medical and surgical aspects of disabling conditions will be explained in relation to rehabilitation.
d. Formulation of appropriate goals (long & short term) in treatment & rehabilitation will be discussed.
e. Student should be able to prescribe, check - out and train the uses of various rehabilitation aids.

SYLLABUS

Unit	Topics	Didadic Hours	Clinical /practical Hours	Total Hours
I	INTRODUCTION TO REHABILITATION i. Definition ii. Aims and principles iii. Scope of rehabilitation iv. Impairment,Disablement,Handicap v. Rehabilitation team and its members vi. Physiotherapy in rehabilitation	10	-	10
II	COMMUNICATION AND BEHAVIOURAL DISORDERS 1. Communication disorders- Definition, types, principles of management 2. Behavioural disorders- Definition, types, principles of management	5	-	5
III	PHYSICAL DYSFUNCTION i. Methods of evaluation for physical dysfunction ii. Management of disabilities with reference to: a. Spinal cord injury b. Poliomyelitis, muscular dystrophy c. Cerebral palsy d. Stroke, Burns, Arthritis e. Peripheral nerve lesions f. Sports injuries & cardio respiratory dysfunction.	10	-	5

IV	ORTHOTICS , PROSTHETICS AND MOBILITY AIDS i. Definition & types of Orthosis ii. Principles of orthotic devices iii. Various orthotic devices & its functional uses iv. Indications & Contra-indications for orthotics, calipers etc. v. Definition, types & functions of prosthesis vi. Various types of artificial limbs vii. Definition, types & functions of mobility aids viii. Indications for different types of mobility aids	15	-	15
V	ARCHITECTURAL BARRIERS i. Definition Architectural components as barriers for the impaired. ii. Principles of modifications with reference to a. Rheumatoid arthritis b. Cerebro vascular accident c. Spinal cord injury and other disabling conditions.	10	-	10

Recommended books:

1. S. Sunder, "Textbook of Rehabilitation "Jaypee brothers, 4th edition, 2019.
2. O'Sullivan, Susan B, Physical rehabilitation: Assessment and treatment, F. A. Davis 7th edition 2020.
3. Leslie Rydberg, Sarah Hwang, "Physical Medicine and Rehabilitation Pocketpedia" Demos Medical, 4th edition, 2023.
4. Frederic J. kottke, Frank H.Krusen, " Krusens handbook of physical Medicine and Rehabilitation", Saunders, 4th edition, 1990.

Referred books:

1. Michelle M. Lusardi**, "**Orthotics and Prosthetics in Rehabilitation" Saunders, 3rd edition, 2012.
2. David X. Cifu "Braddom's Physical Medicine and Rehabilitation" Elsevier, 6th edition, 2020.

Web reference:

1. https://rehabcouncil.nic.in/
2. https://www.physio-pedia.com/Rehabilitation
3. https://musculoskeletally.com
4. https://now.aapmr.org
5. https://www.stroke.org

CLINICAL REASONING AND EVIDENCE BASED PRACTICE

COURSE CODE: U21BPTP850 TOTAL HOURS**:** 30 hours

COURSE DESCRIPTION:

The subject serves as knowledge and the generation of hypothesis of factorsassumed to underlying limitations of physical capacity and performance.

COURSE OBJECTIVES:

The objective of the course is to enable the student to understand best available evidence with clinical judgment of patient‟s preferences, values, and that further considers the larger social context in which physical therapy services are provided to optimize patient outcomes.

SYLLABUS

Unit	Content	Didactic Hours	Practical Hours	Total Hours
I	1. EVIDENCE BASED PRACTICE: Definitions, introduction. 2. CONCEPTS OF EVIDENCE BASED PHYSIOTHERAPY: Awareness, Consultation, Judgement, and Creativity	4	0	4
II	1. DEVELOPMENT OF EVIDENCE BASED KNOWLEDGE: The Individual Professional, Professionals within a discipline, and Professionals across disciplines. 2. EVIDENCE BASED PRACTITIONER: The Reflective Practitioner, The E Model and its implication	4	0	4
III	1. FINDING THE EVIDENCE: Measuring outcomes in evidence based practice, measuring health outcomes, measuring clinical outcomes, inferential	12	0	12

	statistics and causation 2. SEARCHING FOR THE EVIDENCE: Asking Questions, Identifying different sources of evidence, electronic bibliographic databases and World Wide Web, conducting a literature search. step by-step search for evidence 3. ASSESSING THE EVIDENCE: EVALUATING THE EVIDENCE; Levels of evidence in research using quantitative methods, levels of evidence classification system, outcome measurement, biostatistics, the critical review of research using qualitative methods 4. SYSTEMATICALLY REVIEWING THE EVIDENCE: Stages of systematic reviews, Meta-analysis, the Cochrane collaboration 5. ECONOMIC EVALUATION OF THE EVIDENCE: Types of economic evaluation, conducting economic evaluation, critically reviewing economic evaluation, locating economic evaluation in the literature 6. USING THE EVIDENCE: Building evidence in practice; Critically Appraised Topics (CATs), CAT format, using CATs, drawbacks of CATs			
IV	1. PRACTICE GUIDELINES, ALGORITHMS, AND CLINICAL PATHWAYS: Recent trends in health care, Clinical Practice Guidelines (CPG), Algorithms, Clinical pathways, Legal implications in clinical pathways and CPG, Comparison of CPGs, Algorithms and Clinical Pathways 2. COMMUNICATING EVIDENCE TO CLIENTS, MANAGERS AND FUNDERS: Effectively communicating evidence, Evidence based communication in the face of uncertainty; Evidence based communication opportunities in everyday practice.	7	0	7
V	RESEARCH DISSEMINATION AND TRANSFER OF KNOWLEDGE: Models of research transfer, Concrete research transfer strategies, Evidence based policy.	3	0	3

RECOMMENDED TEXT BOOKS

1. Dianne V. Jewell , "*Guide to Evidence-Based Physical Therapist Practice*", Jones and Bartlett Publishers, 2nd Edition, 2010.
2. Mark A. Jones; Darren A Rivett, "*Clinical Reasoning in Musculoskeletal Practice*", Elsevier Publishers, 2nd Edition, 2018.
3. Eileen D. Gambrill, "*Critical Thinking and the Process of Evidence-Based Practice*", OUP USA Publishers, 1st Edition, 2018.

RECOMMENDED REFERENCE BOOKS

1. Scot Raab; Deborah I. Crai , "*Evidence-Based Practice in Athletic Training*", Human Kinetics Publishers, 1st Edition, 2016.
2. William E. Amonette; Kirk L. English; William J. Kraemer, "*Evidence-Based Practice in Exercise Science*", Human Kinetics Publishers, 1st Edition, 2015.
3. Brent L. Arnold; Brian Schilling , "*Evidence Based Practice in Sport and Exercise*", F.A. Davis Company Publishers, 1st Edition, 2016.

RECOMMENDED WEB REFERENCE

1. https://www.physio-pedia.com/Evidence_Based_Practice(EBP)_in_Physiotherapy
2. https://www.physio-pedia.com/Clinical_Reasoning

EDUCATION TECHNOLOGY

COURSE CODE: U21BPTP852 TOTAL HOURS**:** 30 hours

COURSE DESCRIPTION:

The course aims at introducing teaching methods, curriculum principles, learning process, teaching strategies, class room organization and management including micro-teaching to students.

COURSE OBJECTIVES:

At the completion of the course the students are expected to acquire knowledge about the philosophies of education, values, principles of curriculum, and techniques of teaching. Psychological tools and principles of guidance and counselling.

SYLLABUS

Unit	Topic	Didactic Hours	Clinical Hours	Total Hours
I	EDUCATION AND PHILOSOPHY 1. Relationship between philosophy and Education-Educational aims. 2. Elements of modern and contemporary philosophies of education-idealism, naturalism, pragmatism and existantialism. 3. Concept of values of with reference to Sri Aurobindo, mahatma gandhi, john dewey and rousseau ethical standards. 4. Agencies of education - formal, informal and non formal education.	6	-	6

Unit	Topic	Didactic Hours	Clinical Hours	Total Hours
II	CURRICULUM DESIGN 1. Concept of curriculum 2. Types and approaches of curriculum disciplinary integrated problem based. 3. Determinants of curriculum 4. Organization of curriculum.	5	-	5
III	CONCEPT OF TEACHING AND LEARNING 1. Principles of learning 2. Maxims of teaching 3. Techniques of teaching - planning of teaching lesson Plan, micro teaching, development of skills lecture, group discussion, case study, role play, simulated patient management demonstration.	6	-	6
IV	INSTRUCTURAL MEDIA i. Communication concept theory ii. Selection of media – principles iii. AV aids a) Black board b) Display board c) Over head projector (O.H.P) d) Video e) Power point presentation	6	-	6
V	GUIDANCE AND COUNSELLING 1. Philosophy, principles and concepts 2. Need for guidance objectives of guidance kinds of guidance educational, vocational, personal and social. 3. Types of counseling- directive, non-directive, eclectic and group counselling. 4. Guidance and counselling services for students and faculty 5. Continuing education.	7	-	7

RECOMMENDED TEXT BOOKS

1) Mohanthy.J."Indian Education in the Emerging Society," Sterling publishers, 2nd edition, 2007.
2) Tammer, Daniel & Lanvel N. Thanner "Curriculum Development - Theory into practice," MacMillian Publishers, 4th edition, 2012.
3) Ebel Robert.L. "Measuring Educational Achievement," Prentice Hall of India Publisers, 3rd edition, 2000.
4) Sampath K. & Panneerselvam.S, "Introduction to Educational Technology," 3rd edition, 2010.

RECOMMEDED REFERENCE BOOKS

1) Kumar.K.L. "Educational Technology," New Age International Publishers, 1st edition 2004.
2) Thorndike. R.L, "Measurement and Evaluation in Psychology and Education," John Wiley & Sons Publishers, 8th edition, 2011.
3) Chauhan.S.S. "principles and Techniques of Guidance," Vikas publishing House, 2nd edition, 2009.
4) Indu Dave "The Basic Essentials of Counselling," Sterling publishers, 4th edition, 2007.

RECOMMENDED WEB REFERENCE:

1) www.twinkl.co.in.com
2) www.rand.org.com
3) www.aeseducation.com

CLINICAL TRAINING

COURSE CODE: U21BPTP853 TOTAL HOURS**:** 90 hours

The students are posted in various departments in S.M.V.M.C.& H. such as orthopaedics, neurology, general medicine, general surgery, paediatrics and O.B.&G. on rotational basis.